

A Comprehensive Introduction to Tensors

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Preface

0.1 Prerequisites

0.2 Organization

0.3 On Exercises

0.4 Acknowledgements

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Part I

Foundations

CHAPTER 1

Review of Linear Algebra

1.1 Vector Spaces

Before we describe in detail what a vector space is, we must first understand the two operations that help serve to define a vector space. We take the concept of addition and multiplication that we understand from arithmetic and generalize:

Definition 1 (Addition). *If we have a set V and elements $u, v \in V$, then addition is an operation that assigns an element $u + v \in V$ to those two vectors.*

Similarly, we can construct a definition for multiplication:

Definition 2 (Scalar Multiplication). *If we have a set V and elements $u \in V$ and $\lambda \in \mathbb{F}$, where $\mathbb{F}$ can be either $\mathbb{R}$ or $\mathbb{C}$ then scalar multiplication assigns an element $\lambda u \in V$ to that vector and scalar.*

Note that we have defined multiplication of a vector by a *scalar*. We will discuss the multiplication of two vectors together later, but you may recall from linear algebra that there exist an inner product (a “dot product”), and a cross product, the former of which produces a scalar, and the latter of which produces a vector.

Now we can define a vector space. The purpose of this mathematical concept is to allow us to generalize the properties of addition and multiplication on vectors in Euclidean 2-space or 3-space, or, more concisely, $\mathbb{R}^2$ or $\mathbb{R}^3$. We will find that in our study of tensor calculus, the concept of a vector space will become even more general. The setting for this abstraction is a field, denoted $\mathbb{F}$, which in our applications will usually be $\mathbb{R}$ or $\mathbb{C}$.

Definition 3 (Vector Space). *A vector space is a set V containing two operations, an addition and a scalar multiplication, both on V , and:*

- **Commutativity:** $u + v = v + u \quad \forall u, v \in V$.
- **Associativity:** $(u + v) + w = u + (v + w)$ and $(\lambda\mu)u = \lambda(\mu u) \quad \forall u, v, w \in V$ and $\forall \lambda, \mu \in \mathbb{F}$.
- **Distributivity:** $\lambda(u + v) = \lambda u + \lambda v$ and $(\lambda + \mu)u = \lambda u + \mu u$ for all $\lambda, \mu \in \mathbb{F}$ and all $u, v \in V$.
- **Additive Identity:** *There exists some element $0 \in V$ such that $u + 0 = u$ for all $u \in V$. In other words, there exists a vector that functions very similarly to zero with respect to addition in arithmetic; any vector plus the identity equals that same vector.*
- **Multiplicative Identity:** *There exists some element $1 \in \mathbb{F}$ such that $1u = u$ for all $u \in V$.*
- **Additive Inverse:** *For every $u \in V$, there exists $v \in V$ such that $u + v = 0$. In less formal terms, for every vector in the vector space, there must exist some vector also in the vector space that, when the two are added together, make the additive identity vector; this is similar to how a number plus its negative of that number equals zero.*

You may be familiar with the definition of a vector as a set of 2 or 3 numbers, like $(1, 2, 3)$. This is not entirely false, but it is not general enough for our purposes. With the concept of a vector space in hand, we define the vector as follows:

Definition 4 (Vector). *A vector is considered to be an element of a vector space.*

In certain contexts, it may be useful to further classify vector spaces as either real or complex. We find in linear algebra that real and complex vector spaces exhibit certain distinct properties from each other, although they are very similar in many ways, by the definition of a vector space.

Definition 5 (Real and Complex Vector Spaces). *A vector space where $\mathbb{F} = \mathbb{R}$ is called a real vector space. A vector space where $\mathbb{F} = \mathbb{C}$ is called a complex vector space.*

It is now time to begin proving some basic theorems about this mathematical concept. These proofs should be fairly straightforward to understand and should even be review for the audience of this textbook.

Theorem 1. *A vector space has a unique additive identity.*

Proof. Suppose that we have $a, b \in V$ such that a and b are both additive identities of V . Then we must have

$$a = a + b = b + a = b \quad (1.1)$$

Hence, $a = b$, and so V must have a unique additive identity. $\square$

Theorem 2. *Every element in a vector space has a unique additive inverse.*

Proof. We have a vector $v \in V$. Let u and w be additive inverses. Then

$$u = u + 0 = u + (v + w) = (u + v) + w = 0 + w = w \quad (1.2)$$

Hence $u = w$, and so there must be only one unique additive inverse. $\square$

Theorem 3. *Zero times a vector is equal to the zero vector, or $0v = 0$, for every $v \in V$.*

Proof. We have that

$$0v = (0 + 0)v = 0v + 0v \quad (1.3)$$

Adding $-0v$ to both sides of the equation, we obtain

$$0 = 0v \quad (1.4)$$

$\square$

Theorem 4. *A scalar times the zero vector is equal to the zero vector, or $a0 = 0$, for all $a \in \mathbb{F}$.*

Proof. Given some $a \in \mathbb{F}$, we have

$$a0 = a(0 + 0) = a0 + a0 \quad (1.5)$$

Adding $-a0$ to both sides of the equation, we obtain

$$0 = a0 \quad (1.6)$$

□

Theorem 5. *$(-1)v = -v$ for all $v \in V$, where $-v$ denotes the additive inverse of v .*

Proof. We are given a $v \in V$, and therefore

$$v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0 \quad (1.7)$$

Hence $(-1)v$ must be the additive inverse of v . □

Now that we have a better idea of what vector spaces are, it is time to define another useful mathematical concept: the subspace. Oftentimes, we will deal within vector spaces inside vector spaces, and the concept of the subspace helps us formalize and deal with many of these ideas efficiently.

Definition 6 (Subspace). *A subspace of a vector space is a vector space that is a subset of a vector space; this subspace must itself be a vector space under the same addition and scalar multiplication operations defined on V .*

As the vector space is closed under addition and scalar multiplication by Definitions 1 and 2, it follows that the subspace must be closed under the same operations:

Remark 1. *Given some subspace W and vectors $v, w \in W$, we must have that $v + w \in W$ and $\lambda v \in W$.*

It is time to review some concepts that should hopefully be familiar from linear algebra. For example, the linear combination:

Definition 7 (Linear Combination). *A linear combination of some vectors $u, v, w, \dots \in V$ is an expression*

$$au + bv + cw + \dots \quad (1.8)$$

Theorem 6. *A subspace W where $v, w \in W$ and $\lambda, \mu \in \mathbb{F}$ must contain all linear combinations $\lambda v + \mu w$.*

The proof is left as an exercise for the reader.

We turn our attention briefly toward matrices, and find that we can apply our concept of vector spaces to describe the properties of matrices in greater detail:

Definition 8 (Column Space). *Given a matrix A , the column space of A contains all the linear combinations of the columns of A .*

We introduce a couple more concepts from linear algebra, for the sake of review:

Definition 9 (Linear Independence). *A set of vectors $v_1, \dots, v_n \in V$ is linearly independent if*

$$a_1 v_1 + \dots + a_n v_n = 0 \implies a_1 = \dots = a_n = 0 \quad (1.9)$$

Definition 10 (Span). *A set of vectors $v_1, v_2, \dots \in V$ spans a vector space V if any vector $u \in V$ can be represented as a linear combination of $v_1, v_2, \dots$.*

Theorem 7. *The system $Ax = b$ is solvable if and only if b is in the column space of A .*

Proof. If b is not in the column space of A , then no linear combination of the columns of A will equal b .

Remark 2. *The column space of A can also be described by all vectors Ax , where A and x have appropriate dimensions.*

Therefore, if b is not in the column space of A , we must have that $Ax = b$ has no solution. Therefore, $Ax = b$ is solvable if b is in the column space of A . Similarly, we conclude that if $Ax = b$ is solvable, then there must exist some linear combination of the columns of A that equal b (see the remark). Hence, we have proved the claim in both directions. $\square$

The concept of the basis is an incredibly useful one, because with the basis, we are able to view vector spaces similarly to how we can view Euclidean n -space: with components. The vector $(1, 2, 3)$, for instance, simply shows its components with respect to the standard basis in that expression. The concept of a basis also allows us to formalize the concept of a dimension mathematically. This will prove invaluable in our study of manifolds and tensors later on in the book.

Definition 11 (Basis). *The vectors $v_1, v_2, \dots, v_n \in V$ form a basis of V if $v_1, v_2, \dots, v_n$ are linearly independent and span V .*

Definition 12 (Dimension). *The dimension is the number of vectors in a basis.*

Definition 13 (Components). *Let $v_1, \dots, v_n$ be a basis of V . Then a vector $u \in V$ can be described by its components $u = (a_1, a_2, \dots, a_n)$, where*

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = u \quad (1.10)$$

1.2 Dual Spaces

The idea of a dual vector space will become paramount to understanding the indices of tensors, but for now it will serve us well to define what a dual space is. For this, we turn to the mathematical tool of linear functionals, which simply take a vector and map it to a scalar. In this way, it can be thought of as the “opposite” of a vector.

Definition 14 (Linear Functional). *A linear functional is a linear transformation from V to $\mathbb{F}$.*

In quantum mechanics, the linear functional is called a “bra,” whereas a vector is called a “ket.” This is because, when you take the linear functional of a vector, you obtain a scalar; this is another way to view the inner product, as a product between a bra and a ket, or the linear functional of a vector.

Using the concept of linear functionals, we define the dual space:

Definition 15 (Dual Space). *The dual space V^* is the set of all linear functionals from V to $\mathbb{F}$.*

The introduction of the concept of the dual space begs the question of what the dimension of the dual space necessarily is. And it turns out that it is simply the dimension of the vector space it is the dual of:

Theorem 8. *If we have a finite-dimensional V (i.e. the dimension is not infinity), then V^* has the same dimension as V .*

Proof. The following lemma is stated, with the proof being a simple exercise in linear algebra.

Lemma 1. *The set of linear maps from a vector space V to a vector space W has a dimension that is the product of $\dim V$ and $\dim W$.*

Therefore, we have

$$\dim V^* = \dim V \dim \mathbb{F} = \dim V \quad (1.11)$$

Therefore, V^* has the same dimension as V . $\square$

It is time to introduce ourselves to various different kinds of functionals that will serve greater importance. We consider how a vector v can be represented by coordinates $x_1, \dots, x_n$. Therefore, it is useful to define a sort of “coordinate function” that maps a given vector to one of its coordinates, like so:

Definition 16 (Coordinate Functions). *The coordinate functions are functionals φ_i such that, given a vector $v = (x_1, \dots, x_n)$,*

$$\varphi_i(v) = x_i \quad (1.12)$$

Additionally, since we have discussed bases in vector spaces before, the question naturally arises: can we have a basis of a dual space? We can, and it is called the dual basis. It is induced from the basis of the vector space:

Definition 17 (Dual Basis). *The dual basis is a set of functionals $\varphi_1, \dots, \varphi_n$ where, given a basis $v_1, \dots, v_n$ of V , we have that*

$$\varphi_i(v_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad (1.13)$$

It may serve us well to condense this into the following notation:

Definition 18 (Kronecker Delta). *The Kronecker Delta is a function as follows:*

$$\delta_j^i = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad (1.14)$$

We clearly see that the dual basis of a basis vector is equivalent to the Kronecker Delta function.

This Kronecker Delta will surely come up again in our studies of tensors at various points, so it will be useful to define it now. As it turns out, the Kronecker Delta has a myriad of applications in differential geometry, tensor calculus, and general relativity, and is somewhat analogous to the identity matrix, and certainly equal in importance.

Below are a couple useful theorems related to dual bases. These theorems will illuminate certain properties of the dual basis as well as its relationship to the basis of the vector space.

Theorem 9. *If we have a basis $v_1, \dots, v_n \in V$ and a dual basis $\varphi_1, \dots, \varphi_n$, then*

$$v = \varphi_1(v)v_1 + \dots + \varphi_n(v)v_n \quad (1.15)$$

Proof. We have that $v = c_1v_1 + \dots + c_nv_n$, as all vectors in a vector space are linear combinations of the basis. It follows by the definition of a dual basis that $\varphi_i(v) = c_i$. Hence, we obtain our desired result:

$$v = \varphi_1(v)v_1 + \dots + \varphi_n(v)v_n \quad (1.16)$$

□

Theorem 10. *The dual basis of a basis of V must be a basis of V^* .*

Proof. We have a basis $v_1, \dots, v_n \in V$, and a dual basis $\varphi_1, \dots, \varphi_n \in V^*$. Let us suppose that there exists some $a_1, \dots, a_n \in \mathbb{F}$ such that

$$a_1\varphi_1 + \dots + a_n\varphi_n = 0 \quad (1.17)$$

Since $(a_1\varphi_1 + \dots + a_n\varphi_n)v_k = a_k$, then we have that the coefficients of the dual basis must all vanish. Therefore, the dual basis is linearly independent. Note that the dual basis also spans V^* , and so we conclude that the dual basis is a basis of V^* . $\square$

The Dirac functional is a very special type of functional. It is defined only when the vector space V contains functions, as defined below.

Definition 19 (Dirac Functional). *Let V be the vector space of all continuous real functions on the interval $[0, 1]$. Given a vector $v(x) \in V$, The Dirac Functional is defined as*

$$\delta(v) = v(0) \quad (1.18)$$

We have already established that the dimension of the vector space is equal to the dimension of the dual space, but can we establish an isomorphism between the two spaces? The answer is straightforward.

Theorem 11. *Let φ be the dual vector corresponding to v ; that is, the functional $\varphi(w) = \langle v, w \rangle$, where $\langle \cdot, \cdot \rangle$ denotes the inner product (see Appendix C). We have that the map*

$$v \mapsto \varphi \quad (1.19)$$

must be an isomorphism between V and V^ .*

Proof. Let us begin by proving that this map is linear. Consider the vector $u + v$. Then the mapping gives:

$$u + v \mapsto \varphi_{u+v} = \langle u + v, \cdot \rangle = \langle u, \cdot \rangle + \langle v, \cdot \rangle = \varphi_u + \varphi_v \quad (1.20)$$

Then, we consider the vector λv .

$$\lambda v \mapsto \varphi_{\lambda v} = \langle \lambda v, \cdot \rangle = \lambda \langle v, \cdot \rangle = \lambda \varphi_v \quad (1.21)$$

Hence, the map must be linear. Now, if $\varphi = 0$, then it follows that $\langle v, w \rangle = 0$ for all w , and so $v = 0$. Therefore, $v \mapsto \varphi$ is injective, and since $\dim V = \dim V^*$, the map is also an isomorphism. $\square$

One final topic that may be of importance is that of the annihilator. The annihilator of a given subset of the vector space is a subset of the dual space where each functional takes every member of the subset of the vector space to zero. In more formal terms, the annihilator is defined like so:

Definition 20 (Annihilator). *We have a subset $U \subseteq V$. The annihilator $U^0 \subseteq V^*$ is the set of linear functionals φ such that $\varphi(u) = 0$ for all $u \in U$.*

We will eventually know the vectors and dual vectors by different names (if you are curious, the names are contravariant and covariant, respectively), but it is important to establish a strong knowledge base in vector spaces and dual spaces in order to provide the groundwork for our studies in tensors later.

1.3 Bilinear Forms

It is now useful to study a field of mathematics known as multilinear algebra, which studies those kinds of maps with multiple different arguments. We begin with the simplest multilinear map, that being the bilinear map. Below is a quick definition of the Cartesian product for the reader's convenience:

Definition 21 (Cartesian Product). *The Cartesian product of two sets U and V is the set $U \times V$ of all ordered pairs (u, v) , where $u \in U$ and $v \in V$.*

As it turns out, this definition is crucial to understand the concept of the bilinear map, for the Cartesian product is the method by which we are able to account for multiple different

arguments in a multilinear map. The definition of a bilinear map is simple: we take the Cartesian product of two vector spaces and map it to another vector space. The bilinear map must also have certain special properties, that being that it is linear in each argument. A more formal definition is illustrated below.

Definition 22 (Bilinear Map). *A bilinear map T is a function mapping the Cartesian product of two vector spaces $V \times W$ to another vector space U such that the following properties hold:*

- For any $\lambda \in \mathbb{F}$, then given a $v \in V$ and $w \in W$,

$$T(\lambda v, w) = T(v, \lambda w) = \lambda T(v, w) \quad (1.22)$$

- For all $v, v_1, v_2 \in V$ and $w, w_1, w_2 \in W$, then

$$T(v_1 + v_2, w) = T(v_1, w) + T(v_2, w) \quad (1.23)$$

$$T(v, w_1 + w_2) = T(v, w_1) + T(v, w_2) \quad (1.24)$$

Now that we are familiar with the formal definition of a bilinear map, we are able to prove several elementary theorems concerning this concept. We begin by proving that if the zero vector is an argument in the bilinear map, then the resulting vector must be the zero vector. This theorem follows shortly from the fundamental properties of the bilinear map.

Theorem 12. *If $T : V \times W \rightarrow U$ is bilinear, then if $v \in V$ and if $w \in W$, then $T(0, w) = 0$ and $T(v, 0) = 0$.*

Proof. The former claim can be proven like so:

$$T(0, w) = T(0v, w) = 0T(v, w) = 0 \quad (1.25)$$

The latter claim can be proven in a similar manner:

$$T(v, 0) = T(v, 0w) = 0T(v, w) = 0 \quad (1.26)$$

Therefore, we have reached our desired results. $\square$

Consider the set of all maps from $V \times W$ to U . It is obvious that the set of all bilinear maps from $V \times W$ to U is a *subset* of the previous set, but we can make a stronger claim. We can prove that the set of bilinear maps is a subspace of all the maps from $V \times W$ to U . The proof is left to the reader.

Theorem 13. *The set of all bilinear maps from $V \times W$ to U is a subspace of all maps (not necessarily linear maps) from $V \times W$ to U .*

The reader who is familiar with the properties of spaces of maps may be able to prove this as an exercise.

Our final theorem for now concerns the dimension of the set of bilinear maps. We prove that the dimension of the set of bilinear maps is fairly straightforward: simply the product of the dimension of all relevant vector spaces.

Theorem 14. *Given finite-dimensional vector spaces U, V, W , then the set of all bilinear maps from $V \times W$ to U is finite-dimensional.*

Proof. Let $\{v_1, \dots, v_m\}$, $\{w_1, \dots, w_n\}$, and $\{u_1, \dots, u_p\}$ be bases for V , W , and U , respectively. A bilinear map $B: V \times W \rightarrow U$ is completely determined by its values on pairs (v_i, w_j) . For each pair (i, j) , we can write

$$B(v_i, w_j) = \sum_{k=1}^p a_{kij} u_k, \quad (1.27)$$

where the a_{kij} are scalars. Thus, B is determined by mnp scalars, and the space of all such maps is isomorphic to $\mathbb{F}^{mnp}$. Therefore, its dimension is $(\dim V)(\dim W)(\dim U)$. $\square$

We now approach a new important concept: that of the bilinear form. The bilinear form is a special kind of bilinear map, in that the resulting vector space is rather the set of scalars, or $\mathbb{F}$. We will see soon that bilinear forms are a kind of tensor, but we will save the specifics for when we have defined the tensor.

Definition 23 (Bilinear Form). *The bilinear form on V over $\mathbb{F}$ is a kind of bilinear map that maps the Cartesian product of V with itself to $\mathbb{F}$. In other words, the bilinear form T is a map*

$$T : V \times V \rightarrow \mathbb{F} \quad (1.28)$$

Given two n -dimensional vectors u and v , we can state that the bilinear form of vectors u and v can be represented by a matrix A_{ij} such that

$$T(u, v) = \sum_{i=1}^n \sum_{j=1}^n u_i^T A_{ij} v_j \quad (1.29)$$

where T indicates the transpose of a vector (i.e. transforming it from a column vector to a row vector).

When considering bilinear forms, we must take into account certain categories of bilinear forms that will serve to be important classifications of bilinear forms. We examine the following types of bilinear forms:

Definition 24 (Types of Bilinear Forms). *A bilinear form can be the following:*

- **Symmetric:** $T(u, v) = T(v, u)$ for all $u, v \in V$
- **Alternating:** $T(v, v) = 0$ for all $v \in V$
- **Antisymmetric:** $T(u, v) = -T(v, u)$ for all $u, v \in V$.

The reader should be familiar with the concept of an inner product on $\mathbb{R}$ or $\mathbb{C}$, but to sufficiently examine more general kinds of spaces, we must generalize our knowledge of the inner product. As it turns out, bilinear forms can be defined as inner products, as in the below definition:

Definition 25 (Inner Product). *A bilinear form that is one of the above types may be thought of as an inner product, and a vector space V along with an inner product $\langle \cdot, \cdot \rangle$ is called an inner product space.*

A given inner product may naturally induce various kinds of geometries on a vector space. Below are two well-known examples of geometries that arise from inner product spaces:

Definition 26 (Orthogonal and Symplectic Geometries). *The following geometries can be defined on an inner product space.*

- **Orthogonal:** *An inner product space with a symmetric form must have an orthogonal geometry.*
- **Symplectic:** *An inner product space with an alternating form must have a symplectic geometry.*

Bilinear forms are also naturally connected to quadratic forms, which *prima facie* may seem markedly distinct from the former. However, we will find that the two mathematical concepts share a close relationship.

Definition 27 (Quadratic Form). *A quadratic form is a polynomial where every term is of degree two, for instance, $x^2 + xy + y^2$. It can also be thought of as a map $V \rightarrow \mathbb{F}$ where $v \mapsto T(v, v)$.*

To understand this relation more deeply, consider the fact that a bilinear form can be represented by a matrix. If we have the bilinear matrix

$$T = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \tag{1.30}$$

Then the quadratic form maps a vector $\begin{pmatrix} x \\ y \end{pmatrix}$ to $x^2 + 2xy + 2yx + 4y^2 = x^2 + 4xy + 4y^2$ through the following matrix multiplication:

$$T((x, y), (x, y)) = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x^2 + 4xy + 4y^2 \tag{1.31}$$

To end our excursion in bilinear forms, we consider a generalization of the polarization identities. In linear algebra, the polarization identities defines the inner product in terms of the norm. We generalize this identity so we recover a bilinear form in terms of its associated quadratic form:

Theorem 15 (Polarization Identities). *Consider the quadratic form $f(v) = T(v, v)$, where T is symmetric. Then we must have*

$$2T(u, v) = f(u + v) - f(u) - f(v) \quad (1.32)$$

$$2T(u, v) = f(u) + f(v) - f(u - v) \quad (1.33)$$

$$4T(u, v) = f(u + v) - f(u - v) \quad (1.34)$$

Proof. We have that

$$f(u + v) = T(u + v, u + v) = T(u, u) + 2T(u, v) + T(v, v) \quad (1.35)$$

$$\implies 2T(u, v) = f(u + v) - f(u) - f(v) \quad (1.36)$$

Then

$$f(u - v) = T(u - v, u - v) = T(u, u) - 2T(u, v) + T(v, v) \quad (1.37)$$

$$\implies 2T(u, v) = f(u) + f(v) - f(u - v) \quad (1.38)$$

Finally, adding 1.34 and 1.36 together, we obtain

$$4T(u, v) = f(u + v) - f(u - v) \quad (1.39)$$

□

1.4 Multilinear Maps

We have reached the point in multilinear algebra where we are able to study multilinear maps. The reader will find multilinear maps to be a reasonable generalization of the bilinear map, as follows:

Definition 28. *A multilinear map from $V_1, \dots, V_n$ to W is a function $T : V_1 \times \dots \times V_n \rightarrow W$ such that the following properties hold; given $a v_i, u_i \in V_i$ for all $i \in [1, n]$ and $\lambda \in \mathbb{F}$:*

$$\begin{aligned} & T(v_1, \dots, v_i + u_i, \dots, v_n) \\ &= T(v_1, \dots, v_i, \dots, v_n) + T(v_1, \dots, u_i, \dots, v_n) \end{aligned} \quad (1.40)$$

and

$$T(v_1, \dots, \lambda v_i, \dots, v_n) = \lambda T(v_1, \dots, v_i, \dots, v_n) \quad (1.41)$$

Similar to the bilinear map, we are able to make the following statement:

Theorem 16. *If one of the arguments of a multilinear map is the zero vector, then the arguments are mapped to the zero vector.*

Proof. Consider $T(v_1, \dots, 0, \dots, v_n)$. We have

$$T(v_1, \dots, 0v_i, \dots, v_n) = 0T(v_1, \dots, v_i, \dots, v_n) = 0 \quad (1.42)$$

□

An example of a multilinear map would be the $n \times n$ determinant, which maps $V \times \dots \times V$ to $\mathbb{F}$, where $V \times \dots \times V$ consists of n copies of V . In this case, we call the determinant n -linear. The determinant is an example of a multilinear form:

Definition 29. *A multilinear form, or n -linear form, is a type of multilinear map from V^n to $\mathbb{F}$, where V^n denotes the Cartesian product of n copies of V .*

The multilinear form will be given special consideration for the remainder of this chapter, for it is related closely to the study of tensors, as we will soon see. But first, we begin to categorize multilinear forms just as we have done with bilinear forms by certain properties that they have. We once again use the terms symmetric, anti-symmetric, and alternating in the same way that the reader expects based on section 1.3.

Definition 30 (Types of Multilinear Forms). *A multilinear form can be the following:*

- **Symmetric:**

$$\begin{aligned} T(v_1, \dots, v_i, \dots, v_j, \dots, v_n) \\ = T(v_1, \dots, v_j, \dots, v_i, \dots, v_n) \end{aligned} \quad (1.43)$$

- **Anti-Symmetric:**

$$\begin{aligned} T(v_1, \dots, v_i, \dots, v_j, \dots, v_n) \\ = -T(v_1, \dots, v_j, \dots, v_i, \dots, v_n) \end{aligned} \quad (1.44)$$

- **Alternating:** If $v_i = v_j$ for some $i \neq j$, then

$$T(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = 0 \quad (1.45)$$

We find that alternating multilinear forms carry certain special properties. For example, we have that the multilinear form is zero if its arguments are linearly dependent. The proof looks fairly intimidating in terms of expressions, but is conceptually very simple.

Theorem 17. *Consider the alternating multilinear form T . If $v_1, \dots, v_m$ is linearly dependent, then $T(v_1, \dots, v_m) = 0$.*

Proof. If $v_1, \dots, v_m$ is linearly dependent, then there must be some v_k such that

$$v_k = \sum_{i=1}^{k-1} a_i v_i + \sum_{i=k+1}^m a_i v_i \quad (1.46)$$

Therefore, we must have

$$\begin{aligned} & T(v_1, \dots, v_k, \dots, v_m) \\ &= T\left(v_1, \dots, \sum_{i=1}^{k-1} a_i v_i + \sum_{i=k+1}^m a_i v_i, \dots, v_m\right) \end{aligned} \quad (1.47)$$

$$\begin{aligned} &= \sum_{i=1}^{k-1} T(v_1, \dots, v_i, \dots, v_m) + \sum_{i=k+1}^m T(v_1, \dots, v_i, \dots, v_m) \\ &= 0 \end{aligned} \quad (1.48)$$

as $v_i = v_j$ if $i \neq j$ implies $T(v_1, \dots, v_i, \dots, v_j, \dots, v_n)$ vanishes. $\square$

We can prove a weaker version of the inversion like so.

Theorem 18. *If T is a nonzero alternating multilinear form on V and $v_1, \dots, v_n \in V$ which span V and are linearly independent, then $T(v_1, \dots, v_n) \neq 0$.*

Proof. We understand that $v_1, \dots, v_n$ are bases, since they span V and are linearly independent. There must exist some $w_1, \dots, w_n \in V$ such that $T(w_1, \dots, w_n) \neq 0$, as T is nonzero. $w_i \neq 0$ for all $i \in [1, n]$, as well, because if $w_i = 0$, then $T(w_1, \dots, 0, \dots, w_n) = 0T(v_1, \dots, v_i, \dots, v_n) = 0$. Hence we have a nonzero $w_1, \dots, w_n$ such that $T(w_1, \dots, w_n) \neq 0$. Letting $v_i = a_i w_i$ gives us $T(v_1, \dots, v_n) \neq 0$. $\square$

It is also an interesting property of alternating multilinear forms that every alternating multilinear form is antisymmetric as well. One may accomplish this by considering $T(v_1, \dots, v_i - v_i, \dots, v_n)$.

Theorem 19. *All alternating multilinear forms are antisymmetric.*

The proof is (once again) left as an exercise to the reader. The reader who has completed exercise 13 will find this exercise to be not difficult.

We consider some more nomenclature before ultimately arriving at the concept of the tensor product; one will hear of the n -covector inevitably in their study of tensors, and it is important to lay out a definition of such when possible.

Definition 31 (n -Covector). *An n -covector is a multilinear form on V (with n copies) over $\mathbb{R}$.*

At last, we have come to the concept of the tensor product. This is the first time that we will (tangentially) be dealing with tensors, although we have not yet rigorously defined what a tensor is. That is reserved for chapter 2. One will find that the tensor product is similar to the Cartesian product, although dissimilar in the fact that it preserves multilinearity (or bilinearity in the case of the tensor product of two vector spaces). Now, we will formally define the tensor product:

Definition 32 (Tensor Product). *The tensor product (a vector space T) of two vector spaces $U \otimes V$ is the set of vectors $t \in T$ such that there exists a bilinear map $(u, v) \mapsto t$ where $u \in U$ and $v \in V$. $t = u \otimes v$ is considered to be the tensor product of u and v .*

Some properties of the tensor product include the following:

$$c(u \otimes v) = cu \otimes v = u \otimes cv \quad (1.49)$$

$$(u + v) \otimes (w + x) = u \otimes w + v \otimes w + u \otimes x + v \otimes x \quad (1.50)$$

It is easy to see how the tensor product is associated with multilinearity. We are able to multiply more than two vector spaces in a similar way, like so:

$$v_1 \otimes v_2 \otimes \dots \otimes v_n = t \quad (1.51)$$

where t is the tensor product of all $v_1, \dots, v_n \in V$. Therefore, $t \in V \otimes \dots \otimes V$. Some of the properties of this tensor product will be left to the reader to verify.

Consider the Cartesian product of u and v , $u \times v$. It is well-known that $c(u \times v) = cu \times cv$. However, we see that this is obviously not the case with the tensor product. Although the tensor product and Cartesian product are built from both vector spaces, they are different in terms of properties. Knowing these differences will be crucial conceptually to distinguishing the difference between these products.

One may find the Cartesian product to be a more “natural” product of two vector spaces, but as we deepen our study of tensors, we will see how the tensor product is well-motivated as well. For instance, tensors can be “decomposed” into their constituent factors based on the concept of the tensor product, and an (m, n) tensor multiplied by a (p, q) tensor gives us an $(m + p, n + q)$ tensor. But that is enough of a sneak peek into chapters 2-3 of this book. Let us talk about the motivation for why we even want to study the concept of tensors in the first place.

1.5 Motivation for Tensors

What is exactly the point of introducing the concept of tensors? We find that tensors have a couple key properties that allow us

to make certain applications much easier. The two fundamental properties of tensors that we are interested in are described below:

1. Tensors are *coordinate-invariant*; that is, under a coordinate transformation (think Cartesian to polar coordinates), tensors remain unchanged. The components of a tensor may change, but the tensor itself does not change.
2. Tensors allow us to condense notation in such a way that multiple different equations can be written in one. This has profound implications in applied mathematics, such as in physics, where we can write even dozens of different equations in terms of one equation that uses a tensor.

We will begin to explore some of the motivations behind tensors, but first, it will serve us well to review one key concept from multivariable calculus:

Definition 33 (Partial Derivative). *The partial derivative of a function $f(x_1, \dots, x_n)$ with respect to a variable x_i can be described using the following equation:*

$$\frac{\partial f}{\partial x_i} = \partial_i f = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h} \quad (1.52)$$

Surely the reader is familiar with the single-variable derivative; the partial derivative essentially amounts to taking the derivative of a function with respect to one variable while keeping all others constant. One may practice this in the exercises; it will become incredibly useful to be proficient in these calculations when we approach more complicated multivariable derivatives.

The gradient is an important concept from multivariable calculus, that is defined as follows:

Definition 34. *The gradient of a function $f(x_1, \dots, x_n)$ is the vector with components*

$$\nabla f = (\partial_1 f, \partial_2 f, \dots, \partial_n f) \quad (1.53)$$

Now, what sort of basis will, when combined with the above components, make the gradient? One would be inclined to say that the gradient is always $\nabla f = (\partial_1 f)e_1 + \dots + (\partial_n f)e_n$, where $e_1, \dots, e_n$ is the basis of the coordinate system. But that would be incorrect. To provide a sketch of the reasoning why, notice that the derivative changes at twice the rate when the basis becomes twice as big, and the derivative changes at half the rate when the basis becomes half as big.

Remark 3. *The gradient of a function must remain invariant under a coordinate transformation; the components of the gradient may change in a certain way, however.*

I will leave it to the reader to verify for one simple function that the assumption made earlier conflicts with the above remark.

As it turns out, the gradient of a function is a tensor! Specifically, it is a $(0, 1)$ tensor. As we will learn soon, it is a special case of applying the exterior derivative to a scalar function. You are not expected to know these words quite yet, but we will rigorously define and apply these concepts soon.

Tensors are also extremely important for condensing multiple equations into one simpler equation. This economy of expression is paramount in applied mathematics, where we look for the simplest expressions describing a given system. Before we consider our example, Maxwell's equations, we define a couple new concepts:

Definition 35 (Divergence and Curl). *The divergence $\nabla \cdot f$ of a vector function $f(x_1, \dots, x_3) = (f_1, \dots, f_3)$ is defined as*

$$\nabla \cdot f = (\partial_1 f_1) + (\partial_2 f_2) + (\partial_3 f_3) \quad (1.54)$$

The curl $\nabla \times f$ is defined as

$$\nabla \times f = \begin{vmatrix} e_1 & e_2 & e_3 \\ \partial_1 & \partial_2 & \partial_3 \\ f_1 & f_2 & f_3 \end{vmatrix} \quad (1.55)$$

This definition is not very general, but in Euclidean 3-space with the standard basis, it works. It can be demonstrated in a

similar way as the gradient that this is not always the case in more general coordinate systems, but we are getting ahead of ourselves. Now we have all of the tools necessary to understand our example of how tensors can conserve the amount of equations necessary for describing something: Maxwell's equations.

Definition 36 (Maxwell's Equations). *Maxwell's four equations are written here in full:*

$$\nabla \times \mathbf{B} = \mu_0(\mathbf{J} + \epsilon_0 \partial_t \mathbf{E}) \quad (1.56)$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (1.57)$$

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B} \quad (1.58)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (1.59)$$

We have that $\mathbf{B}$ represents the magnetic field (a vector quantity), μ_0 represents the vacuum permeability (a constant), $\mathbf{J}$ represents the current density (a vector), ϵ_0 represents the vacuum permittivity (a constant), $\mathbf{E}$ represents the electric field (a vector), and ρ represents the charge density (a scalar).

We will demonstrate that tensor notation will be able to quite sufficiently condense the number of equations that we have to deal with. We begin by defining the electromagnetic tensor, which is simply a matrix:

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix} \quad (1.60)$$

where the E and B components are exactly what one would expect to be. As we have not yet learned about tensors, I will not go into great detail, but the end result is that we have two more general equations:

$$\partial_\mu F^{\nu\mu} = J^\nu \quad (1.61)$$

$$\partial_{[\mu} F_{\nu\lambda]} = 0 \quad (1.62)$$

This is incredible news for the physicist, who is looking to find the most elegant relationship between the electric and magnetic fields. And as it turns out, the utility of the tensor does not stop at Maxwell's equations; the even more complicated Einstein field equations bring dozens of equations together into one. But we will get there in several chapters. There is one more piece of motivation that we are going to discuss before we begin to modify our notation slightly.

The study of tensor calculus is deeply intertwined with that of differential geometry, and therefore, it follows that to understand tensors well, we must understand manifolds well, and vice versa. Tensors are able to shed much light on manifolds. These mathematical objects end up describing properties of manifolds that would otherwise be incredibly complicated to define. For instance, curvature. Qualitatively, everyone knows what curvature is, but it is difficult to rigorously define curvature. However, we may (in part) utilize a tensor known as the Riemann tensor, $R_{\mu\nu\rho\sigma}$ to gain insight into this descriptor of a manifold. Now concerning distance between two points on a manifold, there is yet another tensor that does the job of defining distance quite well. It is none other than the metric tensor, $g_{\mu\nu}$. We find that, in Euclidean space, we have

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \iff ds^2 = dx^2 + dy^2 \quad (1.63)$$

The connection between the distance formula and the metric tensor will become clearer as we continue our study of tensors, but this will suffice in terms of motivation. Now we move onto notation.

1.6 Notation

My choice to leave vectors unbolded and without arrows may leave some readers scratching their heads, but this is where that choice should theoretically pay off, for now we will add indices to vectors, functionals, and forms.

We begin with the scalar. We will leave the scalar as is, without any indices. For example, we have a scalar m for rest mass. No indices are necessary, because a scalar only has one component.

We move onto the vector. Vectors will contain one index as a superscript; these describe the components of a vector. Note that if the superscript is a Greek letter, we start from 0 to 3, whereas if the superscript is a Latin letter, we start from 1 to 3. For example, $V^a = (V^1, V^2, V^3)$, whereas $V^\mu = (V^0, V^1, V^2, V^3)$.

Next, we consider the basis vector. The basis vector also has one index, but this index is a subscript, not a superscript. This is similar to the linear functional, which also has one index as a subscript. One would naturally ask the question why the basis vector and the dual vector have the same notation, whereas the basis vector and the vector do not. When we cover contravariance and covariance in the next chapter, those questions will be answered shortly. Given what we know about the basis and the dual vector, it is only natural that the dual basis is expressed in terms of one index as a superscript.

We examine the bilinear form, which maps $V \times W \rightarrow \mathbb{F}$. Since the dual vector maps $V \rightarrow \mathbb{F}$, one may conclude that the bilinear form can be represented by two indices, both subscripts, like so: $g_{\mu\nu}$.

Similarly, the multilinear n -linear form mapping $V \times \dots \times V \rightarrow \mathbb{F}$ will have n indices, all as subscripts, like $K_{\mu\nu\rho\sigma\pi}$ (I made this tensor up).

Now, it is time to consider the Einstein summation convention, which is a convention that at first may seem annoying and unwarranted, but as we deepen our knowledge of tensors, it will become more natural and motivated. It is defined like so:

Definition 37 (Einstein Summation Convention). *When we have an expression containing two of the same index, one superscript, and one subscript, it is generally the case to sum the expression over that particular index.*

This definition is admittedly abstract, so let us look at a couple of examples. Consider multiplying each component of a vector by its respective basis vector:

$$y = v^1 e_1 + v^2 e_2 + v^3 e_3 = \sum_{i=1}^3 v^i e_i \quad (1.64)$$

Since there is one i as a superscript and one i as a subscript, we can remove the sigma notation, as it is implied by the Einstein summation convention:

$$\sum_{i=1}^3 v^i e_i = v^i e_i \quad (1.65)$$

Let us look at a more complicated expression. Consider the following:

$$\Gamma_{\mu\nu}^\lambda \partial_s x^\mu \partial_s x^\nu \quad (1.66)$$

There are two indices, μ and ν , which feature both as a superscript and a subscript. Therefore, this expression must be summed over both μ and ν , like so:

$$\Gamma_{\mu\nu}^\lambda \partial_s x^\mu \partial_s x^\nu = \sum_{\mu=0}^3 \sum_{\nu=0}^3 \Gamma_{\mu\nu}^\lambda \partial_s x^\mu \partial_s x^\nu \quad (1.67)$$

It takes practice to identify when the Einstein summation convention is applicable, so there are several exercises that will allow you to practice that skill. With that being said, we have reached the end of chapter 1. We now have acquired all the background knowledge we need to rigorously define the tensor.

1.7 Exercises

Exercise 1: Prove that a subspace containing v and w must contain all linear combinations $\lambda v + \mu w$.

Exercise 2: Are all Euclidean n -spaces $\mathbb{R}^n$ linear spaces? Prove your claim.

Exercise 3: Prove that the dimension of a subspace $U \subseteq V$ is less than or equal to the dimension of the vector space V .

Exercise 4: A matrix A is considered *invertible* if there exists a matrix A^{-1} such that $A^{-1}A = AA^{-1} = I$, where I is the identity matrix. Prove that if a matrix A is invertible, then the dimension of the column space is the number of columns of A .

Exercise 5: Prove that the annihilator U^0 of a subspace $U \subseteq V$ is a subspace itself of V^* .

Exercise 6: Given a vector space V and a functional φ that acts on V , prove that the dimension of V is equal to the dimension of the null space of φ plus the dimension of the range of φ .

Exercise 7: When is a given linear functional on V injective? When is it surjective? Both? Neither?

Exercise 8: Consider the Cartesian product of two vector spaces $V \times W$. Let $v_1, \dots, v_m$ be a basis for V and let $w_1, \dots, w_n$ be a basis for W . Find a dual basis of $(V \times W)^*$. Is it the same as the dual basis of $V^* \times W^*$?

Exercise 9: Prove that the inner product over real numbers is bilinear. Does the same hold for the inner product over complex numbers? Justify.

Exercise 10: The cross product in $\mathbb{R}^3$ takes two vectors in $\mathbb{R}^3$ and maps it to another vector in $\mathbb{R}^3$. Does this function constitute a bilinear map?

Exercise 11: Prove that the set of all bilinear maps from $V \times W$

to U is a *subspace* of all maps from $V \times W$ to U .

Exercise 12: Can a bilinear form be both symmetric and antisymmetric at the same time?

Exercise 13: Prove that an alternating bilinear form must be antisymmetric as well.

Exercise 14: Can you think of an inner product space that contains both an orthogonal and symplectic geometry over $\mathbb{R}$?

Exercise 15: Find $\dim V^n$.

Exercise 16: Show that all alternating multilinear forms must be anti-symmetric.

Exercise 17: Construct a 3-linear form $T : \mathbb{F}^3 \times \mathbb{F}^3 \times \mathbb{F}^3 \rightarrow \mathbb{F}$.

Exercise 18: Consider an n -linear form from $V \times \dots \times V \rightarrow \mathbb{F}$, where we have $T(v_1, \dots, v_n) = a$. Find $T(v_n, \dots, v_1)$. Consider all cases of n .

Exercise 19: Verify that $c(v_1 \otimes \dots \otimes v_n) = v_1 \otimes \dots \otimes cv_i \otimes \dots \otimes v_n$.

Exercise 20: Verify that $v_1 \otimes \dots \otimes (v_i + v_{i1}) \otimes \dots \otimes v_n = v_1 \otimes \dots \otimes v_i \otimes v_n + v_1 \otimes \dots \otimes v_{i+1} \otimes \dots \otimes v_n$.

Exercise 21: Find *all* partial derivatives of the following functions:

(a) $f(x, y, z) = 4xy$

(b) $f(x, y, z) = x^2 + y^2 + z^2$

- (c) $f(x, y, z) = y + yz$
- (d) $f(x, y, z) = \sin(xy) + \cos z$
- (e) $f(x, y, z) = e^{x^2 y^2 z}$
- (f) $f(x, y, z, w) = -w^2 + x^2 + y^2 + z^2$
- (g) $f(x_1, \dots, x_n) = \sum_{i=1}^n \frac{x_i}{i}$

Exercise 22: Find the gradient of all functions in Exercise 21.

Exercise 23: Verify that Remark 3 is incompatible with the assumption that $\nabla f = \sum_{i=1}^n (\partial_i f) e_i$ for every coordinate system, using the function $f(x, y) = x^2 + y^2$.

Exercise 24: (Gauss' Law.) Gauss' Law can be derived from Maxwell's equations. Derive the following:

$$\oiint_S E \cdot dA = \frac{q}{\epsilon_0} \quad (1.68)$$

Hint: We can use the identity

$$\oiint_S E \cdot dA = \iiint_V (\nabla \cdot E) dV \quad (1.69)$$

Exercise 25: Identify the different mathematical objects based on index:

- (a) a
- (b) x
- (c) V^μ
- (d) e_x
- (e) φ_i
- (f) $\varphi_i v^i$
- (g) $E_{\mu\nu}$
- (h) $\epsilon_{\mu\nu\rho}$

Exercise 26: Write if the Einstein summation convention is applicable to each expression, and if so, write the full form of each expression (with sums):

- (a) $x^i e_i$
- (b) $v^i v_i$
- (c) $a^\mu b_\nu$
- (d) $g_{\mu\nu} dx^\mu dx^\nu$
- (e) $\Lambda_{\mu}^{\mu'} x^\mu$
- (f) $\epsilon_{\mu\nu\rho} x^\mu x^\nu x^\rho x^\sigma$
- (g) $a^i b^i c_i d_i$
- (h) $\Gamma_{\mu\nu}^\lambda \nabla_\lambda v^\sigma$

Challenge 1: A *Hilbert space* H is defined as an inner product space that is considered to be a complete; that is, every Cauchy sequence of points in the space has a limit that is also in the space.

Definition 38 (Cauchy Sequence). *A Cauchy sequence is a sequence of elements $x_1, x_2, \dots$ such that for every ε , there must exist some number N such that for every $m, n > N$, then*

$$\|x_m - x_n\| < \varepsilon \quad (1.70)$$

A Hilbert space is also considered to be a metric space:

Definition 39 (Metric Space). *A metric space M is a set equipped with a distance function $d : M \times M \rightarrow \mathbb{R}$ with a couple key properties:*

- $d(x, x) = 0 \quad \forall x \in M$.
- $x \neq y \implies d(x, y) > 0$.
- $d(x, y) = d(y, x)$.
- $d(x, z) \leq d(x, y) + d(y, z)$.

(a) Is $\mathbb{R}^n$ considered to be a Hilbert space? Justify using the definition of a Hilbert space.

(b) A sequence x_n is weakly convergent to some vector x when

$$\lim_{n \rightarrow \infty} \langle x_n, v \rangle = \langle x, v \rangle \quad \forall v \in H \quad (1.71)$$

Prove that any orthonormal sequence converges weakly to 0.

(c) Define the dual space of H , H^* . Define a norm on H^* , and prove that it is a valid norm. Does it satisfy the parallelogram law?

Definition 40 (Parallelogram Law). *On an inner product space, the parallelogram law is stated like so:*

$$2\|x\|^2 + 2\|y\|^2 = \|x + y\|^2 + \|x - y\|^2 \quad \forall x, y \in H^* \quad (1.72)$$

Challenge 2: Consider three topological vector spaces X, Y, Z . We let $T : X \times Y \rightarrow Z$ be bilinear and separately continuous.

Definition 41 (Separately Continuous). *A map $X \times Y \rightarrow Z$ is separately continuous if $\forall x \in X$, then $y \mapsto T(x, y)$ is continuous, and if $\forall y \in Y$, then $x \mapsto T(x, y)$ is continuous.*

(a) Prove that if a bilinear map is continuous at $(0,0)$, then it must be continuous everywhere.

(b) Let X be a Baire space:

Definition 42 (Baire Space). *A topological space X is a Baire space if a countable number of unions of closed sets with no interior also have an empty interior.*

Let Y be metrizable; that is, it is homeomorphic to a metric space. Prove that T must be continuous, if it is separately continuous.

(c) Some bilinear maps that are separately continuous are continuous. Can you think of a bilinear map that is separately continuous but not continuous?

Challenge 3: The Kronecker tensor product is defined as follows, given two matrices A, B , where A is $m \times n$:

Definition 43 (Kronecker Tensor Product).

$$A \otimes B = \begin{pmatrix} a_{11}B & \dots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \dots & a_{mn}B \end{pmatrix} \quad (1.73)$$

Construct a working iteration of the Kronecker tensor product in MATLAB. Explain how it relates to the tensor product that was discussed in section 1.4.

Challenge 4: Consider the following definition:

Definition 44 (Commutative Ring). *A set with two operations, addition and multiplication, is a commutative ring if addition is associative and commutative, there exists an additive identity, there exists an additive inverse for every element, multiplication is associative and commutative, there exists a multiplicative identity, and multiplication distributes over addition.*

We examine a multilinear function T that takes the columns of a given $n \times n$ matrix over some commutative ring, where we have

$$T(A) = T(a_1, \dots, a_n) \quad (1.74)$$

(a) Prove that $T(A)$ is uniquely determined by how T operates on the identity matrix.

(b) If T is symmetric in its column arguments, show that the expansion reduces to the permanent-like sum (no signs). Compare with the alternating case (determinant).

(c) Prove that every multilinear alternating form T satisfying $T(I) = 1$ equals the determinant. First prove this case over a field, then remark on rings.

CHAPTER 2

Basic Definitions

2.1 Contravariance

At last we reach the chapter where we define a tensor. But first, it is important to introduce a couple key terms: contravariance and covariance. Although these terms may sometimes be considered outdated today, I believe that it is important to at least gain some understanding of the meaning behind these terms.

We define the contravariant indices of a vector v^i like so:

$$v = v^i e_i \implies v^i = v \cdot e_i \quad (2.1)$$

We recall that e_i are all the basis vectors, and $v^i e_i$, by the Einstein summation convention, implies that we really have

$$v^i e_i = \sum_i v^i e_i \quad (2.2)$$

Note that the dot product $v \cdot e_i$ indicates the projection of v onto e_i , which gives us the component v^i . The contravariant components of a vector are generally considered the “typical” components of a vector; that is, when we think of a vector as three numbers in $\mathbb{R}^3$ in high school, those numbers are the contravariant components of the vector.

Now, we consider how the contravariant components transform under a coordinate transformation. It is obvious that when the basis changes, the components of the vectors also change, yet the vector itself does not change. This is what we mean when we say that a vector, and soon, a tensor, is coordinate-invariant. The components may change, but the vector/tensor does not.

The components of a vector always change via a transformation matrix; that is, given two different sets of contravariant

components of a vector v^j and v^i , we can relate them by the following equation:

$$v^j = \Lambda_i^j v^i = \sum_i \Lambda_i^j v^i \quad (2.3)$$

where Λ_i^j is some transformation matrix that depends on the coordinate transformation at hand. Recall that the Einstein summation convention applies here.

Now that we have developed the basic properties of a contravariant vector, it is time to answer the following question: what does contravariant even mean?

Definition 45 (Contravariance). *A vector with given components is considered to be contravariant if its components transform against the basis vectors (hence the term “contra”). That is to say, the manner in which the components transform is inverse to the manner in which the coordinate basis transforms.*

This is a nice definition, but it may confuse the reader even more. So, to illustrate what I mean by this, I shall give a simple example: consider a vector v and a basis e_i . If the basis undergoes a rotation to a new basis e_j , the vector will appear from the basis' point of view as rotating in the opposite direction; hence, the components of the vector will change accordingly. This is what is meant by contravariant; the components transform *against* the basis vectors.

Now, we shall consider one of the most pertinent applications of the contravariant vector: its importance in 4-dimensional Minkowski spacetime. In special relativity, the contravariant vector is often called a four-vector; let us define what that is:

Definition 46 (Four-Vector). *A four-vector is a contravariant vector with four indices from 0 to 3, 0 representing the time axis and 1-3 representing the spatial axes, transforming accordingly:*

$$v^\nu = \Lambda_\mu^\nu v^\mu \quad (2.4)$$

Did you catch the Einstein summation convention used here? Hopefully this is good practice. The four-vector really is just

a vector with four indices: (v^0, v^1, v^2, v^3) , within the setting of Minkowski spacetime:

Definition 47 (Minkowski Geometry). *Minkowski geometry is a type of geometry where the distance formula, given spatial axes x, y, z and temporal axis t , is as follows:*

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 \quad (2.5)$$

It is interesting to note that dt^2 is negative. The reader can speculate about why this is so as an exercise. Unfortunately, we cannot do much with this application until we introduce covariance, so we will leave our discussion of Minkowski spacetime here and transition to the tangent space.

The tangent space will be treated informally and qualitatively in this section of the textbook; we will consider more rigorous definitions and properties later.

Definition 48 (Tangent Space). *The tangent space $T_p M$ is the set of all vectors tangent to a given manifold M at a point p .*

Remark 4. *The vectors comprising the tangent space of a manifold at a point are all contravariant.*

All tangent spaces for every point on the manifold can be combined into a mathematical object known as the tangent bundle:

Definition 49 (Tangent Bundle). *The tangent bundle TM is the set of all tangent spaces of a manifold M .*

It is easy to see that the vectors comprising the tangent bundle are also contravariant, given the prior remark. Now that we have considered the contravariant vector, let us move on to the contravariant tensor. Although we have not defined the tensor yet, we will begin to gain some idea of what a tensor is.

Remark 5. *The tangent product of a vector space V , $T^p(V) = V \otimes \dots \otimes V$, where V is multiplied n times, is the space of tensors with n contravariant (superscript) indices. These are called contravariant tensors.*

It is worth noting that these tensors are also labeled $(n, 0)$ tensors, the reason for which we will soon discover. If we recall the definition of a tangent product from Chapter 1, we find that this tangent product T^p is simply the set of all multilinear functionals from $V^* \times \dots \times V^* \rightarrow \mathbb{F}$, where V^* is multiplied n times.

2.2 Covariance

Now that we understand the concept of contravariance, it is only natural to move on to the concept of covariance. The covariant vector is denoted by v_i , and is also called the one-form, the reason for which will be fully uncovered in Chapter 9. The relationship between the vector v , the covariant vector, and the dual basis e^i is shown below, similar to that of the contravariant vector:

$$v = v_i e^i \implies v_i = v \cdot e^i \quad (2.6)$$

Yes, we are still using the Einstein summation convention. It is also important to note that the covariant vector may differ in terms of the transformation matrix than the contravariant vector; that is to say, given two contravariant vectors v^i and v^j and their associated covariant vectors v_i and v_j , we have that

$$v^j = \Lambda_i^j v^i, v_j = \overline{\Lambda_j^i} v_i \quad (2.7)$$

where

$$\Lambda_i^j \neq \overline{\Lambda_j^i} \quad (2.8)$$

In this case, the bar does not indicate a complex conjugate; it simply indicates a different transformation matrix. This is because covariant vectors transform *with* the basis, instead of against the basis. Although it is difficult to incorporate a geometric argument as to how covariant vectors transform, it is important to recall that the covariant vector is also known as the dual vector, so one can imagine how the dual vector would transform opposite to the vector itself, especially when you consider how the vector and a dual vector map to a scalar, $v_i w^i = x$, by the Einstein summation convention.

One may wonder how the contravariant vector and covariant vector are related, and as it turns out, there is an easy relationship between them. However, first, we must define one important term:

Definition 50 (Metric Tensor). *The metric tensor is a bilinear form $g_{\mu\nu} : V \times V \rightarrow \mathbb{F}$ that calculates the “distance” between two points in space.*

The metric tensor may take the form of a matrix, which depends on the space. For example, in Minkowski geometry, the metric tensor takes the form

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2.9)$$

In this case, we have the following:

Remark 6. *The relationship between the metric tensor, covariant vector, and contravariant vector can be described like so:*

$$v_\mu = g_{\mu\nu} v^\nu \quad (2.10)$$

One may use this identity to show the following:

Theorem 20. *If the indices of a contravariant vector are described by (v^0, v^1, v^2, v^3) , then the indices of a covariant vector are described by $(-v^0, v^1, v^2, v^3)$.*

The proof of this is left to the reader as an exercise.

Now consider the following. An example of a covariant vector is actually the gradient! To see this, we consider a scalar function f , and its derivative with respect to a coordinate system x^μ and another coordinate system x^ν .

Theorem 21 (Chain Rule). *The Chain Rule for partial derivatives is stated as follows:*

$$\frac{\partial f}{\partial x^j} = \frac{\partial f}{\partial x^i} \cdot \frac{\partial x^i}{\partial x^j} = \sum_i \frac{\partial f}{\partial x^i} \frac{\partial x^i}{\partial x^j} = \partial_i f \partial_j^i \quad (2.11)$$

Did you catch the Einstein summation convention here? It's okay if you didn't. This one was tricky to spot. But keep in mind now, that the partial derivative has a subscript index. Now, by the chain rule, we have

$$\partial_\nu f = \frac{\partial f}{\partial x^\nu} = \frac{\partial f}{\partial x^\mu} \cdot \frac{\partial x^\mu}{\partial x^\nu} = \partial_\mu f \partial_\nu^\mu = \overline{\Lambda}_\nu^\mu \partial_\mu f \quad (2.12)$$

We see that the partial derivative therefore transforms covariantly, and therefore the gradient must transform like so as well.

As a brief aside, we will observe the utility of the covariant vector in applications to relativistic quantum physics. There will be a theorem presented below, with proof:

Theorem 22 (Klein-Gordon Equation). *The Klein-Gordon Equation is*

$$-\frac{1}{c^2}(\partial_0)^2\psi + \nabla^2\psi = \left(\frac{mc}{\hbar}\right)^2\psi \quad (2.13)$$

In this case, ψ represents the wavefunction of a particle (think of it as a sort of vector), c represents the speed of light, m represents the rest mass, and $\hbar$ is Planck's reduced constant.

Proof. We start with the following lemma:

Lemma 2. $p^\mu p_\mu - m^2 c^2 = 0$, where p is the momentum operator.

In this proof, we are assuming that the potential energy is zero throughout, so the particle is free. Momentum is defined as

$$p_\mu = i\hbar\partial_\mu \quad (2.14)$$

Therefore, we must have that

$$p^\mu p_\mu = -\hbar^2 \partial^\mu \partial_\mu \equiv -\frac{\hbar^2}{c^2} \partial_0^2 + \hbar^2 \nabla^2 \quad (2.15)$$

Definition 51 (Laplacian). *The Laplacian operator on Minkowski spacetime is defined as*

$$\nabla^2 f = (\partial_i)^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \quad (2.16)$$

Therefore, we obtain

$$-\frac{\hbar^2}{c^2}(\partial_0)^2\psi + \hbar^2\nabla^2\psi = m^2c^2\psi \quad (2.17)$$

$$\implies -\frac{1}{c^2}(\partial_0)^2\psi + \nabla^2\psi = \left(\frac{mc}{\hbar}\right)^2\psi \quad (2.18)$$

giving us our desired result. $\square$

The Klein-Gordon equation is one of the most important equations in relativistic quantum physics, as it is related to the Schrodinger equation in that it governs the behavior of a wavefunction (particle) over time. However, we see that the contravariant and covariant vectors play a prominent role in the formation of this equation, through the momentum operators.

Now that we have seen a simple application of the covariant vector, we turn our attention to the cotangent bundle. One can deduce that the tangent bundle is a vector space, as it contains contravariant vectors in the tangent vectors and satisfies various properties of the vector space. This naturally induces a dual space, as every vector space has one, known as the cotangent bundle. It is difficult to understand the relevance of the cotangent bundle, but it can be thought as the set of all differentials from the manifold M to the field $\mathbb{F}$, whereas the tangent bundle is the set of all differentials the other way around. Perhaps we should define the differential now:

Definition 52 (Differential). *The differential, also known as the Jacobian matrix, of a function $f = (f_1, \dots, f_m)$ is the matrix*

$$\begin{pmatrix} \partial_1 f_1 & \dots & \partial_n f_1 \\ \vdots & \ddots & \vdots \\ \partial_1 f_m & \dots & \partial_n f_m \end{pmatrix} \quad (2.19)$$

We see that the differential is an $m \times n$ matrix.

Therefore, we have that given a manifold $M = (x_1, \dots, x_n)$, the differential on $T : M$ to $\mathbb{F}$ (a specific case of the broader definition above) is

$$dT = (\partial_1 T \quad \dots \quad \partial_n T) \quad (2.20)$$

Now, let us finally discuss covariant tensors. We begin in a similar manner to that of last section:

Remark 7. *The tangent product of a dual space V^* , $T^p(V^*) = V^* \otimes \dots \otimes V^*$, where V^* is multiplied n times, is the space of tensors with n covariant (subscript) indices. These are called covariant tensors.*

As perhaps expected, these tensors are labeled $(0, n)$ tensors. And, connecting back to the previous section once more, we may guess that the tangent product is the set of all multilinear functionals from $V \times \dots \times V \rightarrow \mathbb{F}$, where V is multiplied n times. Now that we have examined some special cases of tensors, it is time to stop beating around the bush and time to start defining the tensor in its most general form.

2.3 Tensors as Multilinear Maps

We begin, of course, by defining the tensor:

Definition 53 (Tensor). *A tensor T is a multilinear map $V^* \times \dots \times V^* \times V \times \dots \times V \rightarrow \mathbb{F}$.*

One would notice immediately that scalars, vectors, and dual vectors are all tensors. Linear operators are also tensors as well, like so:

$$T(v, \varphi) = \varphi(Tv) \quad (2.21)$$

In the equation above, T acts as a linear operator on v , returning a vector Tv . Or, in index notation, we have

$$T^\mu_\nu : V^\nu \rightarrow T^\mu_\nu V^\nu \quad (2.22)$$

(Make sure you remember the Einstein summation convention here!) Notice that here, we could treat the tensor as a matrix, or an array of numbers. This is another, less rigorous definition that may assist the reader's comprehension of tensors: the tensor can (sometimes) be represented as a multidimensional array of numbers.

Now, we define a more general object known as the variant:

Definition 54 (Variant). *A variant is some mathematical object that can be constructed using a defined method in different coordinate systems.*

An example of the variant would be the basis, which we can derive (literally) in various coordinate systems by taking the derivative of position. Here is an example to demonstrate: consider the Cartesian coordinate system (x, y, z) . Taking the partial derivatives ∂_x , ∂_y , and ∂_z of the position vector, we obtain

$$\begin{aligned}\partial_x(x, y, z) &= (1, 0, 0), \partial_y(x, y, z) = (0, 1, 0), \\ \partial_z(x, y, z) &= (0, 0, 1)\end{aligned}\tag{2.23}$$

Now, the point of all this is to demonstrate that the tensor is a type of variant; specifically, it can be constructed in the way defined above and it must transform in a specific, defined way; it will be revealed soon how the tensor must transform.

It is fairly obvious that the set of tensors with the same indices can be considered a vector space; I will leave it to the reader to show this. Tensors can be added together and multiplied by scalars, just as how scalars, vectors, and dual vectors can. To illustrate this claim, consider how matrices of the same dimension can be added together and multiplied by scalars.

The tensor with m contravariant indices and n covariant indices can be represented through its components and the tensor product of the basis and dual basis:

$$T = T_{j_1 \dots j_n}^{i_1 \dots i_m} e_1 \otimes \dots \otimes e_m \otimes \varphi^1 \otimes \dots \otimes \varphi^n \tag{2.24}$$

Notice that the tensor T always remains the same, even with different bases; it is just the components $T_{j_1 \dots j_n}^{i_1 \dots i_m}$ that change.

Alternatively, we can write the components of a tensor as a function of the basis and dual basis:

$$T_{j_1 \dots j_n}^{i_1 \dots i_m} = T(\varphi^1, \dots, \varphi^m, e_1, \dots, e_n) \tag{2.25}$$

Remember that the tensor, at its core, is a function itself that maps m dual vectors and n vectors to a scalar. This is what we

mean by that. We can obtain an $m + n$ dimensional array of numbers using this formula like so:

Consider some tensor T with the following outputs:

$$\begin{aligned}
 T_1^1 &= T(\varphi^1, e_1) = 4 \\
 T_2^1 &= T(\varphi^1, e_2) = 3 \\
 T_3^1 &= T(\varphi^1, e_3) = -6 \\
 T_1^2 &= T(\varphi^2, e_1) = 0 \\
 T_2^2 &= T(\varphi^2, e_2) = 5 \\
 T_3^2 &= T(\varphi^2, e_3) = -2 \\
 T_1^3 &= T(\varphi^3, e_1) = -1 \\
 T_2^3 &= T(\varphi^3, e_2) = 2 \\
 T_3^3 &= T(\varphi^3, e_3) = 0
 \end{aligned}$$

Then, we can construct an array T_j^i like so:

$$\begin{aligned}
 &(\varphi_1 \quad \varphi_2 \quad \varphi_3) T \begin{pmatrix} v^1 \\ v^2 \\ v^3 \end{pmatrix} \\
 &= (\varphi_1 \quad \varphi_2 \quad \varphi_3) \begin{pmatrix} 4 & 3 & -6 \\ 0 & 5 & -2 \\ -1 & 2 & 0 \end{pmatrix} \begin{pmatrix} v^1 \\ v^2 \\ v^3 \end{pmatrix} \quad (2.26)
 \end{aligned}$$

where the components of the vector/dual vector are in the basis implied above. Take some time to examine this equation to ensure you understand how I created an array of numbers based on the outputs of the tensor.

Tensors can also be created from other tensors acting on each other; consider the following examples: we have a tensor T_ν^μ and a tensor U_ρ^ν . If we “combine” the two in such a way that one acts on the other, we obtain another tensor:

$$S_\rho^\mu = T_\nu^\mu U_\rho^\nu \quad (2.27)$$

Again, notice the Einstein summation convention here. Tensors can also act on tensors of different dimensions, like so:

$$u_\mu = T_{\mu\nu} v^\nu \quad (2.28)$$

Notice that when the Einstein summation convention applies to a certain index, the index disappears. Hopefully this should make intuitive sense to the reader. To construct a rigorous proof that we can perform this action, we will have to wait until Section 3.2, but let us take a look at a simple case. We consider two tensors $T = T^\mu_\nu e_\mu \otimes \varphi^\nu$ and $U = U^\nu_\rho e_\nu \otimes \varphi^\rho$. Then the resulting tensor must be related to the tensor product:

$$T \otimes U = T^\mu_\nu U^\rho_\sigma e_\mu \otimes e_\rho \otimes \varphi^\nu \otimes \varphi^\sigma \quad (2.29)$$

However, this is not what we are looking for; not completely, at least. We are looking for $T^\mu_\nu U^\nu_\rho$. And as we will soon see, by the Einstein summation convention, we must have

$$\sum_\nu T^\mu_\nu U^\nu_\rho e_\mu \otimes \varphi^\rho = S^\mu_\rho e_\mu \otimes \varphi^\rho \quad (2.30)$$

for some tensor S^μ_ρ . Note that sometimes the indices are spaced to show the order of the indices. This will be important especially when we begin to raise and lower indices of tensors with multiple indices, but for simpler tensors, it is not always necessary.

We are familiar with the metric tensor. We actually have that on spaces with a defined metric tensor, the metric tensor pertains to the inner product:

Definition 55 (Inner Product). *The inner product between two vectors v and w is defined as*

$$v \cdot w = g_{\mu\nu} v^\mu w^\nu \quad (2.31)$$

If the inner product vanishes, the two vectors must be orthogonal. Now, the norm of a vector can be defined like so:

Definition 56 (Norm). *The norm of a vector v is defined as*

$$||v|| = g_{\mu\nu} v^\mu v^\nu \quad (2.32)$$

Note that, unlike in Euclidean geometry, the norm may not always be positive-definite; that is to say, the norm might not always be greater than or equal to zero.

To conclude our short excursion into the study of tensors, we examine the tensor algebra; in order to define the tensor algebra, we need to define the direct sum of vector spaces:

Definition 57 (Direct Sum). *The direct sum of two abelian groups (they have an operation that is commutative and associative and have an identity and inverse) $(A, +)$ and $(B, \cdot)$, where $+$ and $\cdot$ are the defined aforementioned operations of each abelian group, is the set $A \oplus B$ where we have the elements (a_1, b_1) , (a_2, b_2) and the operation $(a_1 + a_2, b_1 \cdot b_2)$. If we have multiple groups that we are summing, we can write*

$$\bigoplus_{i=1}^n A_i = A_1 \oplus \dots \oplus A_n \quad (2.33)$$

Now we consider the tensor algebra:

Definition 58 (Tensor Algebra). *Let $T^k V$ be the tensor product of V k times. Then, the tensor algebra $T(V)$ of a vector space V is*

$$T(V) = \bigoplus_{i=0}^{\infty} T^i(V) = \mathbb{F} \oplus V \oplus (V \otimes V) \oplus \dots \quad (2.34)$$

This is called an algebra, which is a type of mathematical structure. In this algebra, multiplication is defined to be the tensor product. In abstract algebra, tensor algebras will be particularly useful, as they serve as a foundation for other mathematical objects, and they have applications in areas of physics such as quantum mechanics.

In this chapter, we have finally successfully defined the tensor and outlined some of its key properties. We will go further in depth and we will increase the amount of rigor as we move through chapters, but this section and the rest of this chapter essentially serve to provide a gentler introduction to tensors before we move onto a rigorous introduction to tensor algebra.

2.4 Coordinate Transformation Laws

Consider two bases on some vector space, e_i and $e_{i'}$. By the vector transformation rules, we have that

$$e_{i'} = \Lambda_{i'}^j e_j \quad (2.35)$$

Furthermore, we have that

$$e_j = \Lambda_j^{k'} e_{k'} \quad (2.36)$$

Therefore, we can say that the relationship between two basis vectors in the same basis is

$$e_{i'} = \Lambda_{i'}^j \Lambda_j^{k'} e_{k'} \quad (2.37)$$

However, we know that

$$e_{i'} = 0e_{1'} + 0e_{2'} + \dots + 1e_{i'} + \dots + 0e_{n'} \quad (2.38)$$

Therefore, we can say that

$$e_{i'} = \delta_{i'}^{k'} e_{k'} \quad (2.39)$$

(Recall the Einstein summation convention, once again.) In this case, δ_b^a is, as you may remember, the Kronecker delta.

Meanwhile, dual vectors transform in a similar yet slightly different way:

$$\varphi^{i'} = \Lambda_j^{i'} \varphi^j \quad (2.40)$$

and therefore

$$\varphi^{i'} = \delta_{j'}^{i'} \varphi^{j'} \quad (2.41)$$

It really is just a matter of balancing the indices, and luckily for us, a thoughtful application of common sense goes a long way in manipulating these indices, not just with this particular application but as we continue throughout our study of tensors.

We are aware of how vectors transform. But how do tensors transform? We have a tensor T with components $T^{i_1 \dots i_m}_{j_1 \dots j_n} =$

$T(\varphi^1, \dots, \varphi^m, e_1, \dots, e_n)$. Let us examine how these components transform, by exploiting the fact that the tensor is multilinear:

$$T^{i'_1 \dots i'_m}_{j'_1 \dots j'_n} = T(\varphi^{1'}, \dots, \varphi^{m'}, e_{1'}, \dots, e_{n'}) \quad (2.42)$$

$$= T(\Lambda_1^{1'} \varphi^1, \dots, \Lambda_m^{m'} \varphi^m, \dots, \Lambda_1^1 e_1, \dots, \Lambda_n^n e_n) \quad (2.43)$$

$$= \Lambda_{i'_1}^{i'_1} \dots \Lambda_{i'_m}^{i'_m} \Lambda_{j'_1}^{j'_1} \dots \Lambda_{j'_n}^{j'_n} T(\varphi^1, \dots, \varphi^m, e_1, \dots, e_n) \quad (2.44)$$

$$= \Lambda_{i'_1}^{i'_1} \dots \Lambda_{i'_m}^{i'_m} \Lambda_{j'_1}^{j'_1} \dots \Lambda_{j'_n}^{j'_n} T^{i_1 \dots i_m}_{j_1 \dots j_n} \quad (2.45)$$

We have discovered that tensors transform in what is perhaps the expected way based on how vectors and dual vectors transform, via the following relationship:

$$T^{i'_1 \dots i'_m}_{j'_1 \dots j'_n} = \Lambda_{i'_1}^{i'_1} \dots \Lambda_{i'_m}^{i'_m} \Lambda_{j'_1}^{j'_1} \dots \Lambda_{j'_n}^{j'_n} T^{i_1 \dots i_m}_{j_1 \dots j_n} \quad (2.46)$$

We can similarly derive the tensor transformation properties using tensor products. Recall that a tensor T can be defined as follows:

$$T = T^{i_1 \dots i_m}_{j_1 \dots j_m} e_1 \otimes \dots \otimes e_m \otimes \varphi^1 \otimes \dots \otimes \varphi^n \quad (2.47)$$

$$= T^{i_1 \dots i_m}_{j_1 \dots j_m} (\Lambda_1^{1'} e_{1'}) \otimes \dots \otimes (\Lambda_m^{m'} e_{m'}) \otimes (\Lambda_1^1 \varphi^{1'}) \otimes \dots \otimes (\Lambda_n^n \varphi^{n'}) \quad (2.48)$$

$$= T^{i_1 \dots i_m}_{j_1 \dots j_n} \Lambda_{i'_1}^{i'_1} \dots \Lambda_{i'_m}^{i'_m} \Lambda_{j'_1}^{j'_1} \dots \Lambda_{j'_n}^{j'_n} e_{1'} \otimes \dots \otimes e_{m'} \otimes \varphi^{1'} \otimes \dots \otimes \varphi^{n'} \quad (2.49)$$

$$= T^{i'_1 \dots i'_m}_{j'_1 \dots j'_n} e_{1'} \otimes \dots \otimes e_{m'} \otimes \varphi^{1'} \otimes \dots \otimes \varphi^{n'} \quad (2.50)$$

$$\implies T^{i'_1 \dots i'_m}_{j'_1 \dots j'_n} e_{1'} = T^{i_1 \dots i_m}_{j_1 \dots j_n} \Lambda_{i'_1}^{i'_1} \dots \Lambda_{i'_m}^{i'_m} \Lambda_{j'_1}^{j'_1} \dots \Lambda_{j'_n}^{j'_n} \quad (2.51)$$

We recall the definition of a differential. We find that, as it turns out, vectors and tensors transform exactly according to the differential:

$$v^{i'} = \frac{\partial x^{i'}}{\partial x^i} v^i \quad (2.52)$$

$$T^{i'_1 \dots i'_m}_{j'_1 \dots j'_n} = \frac{\partial x^{i'_1}}{\partial x^{i_1}} \dots \frac{\partial x^{i'_m}}{\partial x^{i_m}} \frac{\partial x^{j_1}}{\partial x^{j'_1}} \dots \frac{\partial x^{j_n}}{\partial x^{j'_n}} T^{i_1 \dots i_m}_{j_1 \dots j_n} \quad (2.53)$$

Proof. We recall that a tensor is a multilinear map $T : V^* \times \dots \times V^* \times V \times \dots \times V \rightarrow \mathbb{F}$. On a point p of a manifold, the vector space must be the tangent space, and the dual space must therefore be the cotangent space:

$$T : T_p^*M \times \dots \times T_p^*M \times T_pM \times \dots \times T_pM \rightarrow \mathbb{F} \quad (2.54)$$

In order to understand what the bases for each of the spaces we are dealing with are, we suppose several remarks that will be left to the reader to intuit:

Remark 8. *The basis of the tangent space of a manifold is the set $\frac{\partial}{\partial x^i} = \partial_i$.*

Remark 9. *The basis of the cotangent space of a manifold is the set of differentials dx^i .*

We will have to leave a more rigorous justification for this to Part 2, on tensor calculus. That being said, it is easy to see how both transform according to the Jacobian:

$$\partial_{i'} = \frac{\partial x^i}{\partial x^{i'}} \partial_i, dx^{i'} = \frac{\partial x^{i'}}{\partial x^i} dx^i \quad (2.55)$$

Factoring the property of multilinearity of tensors into the proof gives us the desired result. $\square$

For practice, let us consider one simple contrived example. We consider a change of coordinates with Jacobian matrix

$$\begin{pmatrix} \partial_{1'} x^1 & \partial_{2'} x^1 \\ \partial_{2'} x^1 & \partial_{2'} x^2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad (2.56)$$

Consider the tensor

$$T_i = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (2.57)$$

Then

$$T_{1'} = (\partial_{1'} x^1) T_1 + (\partial_{1'} x^2) T_2 = 1(1) + 2(2) = 5 \quad (2.58)$$

$$T_{2'} = (\partial_{2'} x^1) T_1 + (\partial_{2'} x^2) T_2 = 3(1) + 4(2) = 11 \quad (2.59)$$

Therefore, we have that

$$T_{i'} = \begin{pmatrix} 5 \\ 11 \end{pmatrix} \quad (2.60)$$

It can sometimes be difficult to understand how to apply coordinate transformations in practice, so some exercises provided in 2.8 should assist in that regard. For the physicist or engineer, computations like that will be of great importance; for the mathematician, a greater emphasis will be placed on abstraction, which we will find plenty of in the coming chapters.

2.5 Tensor Rank

Terminology in tensor algebra can be rather complicated. For instance, the term “tensor rank” refers to two different objects, one straightforward, one less so. It is important to understand when a source is talking about one or the other, and what each means, so I shall define both in this section.

The first definition of tensor rank is easy to understand. Given a tensor with indices $i_1, \dots, i_m, j_1, \dots, j_n$, the tensor rank is simply the number of total indices that the tensor in question has:

Definition 59 (Tensor Rank (1)). *The tensor rank, also known as the order, of a tensor $T_{j_1 \dots j_n}^{i_1 \dots i_m}$ is the number $m+n$, or the total number of indices that the tensor contains, both contravariant and covariant.*

For example, a scalar has a rank 0; vectors and dual vectors have rank 1; various matrices, bilinear functionals, and linear operators have rank 2; and so on. Finding the rank of a tensor in this context is a rather trivial exercise, and boils down simply to counting indices.

However, there exists another definition of the rank, which is more complicated. Recall the tensor product. We consider a specific example of the tensor product, known as the outer product of two vectors:

Definition 60 (Outer Product). *The outer product of two vectors $(u^1, \dots, u^i)$ and $(v^1, \dots, v^j)$ is the resulting matrix from the components of both vectors:*

$$u \otimes v = \begin{pmatrix} u^1 v^1 & \dots & u^1 v^n \\ \vdots & \ddots & \vdots \\ u^m v^1 & \dots & u^m v^n \end{pmatrix} \quad (2.61)$$

Theorem 23. *The outer product is a specific example of the tensor product for two vectors.*

Proof. We need to show that there exists a bilinear map $U \times V \rightarrow T$, where $u \in U$, $v \in V$, and $u \otimes v \in T$. To do this, we show the map has all bilinear properties. We consider that

$$(u + w) \otimes v = \begin{pmatrix} (u^1 + w^1)v^1 & \dots & (u^1 + w^1)v^n \\ \vdots & \ddots & \vdots \\ (u^m + w^m)v^1 & \dots & (u^m + w^m)v^n \end{pmatrix} \quad (2.62)$$

$$= \begin{pmatrix} u^1 v^1 & \dots & u^1 v^n \\ \vdots & \ddots & \vdots \\ u^m v^1 & \dots & u^m v^n \end{pmatrix} + \begin{pmatrix} w^1 v^1 & \dots & w^1 v^n \\ \vdots & \ddots & \vdots \\ w^m v^1 & \dots & w^m v^n \end{pmatrix} \quad (2.63)$$

$$= u \otimes v + w \otimes v \quad (2.64)$$

We can repeat a similar process for the second argument. Then, given $a \in \mathbb{F}$ we have

$$au \otimes v = \begin{pmatrix} au^1 v^1 & \dots & au^1 v^n \\ \vdots & \ddots & \vdots \\ au^m v^1 & \dots & au^m v^n \end{pmatrix} = u \otimes av \quad (2.65)$$

Based on the fact that the outer product follows the aforementioned map follows the bilinear properties, the map must therefore be bilinear, and by the definition of a tensor product, the outer product $u \otimes v$ is an example of a tensor product. $\square$

Before we define the second definition of a tensor rank, we consider the rank of a matrix A^{ij} .

Remark 10. *A matrix is considered to have rank one if it can be written as the outer product of two vectors:*

$$A^{ij} = u^i \otimes v^j \quad (2.66)$$

For instance, we consider the matrix

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \otimes \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 10 & 6 \end{pmatrix} \quad (2.67)$$

This matrix clearly has rank one. Now we shall define the rank of a matrix:

Definition 61 (Matrix Rank). *The rank of a matrix is the number of outer products that must be summed to obtain the matrix; that is, given a matrix*

$$A^{ij} = u^{i_1} \otimes v^{j_1} + \dots + u^{i_n} \otimes v^{j_n} \quad (2.68)$$

then the matrix A^{ij} has rank n .

Note that this is equivalent to the definition of matrix rank as the maximum number of linearly independent columns or rows of the matrix. We clearly see that the matrix

$$\begin{pmatrix} 5 & 3 \\ 10 & 6 \end{pmatrix} \quad (2.69)$$

has a maximum of one linearly independent column. Hence, the rank is one. The proof of these definitions being equivalent will be left to the reader if they would find it satisfactory to prove such a claim.

Given this, we can generalize:

Definition 62 (Tensor Rank (2)). *The tensor rank of a tensor $T_{j_1 \dots j_n}^{i_1 \dots i_m}$ is the number of outer/tensor products that must be summed in order to obtain the tensor; that is, given a matrix*

$$\begin{aligned} T_{j_1 \dots j_m}^{i_1 \dots i_m} &= v_{(1)}^{i_{11}} \otimes \dots \otimes v_{(m)}^{i_{1m}} \otimes \varphi_{j_{11}}^{(1)} \otimes \dots \otimes \varphi_{j_{1n}}^{(2)} + \dots \\ &+ v_{(1)}^{i_{p1}} \otimes \dots \otimes v_{(m)}^{i_{pm}} \otimes \varphi_{j_{p1}}^{(1)} \otimes \dots \otimes \varphi_{j_{pn}}^{(2)} \end{aligned} \quad (2.70)$$

Then the tensor has rank p . Note that we have p terms in this expression, each term containing the tensor product of m vectors and n dual vectors.

Note that, in order for the tensor to have order $m + n$, there must be m vectors and n dual vectors in each outer/tensor product.

When it comes to the differing definitions of rank, the reader will surely encounter both types of rank in future studies regarding tensors. In more mathematical contexts, you may see the second definition of rank more often, and the study of the decomposition of tensors by rank is an area of research that may be of interest to some readers. This topic will be covered in greater detail in Chapter 4.

2.6 Tensor Type

The reader will find that the tensor type conceptually is more straightforward than that of the tensor rank. We define tensor type as follows:

Definition 63 (Tensor Type). *The type of a tensor with m contravariant indices and n covariant indices is (m, n) .*

We examine the following tensors as examples. Consider the scalar a . Since the scalar has neither a contravariant index nor a covariant index, we say that the scalar is a tensor of type $(0, 0)$.

The vector v^i appears to have one contravariant index, yet no covariant indices. Therefore, the vector must be a tensor of type $(1, 0)$.

The dual vector, or linear functional, φ_i , has one covariant index and no contravariant indices. Therefore, this mathematical object is a tensor of type $(0, 1)$.

With matrices, it is not often clear at first glance what type the tensor in question is. It could be either $(2, 0)$, $(1, 1)$, or $(0, 2)$. For example, many matrices that are equivalent to bilinear forms are of type $(0, 2)$, since they transform covariantly by virtue of taking in two vectors and outputting a scalar.

Consider the metric tensor, $g_{\mu\nu}$, where

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2.71)$$

This matrix appears to be a (0,2) tensor by a quick examination of the indices. However, the matrix associated with the Kronecker delta function,

$$\delta_j^i = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (2.72)$$

is a (1,1) tensor, due to the fact that it has one contravariant index and one covariant index. The point is, when dealing with matrices, there is no one set tensor type that all matrices have, unlike vectors, dual vectors, and scalars; we have to examine the indices instead.

Theorem 24. *The tensor product of two tensors $T_{j_1 \dots j_n}^{i_1 \dots i_m}$ and $U_{l_1 \dots l_q}^{k_1 \dots k_p}$ is a tensor of type $(m + p, n + q)$.*

Proof. We have that

$$T \otimes U = T_{j_1 \dots j_n}^{i_1 \dots i_m} U_{l_1 \dots l_q}^{k_1 \dots k_p} e_{i_1} \otimes \dots \otimes e_{i_m} \otimes e_{k_1} \otimes \dots \otimes e_{k_p} \\ \otimes \varphi^{j_1} \otimes \dots \otimes \varphi^{j_n} \otimes \varphi^{l_1} \otimes \dots \otimes \varphi^{l_q} \quad (2.73)$$

$$= S_{j_1 \dots j_{n+q}}^{i_1 \dots i_{m+p}} e_{i_1} \otimes \dots \otimes e_{i_{m+p}} \otimes \varphi^{j_1} \otimes \dots \otimes \varphi^{j_{n+q}} \quad (2.74)$$

which appears to be a tensor of type $(m + p, n + q)$. $\square$

The words order, type, and rank are all often interchanged in literature concerning tensors, and it is unfortunate that such terms are so nebulous when it comes to identifying key properties of tensors. However, for the purposes of this book, we will refer to “order” as the number of indices the tensor has, “rank” as the natural extension of matrix rank, and “type” as the numbers in parentheses identifying the number of contravariant and covariant indices. Just bear in mind that these terms may vary from textbook to textbook, and from paper to paper.

2.7 Examples of Tensors

To supplement our rudimentary understanding of tensors thus far, let us consider some basic examples of tensors. We are already familiar with the metric tensor. In Minkowski geometry, we have that the metric tensor in matrix form is

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2.75)$$

This corresponds with the distance formula in Minkowski space-time,

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 \quad (2.76)$$

through the relation

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (2.77)$$

$$= \begin{pmatrix} dt & dx & dy & dz \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} dt \\ dx \\ dy \\ dz \end{pmatrix} \quad (2.78)$$

In Euclidean space, the metric tensor can be described as follows:

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (2.79)$$

such that

$$g_{ij} dx^i dx^j = \begin{pmatrix} dx & dy & dz \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} \quad (2.80)$$

$$= dx^2 + dy^2 + dz^2 = ds^2 \quad (2.81)$$

We have successfully recovered the distance formula in Euclidean space through the metric tensor. The metric tensor is clearly a

(0,2) tensor, as it is a bilinear form (and it must be in order for the Einstein summation convention to work out).

Now, we consider the Kronecker delta. The Kronecker delta, as we have defined before, is $\delta_j^i = 1$ when $i = j$, and $\delta_j^i = 0$ when $i \neq j$. It is easy to show that this corresponds to the identity matrix

$$\delta_j^i = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \quad (2.82)$$

We move on to the Levi-Civita symbol, where we must first define the concept of even and odd permutations:

Definition 64 (Even and Odd Permutations). *Given some $(x_1, \dots, x_n)$, a permutation of $(x_1, \dots, x_n)$ is considered even if you can arrange it by swapping adjacent indices an even number of times, and a permutation is considered odd if you can arrange it by swapping adjacent indices an odd number of times.*

With that in mind, let us define the Levi-Civita symbol:

Definition 65 (Levi-Civita Symbol). *The Levi-Civita symbol is defined as*

$$\epsilon_{i_1 \dots i_n} = \begin{cases} 1 & (i_1, \dots, i_n) \text{ is an even permutation of } (1, \dots, n) \\ -1 & (i_1, \dots, i_n) \text{ is an odd permutation of } (1, \dots, n) \\ 0 & \text{else} \end{cases} \quad (2.83)$$

The Levi-Civita symbol can be defined as follows:

$$\epsilon_{i_1 \dots i_n} = \epsilon(e_{i_1}, \dots, e_{i_n}) \quad (2.84)$$

where $e_{i_1}, \dots, e_{i_n}$ is the standard basis for $\mathbb{F}^n$. So in two dimensions, the Levi-Civita symbol would be

$$\epsilon_{ij} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (2.85)$$

We turn once again to quantum mechanics. If you have studied quantum mechanics before, you are well aware that linear operators play a key role in this specific discipline. We have, in fact, defined linear operators as (1,1) tensors. H is often used for quantum mechanics operators because many operators in quantum mechanics are Hermitian.

Definition 66 (Hermitian Operator). *A Hermitian operator is a linear operator where the matrix is equal to its complex conjugate transpose (the Hermitian conjugate); that is,*

$$H = H^\dagger \quad (2.86)$$

An example of a Hermitian operator would be

$$H = \begin{pmatrix} 1 & 1+i & 2+i \\ 1-i & 2 & 1+2i \\ 2-i & 1-2i & 3 \end{pmatrix} \quad (2.87)$$

Therefore, we consider the tensor H_j^i . We have defined $T_j^i = T(\varphi^i, e_j)$. Therefore,

$$H_j^i = H(\varphi^i, e_j) = \varphi^i H(e_j) \quad (2.88)$$

We must define Dirac notation:

Definition 67 (Dirac Notation). *In Dirac notation, a vector is denoted $|\alpha\rangle$, and a dual vector is denoted $\langle\beta|$. The inner product between two vectors $|\alpha\rangle$ and $|\beta\rangle$ is $\langle\beta|\alpha\rangle$.*

Therefore,

$$\varphi^i H(e_j) = \langle e_j | H | e_i \rangle \quad (2.89)$$

It is easy to see that tensors are critical to understanding quantum mechanics at an advanced level, and we will explore some of the applications of tensor analysis to quantum physics in greater detail later in the book.

We examine the moment of inertia tensor, which is defined as follows: the moment of inertia tensor is a bilinear form that, when we input the angular velocity vector in both arguments, we

obtain twice the total kinetic energy of the system. Additionally, the moment of inertia tensor with a raised index (that is, a linear operator) takes the angular velocity vector and outputs the angular momentum.

Definition 68 (Moment of Inertia Tensor). *The moment of inertia tensor I_{ij} satisfies the following two properties:*

$$\frac{1}{2}I_{ij}(\omega^i, \omega^j) = KE \quad (2.90)$$

$$L^i = I_j^i \omega^j \quad (2.91)$$

Perhaps a more detailed description would be as follows:

Remark 11. *The $(2,0)$ moment of inertia tensor is written as follows:*

$$I_{jk} = \sum_i m_i (g_{ii} r^i r^i \delta_{jk} - r_j r_k) \quad (2.92)$$

where r^i is the position vector and m_i is the mass of particle i . Note that we are summing over all particles to capture the moment of inertia for the whole system.

Let us find I_{xx} and I_{xy} . We have that

$$I_{xx} = \sum_i m_i (g_{ii} r^i r^i \delta_{xx} - r_x r_x) = \sum_i m_i (x_i^2 + y_i^2 + z_i^2 - x_i^2) \quad (2.93)$$

$$= \sum_i m_i (y_i^2 + z_i^2) \quad (2.94)$$

Then,

$$I_{xy} = \sum_i m_i (g_{ii} r^i r^i \delta_{xy} - r_x r_y) = \sum_i m_i (-x_i y_i) \quad (2.95)$$

$$= - \sum_i m_i x_i y_i \quad (2.96)$$

We see how we are able to recover many of the standard results from classical physics using basic properties of tensors.

As it turns out, tensors are not only used in mechanics but also in the study of electromagnetism as well; for example, the scalar electric potential Φ of a charge distribution given a position r^i can be written as

$$\begin{aligned} \Phi_i(r^i) = \frac{1}{4\pi} \left(\frac{Q^{(0)}}{\|r^i\|} + \frac{Q_i^{(1)}(r^i)}{\|r^i\|^3} + \frac{1}{2!} \frac{Q_{ij}^{(2)}(r^i, r^j)}{\|r^i\|^5} \right. \\ \left. + \frac{1}{3!} \frac{Q_{ijk}^{(3)}(r^i, r^j, r^k)}{\|r^i\|^7} + \dots \right) \end{aligned} \quad (2.97)$$

Definition 69 (Multipole Moments). *The multipole moments of a charge distribution are the tensors $Q^{(0)}$, $Q_i^{(1)}$, $Q_{ij}^{(2)}$, ..., where these tensors in a Taylor series can add up to the scalar electric potential.*

Another example of a tensor in electrodynamics would be the Maxwell stress tensor, which is defined as follows:

Definition 70 (Maxwell Stress Tensor). *The Maxwell stress tensor of an electromagnetic field is*

$$T_{ij} = E_i \otimes E_j + B_i \otimes B_j - \frac{1}{2}(g_{ij}E^iE^j + g_{ij}B^iB^j)\delta_{ij} \quad (2.98)$$

where E is the electric field, B is the magnetic field, and δ_{ij} is the Kronecker delta with one lowered index (we will explore what that means later in Chapter 3). Note that in this expression, ϵ_0 and μ_0 are all set to 1.

We have already seen the electromagnetic field tensor before, and we have that a result of the electromagnetic field tensor is

Theorem 25 (Lorentz Force Law). *If p^μ is the four-momentum (momentum in four dimensions) of a particle, and τ is the proper time (the amount of time observed by the object itself), q is the charge of a particle and v^ν is its four-velocity, then*

$$\frac{dp^\mu}{d\tau} = qF^\mu_\nu v^\nu \quad (2.99)$$

This is analogous to the classical Lorentz force law:

$$F^i = q(E^i + \epsilon_{ijk} v^j B^k) \quad (2.100)$$

Now we consider some fourth-order tensors, where order denotes the number of indices present. We specifically consider tensors of the form $T^{\mu\rho}{}_{\nu\sigma}$, which is a linear map between second-order tensors such as $S^\mu{}_\nu$.

Definition 71 (Identity Tensor). *The fourth-order identity tensor $J^{\mu\rho}{}_{\nu\sigma}$ is defined as follows; for every tensor $T^\sigma{}_\rho$, we have*

$$J^{\mu\rho}{}_{\nu\sigma} T^\sigma{}_\rho = T^\mu{}_\nu. \quad (2.101)$$

It is clear that

$$J^{\mu\rho}{}_{\nu\sigma} = \delta^\mu{}_\nu \delta^\rho{}_\sigma \quad (2.102)$$

We also have a transposition tensor,

Definition 72 (Transposition Tensor). *The transposition tensor $X^{\mu\rho}{}_{\nu\sigma}$ is a fourth-order tensor where, for every $T^\sigma{}_\rho$ we have*

$$X^{\mu\rho}{}_{\nu\sigma} T^\sigma{}_\rho = T^\mu{}_\nu \quad (2.103)$$

We must have

$$X^{\mu\rho}{}_{\nu\sigma} = \delta^\mu{}_\sigma \delta^\rho{}_\nu \quad (2.104)$$

We have seen many different examples of tensors in this section. Now, it is time to move onto a rigorous examination of some of the algebraic properties of these tensors.

2.8 Exercises

Exercise 1: The distance formula for Minkowski geometry is $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$. Why might the dt^2 term be negative?

Exercise 2: Consider the contravariant vector $(2, 3)$. Find (a) the rotation matrix and (b) the resulting vector if the basis is

rotated by (a) 90 degrees, (b) 180 degrees, (c) 270 degrees, (d) 360 degrees, (e) 45 degrees.

Exercise 3: In your own words, restate why the contravariant vector is named that way.

Exercise 4: Construct an example of a bilinear form and contravariant tensor $t : V^* \times V^* \rightarrow \mathbb{R}$, where $t \in V \otimes V$. Can you do the same for a multilinear contravariant tensor $t : V^* \times V^* \times V^* \rightarrow \mathbb{R}$ where $t \in V \otimes V \otimes V$?

Exercise 5: Prove that if the indices of a contravariant vector are (v^0, v^1, v^2, v^3) , then the indices of a covariant vector are $(-v^0, v^1, v^2, v^3)$.

Exercise 6: We discussed how the gradient ∂_μ is a covariant tensor. What would a theoretical contravariant gradient, ∂^μ , look like? Describe in great detail.

Exercise 7: Consider the metric tensor in Minkowski spacetime. Given a vector $v^\mu = (1, 4, 7, -2)$, what is the covector v_μ ?

Exercise 8: The metric tensor $g_{\mu\nu}$ can be thought of as the inner product $\langle e_\mu, e_\nu \rangle$. Use this to justify why the metric tensor is able to “lower the index” of a vector.

Exercise 9: Show that the set of all tensors of type (m, n) forms a vector space.

Exercise 10: Is the Jacobian matrix/differential a variant? Why or why not?

Exercise 11: Construct a matrix from a tensor in a similar manner as in section 2.3. Feel free to make up your own values for the tensor.

Exercise 12: Write what could potentially be the tensor resulting from the following expressions:

- (a) $T^\mu U_\mu$.
- (b) $T_\nu^\mu U^\nu$.
- (c) $S_\rho^\mu T_\mu$.
- (d) $T^{\mu\nu} S_\sigma^\rho$.
- (e) $T^\mu U_\mu V^\nu W_\nu S^\rho$.

Exercise 13: Find the tensor algebra on $\mathbb{R}^n$ over $\mathbb{R}$. State what this object means intuitively.

Exercise 14: Consider the Cartesian coordinates (x, y, z) and a vector v^i . Consider a rotation about the z -axis θ degrees counterclockwise. Find $v^{i'}$.

Exercise 15: Complete the proof that vectors and tensors transform exactly according to the differential after the step that the bases of the tangent space and cotangent space transform according to the Jacobian.

Exercise 16: Consider a basis $e_{i'}$ such that $e_{1'} = e_1$ and $e_{2'} = 2e_2$. What will happen to the components of any given covariant tensor?

Exercise 17: Attempt to find the components of the vector $(0, 1)$ in polar coordinates by utilizing the Jacobian matrix

$$\begin{pmatrix} \partial_r x & \partial_\theta x \\ \partial_r y & \partial_\theta y \end{pmatrix} \quad (2.105)$$

What happened? Explain.

Exercise 18: Find the outer product of the vectors $(2, 5, 7)$ and $(3, 0, -2)$.

Exercise 19: Find the rank of the following matrix:

$$A^{ij} = \begin{pmatrix} 1 & 2 & 0 & 4 \\ 5 & 0 & -2 & 3 \\ 0 & 3 & 3 & 4 \\ 2 & 5 & 9 & 10 \end{pmatrix} \quad (2.106)$$

Argue that, given a matrix $A \in \mathbb{F}^{m,n}$, the rank cannot exceed m and cannot exceed n .

Exercise 20: Find the tensor rank (as in order) of the following tensors:

- (a) s .
- (b) v^i .
- (c) φ_i .
- (d) T_j^i .
- (e) $T_\rho^{\mu\nu} v^\sigma$.
- (f) $T^{\alpha\beta} v_\beta$.
- (g) $T^{ijkl} U_{ijkl}$.

Exercise 21: Construct the general form of a tensor with 3 covariant indices and 2 contravariant indices of rank 3.

Exercise 22: Write what a $(6,7)$ tensor might look like in index notation. Describe what this tensor looks like as a multilinear map, and as an array of numbers.

Exercise 23: What type of tensor would a trilinear form $V \times V \times V \rightarrow \mathbb{F}$ be? Explain.

Exercise 24: Prove that the type of a tensor does not have any

bearing on the rank of a tensor (that is, the rank is not bound by the type to a specific set of numbers).

Exercise 25: Find the type of the outer product of a 3-dimensional vector and a 5-dimensional dual vector.

Exercise 26: You will learn this later, but the metric tensor can “lower” the index of a tensor like so: $g_{\mu\nu}v^\nu = v_\mu$. There also exists an inverse metric $g^{\mu\nu}$ that can “raise” the index of a tensor in the expected manner. Prove that $g_\nu^\mu = \delta_\nu^\mu$.

Exercise 27: Show that this definition of the Levi-Civita symbol $\epsilon(v_{(1)}, v_{(2)}, v_{(3)}) = (v_{(1)} \times v_{(2)}) \cdot v_{(3)}$ has all the properties of the Levi-Civita symbol defined in Section 2.7. Check that this definition is multilinear.

Exercise 28: Show that the relativistic Lorentz force law using tensor indices is analogous to the classical Lorentz force law.

Exercise 29: Show how the Levi-Civita symbol can be used to obtain the volume of a parallelepiped defined by three vectors u, v, w .

Challenge 1: We are familiar with Minkowski geometry.

(a) Prove that Minkowski space is not a metric space, despite it having a metric tensor. (This shows how mathematical language can sometimes be ambiguous and misleading.)

(b) We define a pseudometric space as the following:

Definition 73 (Pseudometric Space). *A pseudometric space is a set X with a distance function $d : X \times X \rightarrow \mathbb{R}$ that satisfies the following properties:*

$$d(x, y) \geq 0 \tag{2.107}$$

$$d(x, x) = 0 \quad (2.108)$$

$$d(x, y) = d(y, x) \quad (2.109)$$

$$d(x, z) \leq d(x, y) + d(y, z) \quad (2.110)$$

Now, note that the main difference between a metric space and a pseudometric space is that for a metric space, $d(x, y) = 0$ implies that $x = y$. This is not the case for a pseudometric space.

Is Minkowski space a pseudometric space? If not, can you think of a pseudometric space that is not a metric space?

(c) Consider the quantity ds^2 . What happens if $ds^2 < 0$? What if $ds^2 > 0$? What if $ds^2 = 0$? Elaborate in great detail.

(d) Consider the $n + 1$ -dimensional Minkowski space $ds^2 = -dx_0^2 + \sum_{i=1}^n dx_i^2$. We define:

Definition 74 (De Sitter Space). *The n -dimensional de Sitter space of $n + 1$ -dimensional Minkowski geometry is the hyperboloid of one sheet*

$$-x_0^2 + \sum_{i=1}^n x_i^2 = a^2 \quad (2.111)$$

Given that the metric is induced from Minkowski geometry, why must the induced metric be positive-definite, unlike in Minkowski geometry?

Challenge 2: We consider the Klein-Gordon equation.

Definition 75 (Wave Operator). *The wave operator $\square$ is equal to*

$$\square = \frac{1}{c^2}(\partial_0)^2\psi - \nabla^2\psi \quad (2.112)$$

(a) Show that the generalized version of the Klein-Gordon equation with a potential $V(\psi)$ is equal to

$$\square\psi + \frac{\partial V}{\partial\psi} = 0 \quad (2.113)$$

(b) The Klein-Gordon equation is invariant under the transformations

$$\psi(x) \mapsto e^{i\theta}\psi(x) \quad (2.114)$$

$$\bar{\psi}(x) \mapsto e^{-i\theta} \bar{\psi}(x) \quad (2.115)$$

Invoke Noether's theorem to prove that there is a corresponding current to this symmetry; what is this current? Show all work.

(c) Derive the Klein-Gordon equation using the Lagrangian action

$$\mathcal{S} = \int (-\hbar^2 g^{\mu\nu} \partial_\mu \bar{\psi} \partial_\nu \psi - M^2 c^2 \bar{\psi} \psi) d^4x \quad (2.116)$$

(d) Discuss the relationship between the Klein-Gordon equation and the Schrodinger equation. When might the Klein-Gordon equation be used? When would it be more useful to use the Schrodinger equation?

Challenge 3: Let V be a finite-dimensional real vector space. We explore the tensor algebra of V and several structures that arise from it.

(a) Show that $T(V)$ is a graded algebra:

Definition 76 (Graded Algebra). *A graded algebra $T^n(V)$ is an algebra with identity satisfying*

$$T^i(V)T^j(V) \subseteq T^{i+j}(V)$$

for all $i, j \geq 0$.

(b) Prove that $T(V)$ satisfies the following property, based on this definition:

Definition 77 (Algebra Homomorphism). *A linear map $\iota : V \rightarrow A$ into an associative algebra A is said to extend to an algebra homomorphism if there exists a unique algebra homomorphism $\tilde{\iota} : T(V) \rightarrow A$ whose restriction to V equals ι .*

For every associative algebra A and every linear map $\phi : V \rightarrow A$, there exists a unique algebra homomorphism

$$\tilde{\phi} : T(V) \rightarrow A$$

such that $\tilde{\phi}|_V = \phi$.

(c) Prove that I is a two-sided ideal of $T(V)$.

Definition 78 (Two-Sided Ideal). *A subset $I \subseteq T(V)$ is called a two-sided ideal if $T(V)I \subseteq I$ and $IT(V) \subseteq I$. Let I be the ideal generated by all elements of the form*

$$u \otimes v - v \otimes u, \quad u, v \in V.$$

(d) Show that $S(V)$ is a commutative algebra, meaning $ab = ba$ for all $a, b \in S(V)$.

Definition 79 (Symmetric Algebra). *The symmetric algebra of V is the quotient algebra*

$$S(V) = T(V)/I.$$

(e) Let J be the ideal generated by all elements of the form

$$v \otimes v, v \in V.$$

Prove that J is a two-sided ideal.

(f) Show that the multiplication induced on $\Lambda(V)$ satisfies

$$v \wedge v = 0, \quad v \wedge w = -w \wedge v$$

for all $v, w \in V$. (Here $v \wedge w$ denotes the class of $v \otimes w$ in the quotient.)

Definition 80 (Exterior Algebra). *The exterior algebra of V is the quotient*

$$\Lambda(V) = T(V)/J.$$

Challenge 4: We now consider the moment of inertia tensor.

(a) Explain what each component of the moment of inertia tensor entails physically; that is, what does I_{xx} mean? What about I_{xy} ?

(b) Program a simulation where, based on a given system of particles with mass m_i and position vector (x_i, y_i, z_i) , we can find the moment of inertia.

(c) Use this simulation to approximate the moment of inertia of a rod of length L , a sphere of radius r , and a cone of radius r and height h . Assume density is constant.

CHAPTER 3

Tensor Algebra

3.1 Tensor Product

We have already seen the tensor product before. By this point, we should have a conceptual understanding of what exactly the tensor product means. But now, it is time to examine the tensor product with greater rigor and abstraction. Some of these definitions might not be intuitively imaginable, but they still nevertheless make sense logically, so bear this in mind as we examine the many definitions of the tensor product.

Definition 81 (Tensor Product (1)). *The tensor product of a tensor S of order $m+n$ and a tensor T of order $p+q$ is a tensor of order $m+n+p+q$:*

$$S^{i_1 \dots i_m}_{j_1 \dots j_n} \otimes T^{i_1 \dots i_p}_{j_1 \dots j_q} = (S \otimes T)^{i_1 \dots i_{m+p}}_{j_1 \dots j_{n+q}} \quad (3.1)$$

Consider how the tensor is a map from the Cartesian product of vector and dual spaces to a scalar. That means that the images of tensors can be multiplied by each other; this is precisely what the tensor product is as well. We state the definition for two covariant tensors; this definition can easily be extended to mixed and contravariant tensors as well.

Definition 82 (Tensor Product (2)). *The tensor product of two tensors S and T can also be defined as follows: the tensor product of two covariant tensors of order k and order l is as follows:*

$$(S \otimes T)(v_1, \dots, v_{k+l}) = S(v_1, \dots, v_k) \cdot T(v_{k+1}, \dots, v_{k+l}) \quad (3.2)$$

In this case, subscripts do not indicate covariance, but rather indicate the identity of each vector.

This next definition can be rather abstract, but if you are familiar with algebra at the undergraduate level, this should be relatively accessible. The idea behind this definition of the tensor product is that we want to preserve some sort of multilinearity and nothing more. Therefore, we construct a multilinear map, which can be comprised of a map between the Cartesian product and the tensor product, and a map from the tensor product to the range. This is a natural extension of definition 32, if you would like to look back and recall.

Definition 83 (Tensor Product (3)). *Consider the multilinear map $f : V_1 \times \dots \times V_n \rightarrow W$. There must be some linear transformation $\tau : V_1 \otimes \dots \otimes V_n \rightarrow W$ such that*

$$f = \tau \circ t \quad (3.3)$$

We call τ a mediating morphism for f , and t is known as the tensor map. This definition is not used often for our purposes, and generally is used in algebra where maps between spaces like such are more important when defining tensors.

Perhaps it might seem intuitively obvious that if S and T are tensors, then $S \otimes T$ must be a tensor, but we will prove this rigorously. We exploit the fact that tensors transform a certain way, and from there, it is a matter of simple manipulations of transformations.

Theorem 26. *If S and T are tensors, then $S \otimes T$ transforms like a tensor.*

Proof. We have that

$$(S \otimes T)^{i'_1 \dots i'_{m+p} j'_1 \dots j'_{n+q}} = S^{i'_1 \dots i'_m j'_1 \dots j'_n} \otimes T^{i'_{m+1} \dots i'_{m+p} j'_{n+1} \dots j'_{n+q}} \quad (3.4)$$

We transform each of $S^{i'_1 \dots i'_m j'_1 \dots j'_n}$ and $T^{i'_{m+1} \dots i'_{m+p} j'_{n+1} \dots j'_{n+q}}$:

$$= \Lambda_{i_1}^{i'_1} \dots \Lambda_{i_m}^{i'_m} \Lambda_{j'_1}^{j_1} \dots \Lambda_{j'_n}^{j_n} S^{i_1 \dots i_m j_1 \dots j_n} \otimes \Lambda_{i'_1}^{i'_1} \dots \Lambda_{i'_p}^{i'_p} \Lambda_{j'_1}^{j'_1} \dots \Lambda_{j'_q}^{j'_q} T^{i'_1 \dots i'_p j'_1 \dots j'_q} \quad (3.5)$$

We arrange the transformation operators so that they are all in the front of the expression:

$$= \Lambda_{i_1}^{i'_1} \dots \Lambda_{i_m}^{i'_m} \Lambda_{j'_1}^{j_1} \dots \Lambda_{j'_n}^{j_n} \Lambda_{i'_1}^{i_1} \dots \Lambda_{i'_p}^{i_p} \Lambda_{j'_1}^{j_1} \dots \Lambda_{j'_q}^{j_q} S^{i_1 \dots i_m}_{j_1 \dots j_n} \otimes T^{i_1 \dots i_p}_{j_1 \dots j_q} \quad (3.6)$$

And finally, we notice that $S^{i_1 \dots i_m}_{j_1 \dots j_n} \otimes T^{i_1 \dots i_p}_{j_1 \dots j_q} = (S \otimes T)^{i_1 \dots i_{m+p}}_{j_1 \dots j_{n+q}}$:

$$= \Lambda_{i_1}^{i'_1} \dots \Lambda_{i_m}^{i'_m} \Lambda_{j'_1}^{j_1} \dots \Lambda_{j'_n}^{j_n} \Lambda_{i'_1}^{i_1} \dots \Lambda_{i'_p}^{i_p} \Lambda_{j'_1}^{j_1} \dots \Lambda_{j'_q}^{j_q} (S \otimes T)^{i_1 \dots i_{m+p}}_{j_1 \dots j_{n+q}} \quad (3.7)$$

□

Now let us look at some of the fundamental properties of the tensor product. For instance, the tensor product is not commutative. We will not prove this rigorously, but we will argue why. Examine the remark below. We see that, although scalars do commute by multiplication, we have different inputs for different tensors in different arrangements. Therefore, the tensor product cannot be commutative.

Remark 12. *The tensor product is non-commutative; that is, $S \otimes T \neq T \otimes S$. This can be shown by simply considering how, in the case of two covariant tensors, $(S \otimes T)(v_1, \dots, v_{m+n}) = S(v_1, \dots, v_m) \cdot T(v_{m+1}, \dots, v_{m+n})$, whereas $(T \otimes S)(v_1, \dots, v_{m+n}) = T(v_1, \dots, v_n) \cdot S(v_{n+1}, \dots, v_{n+m})$. These expressions are by no means equivalent.*

We now shift our attention to some very simple properties of tensors. The proofs may look intimidating, but do not get lost in the apparent complexity of the expressions; the arguments behind the proofs are rather simple. We shall now present the properties of the tensor product as well as their proofs in full:

Theorem 27. *We have that, for all tensors S_1, S_2, T of appropriate dimension,*

$$(S_1 + S_2) \otimes T = S_1 \otimes T + S_2 \otimes T \quad (3.8)$$

Proof. We have

$$((S_1 + S_2) \otimes T)(\varphi_1, \dots, \varphi_{m+n}, v_1, \dots, v_{p+q}) \quad (3.9)$$

$$\begin{aligned} &= (S_1 + S_2)(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p) \\ &\quad \cdot T(\varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \end{aligned} \quad (3.10)$$

$$\begin{aligned} &= S_1(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p) \cdot T(\varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \\ &\quad + S_2(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p) \cdot T(\varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \end{aligned} \quad (3.11)$$

$$\begin{aligned} &= (S_1 \otimes T)(\varphi_1, \dots, \varphi_{m+n}, v_1, \dots, v_{p+q}) \\ &\quad + (S_2 \otimes T)(\varphi_1, \dots, \varphi_{m+n}, v_1, \dots, v_{p+q}) \end{aligned} \quad (3.12)$$

□

Theorem 28. *We have that, for all tensors S, T_1, T_2 of appropriate dimension,*

$$S \otimes (T_1 + T_2) = S \otimes T_1 + S \otimes T_2 \quad (3.13)$$

Proof. We have

$$(S \otimes (T_1 + T_2))(\varphi_1, \dots, \varphi_{m+n}, v_1, \dots, v_{p+q}) \quad (3.14)$$

$$= S(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p)$$

$$\cdot (T_1 + T_2)(\varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \quad (3.15)$$

$$\begin{aligned} &= S(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p) \cdot T_1(\varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \\ &\quad + S(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p) \cdot T_2(\varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \end{aligned} \quad (3.16)$$

$$\begin{aligned} &= (S \otimes T_1)(\varphi_1, \dots, \varphi_{m+n}, v_1, \dots, v_{p+q}) \\ &\quad + (S \otimes T_2)(\varphi_1, \dots, \varphi_{m+n}, v_1, \dots, v_{p+q}) \end{aligned} \quad (3.17)$$

□

Theorem 29. *We have that, for all tensors S, T of appropriate dimension and every $\lambda \in \mathbb{F}$,*

$$(\lambda S) \otimes T = S \otimes (\lambda T) = \lambda(S \otimes T) \quad (3.18)$$

Proof. We have

$$((\lambda S) \otimes T)(\varphi_1, \dots, \varphi_{m+n}, v_1, \dots, v_{p+q}) \quad (3.19)$$

$$= (\lambda S)(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p) \cdot T(\varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \quad (3.20)$$

$$= S(\lambda\varphi_1, \dots, \lambda\varphi_m, v_1, \dots, v_p) \cdot T(\varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \quad (3.21)$$

$$= (S \otimes T)(\lambda\varphi_1, \dots, \lambda\varphi_m, v_1, \dots, v_p, \varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \quad (3.22)$$

$$= \lambda(S \otimes T)(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p, \varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \quad (3.23)$$

$$= (S \otimes T)(\varphi_1, \dots, \lambda\varphi_m, v_1, \dots, v_p, \varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \quad (3.24)$$

$$= S(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p) \cdot T(\varphi_{m+1}, \dots, \lambda\varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \quad (3.25)$$

$$= S(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p) \cdot (\lambda T)(\varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \quad (3.26)$$

$$= (S \otimes (\lambda T))(\varphi_1, \dots, \varphi_m, v_1, \dots, v_p, \varphi_{m+1}, \dots, \varphi_{m+n}, v_{p+1}, \dots, v_{p+q}) \quad (3.27)$$

□

Theorem 30. *We have*

$$(S \otimes T) \otimes U = S \otimes (T \otimes U) \quad (3.28)$$

Proof. We have

$$((S \otimes T) \otimes U)(\varphi_1, \dots, \varphi_{i+j+k}, v_1, \dots, v_{p+q+r}) \quad (3.29)$$

$$= (S(\varphi_1, \dots, \varphi_i, v_1, \dots, v_p) \cdot T(\varphi_{i+1}, \dots, \varphi_{i+j}, v_{p+1}, \dots, v_{p+q})) \cdot U(\varphi_{i+j+1}, \dots, \varphi_{i+j+k}, v_{p+q+1}, \dots, v_{p+q+r}) \quad (3.30)$$

$$= S(\varphi_1, \dots, \varphi_i, v_1, \dots, v_p) \cdot T(\varphi_{i+1}, \dots, \varphi_{i+j}, v_{p+1}, \dots, v_{p+q}) \cdot U(\varphi_{i+j+1}, \dots, \varphi_{i+j+k}, v_{p+q+1}, \dots, v_{p+q+r}) \quad (3.31)$$

$$= S(\varphi_1, \dots, \varphi_i, v_1, \dots, v_p) \cdot (T(\varphi_{i+1}, \dots, \varphi_{i+j}, v_{p+1}, \dots, v_{p+q}) \cdot U(\varphi_{i+j+1}, \dots, \varphi_{i+j+k}, v_{p+q+1}, \dots, v_{p+q+r})) \quad (3.32)$$

$$= (S \otimes (T \otimes U))(\varphi_1, \dots, \varphi_{i+j+k}, v_1, \dots, v_{p+q+r}) \quad (3.33)$$

□

Finally, we end with a theorem from Spivak's *Calculus on Manifolds* regarding the tensor product. This proof is abstract, so I will add commentary throughout explaining what we are trying to show in each step.

Theorem 31. *If we have a basis $e_1, \dots, e_n \in V$, and a dual basis $\varphi_1, \dots, \varphi_n \in V^*$ such that $\varphi_i(e_j) = \delta_j^i$, then the set of tensor products k times*

$$\varphi_{i_1} \otimes \dots \otimes \varphi_{i_k}, \quad 1 \leq i_1, \dots, i_k \leq n$$

is a basis for the tensor product $V^k = V \otimes \dots \otimes V$, which must have dimension n^k .

Proof. We have that, recalling the definition of the tensor product as well as the fact that $\varphi_i(e_j) = \delta_j^i$,

$$(\varphi_{i_1} \otimes \dots \otimes \varphi_{i_k})(e_{j_1}, \dots, e_{j_k}) = \delta_{j_1}^{i_1} \dots \delta_{j_k}^{i_k} \quad (3.34)$$

Let $v_1, \dots, v_k \in V$ where $v_i = \sum_{j=1}^n a_{ij} e_j$ and $T \in V^k$. Then, considering the multilinearity of the tensor, we must therefore have

$$T(v_1, \dots, v_k) = \sum_{j_1, \dots, j_k=1}^n a_{1j_1} \dots a_{kj_k} T(e_{j_1}, \dots, e_{j_k}) \quad (3.35)$$

The arguments $a_{1j_1}, \dots, a_{kj_k}$ can be equated to the tensor product $\varphi_{i_1} \otimes \dots \otimes \varphi_{i_k}(v_1, \dots, v_k)$ through the fact that $\varphi_{i_k}(v_j) = a_{ji_k}$. Therefore, we can write

$$= \sum_{i_1, \dots, i_k=1}^n T(e_{i_1}, \dots, e_{i_k}) \cdot (\varphi_{i_1} \otimes \dots \otimes \varphi_{i_k})(v_1, \dots, v_k) \quad (3.36)$$

Consider the fact that we can write every tensor in the tensor space as a linear combination of $\varphi_{i_1} \otimes \dots \otimes \varphi_{i_k}$. Then, by the definition of span, it is clear that $\varphi_{i_1} \otimes \dots \otimes \varphi_{i_k}$ spans V^k . Now if we have some numbers $a_{i_1}, \dots, a_{i_k}$ such that

$$\sum_{i_1, \dots, i_k=1}^n a_{i_1} \dots a_{i_k} (\varphi_{i_1} \otimes \dots \otimes \varphi_{i_k}) = 0 \quad (3.37)$$

Then we have, by considering the effect of this tensor on the basis $e_{j_1}, \dots, e_{j_k}$,

$$\sum_{i_1, \dots, i_k=1}^n a_{i_1} \dots a_{i_k} (\varphi_{i_1} \otimes \dots \otimes \varphi_{i_k})(e_{j_1}, \dots, e_{j_k}) = 0 \quad (3.38)$$

which must imply that $a_{j_1} = \dots = a_{j_k} = 0$ if this must be true for all inputs. Therefore, $\varphi_{i_1} \otimes \dots \otimes \varphi_{i_k}$ must be linearly independent. $\square$

3.2 Contraction

You should already be familiar with the Einstein summation convention; one “upstairs” index and one “downstairs” index of the same type indicates that we sum over said index. However, you may be wondering why this is the case. To answer this question sufficiently, we introduce a new operation, known as contraction, which explains exactly what happens when we sum over a contravariant index and a covariant index as in the Einstein summation convention.

Definition 84 (Contraction). *Contraction is an operation on a tensor with at least two indices, one contravariant and one covariant, where for some index i_p and another index j_q , where $1 \leq p \leq m$ and $1 \leq q \leq n$, we have $i_p = j_q$, and so*

$$T^{i_1 \dots i_p \dots i_m}_{j_1 \dots i_p \dots j_n} = T^{i_1 \dots i_{p-1} i_{p+1} \dots i_m}_{j_1 \dots j_{q-1} j_{q+1} \dots j_n} \quad (3.39)$$

In other words, the contraction of an (m, n) tensor gives an $(m - 1, n - 1)$ tensor. Recall that the Einstein summation convention holds here. We generally use the same symbol for contraction; that is, $T^i_{ij} = T_j$; this may be slightly confusing at times but it is notation that the reader may have to acquaint themselves with.

We have already stated that the contraction of a given tensor is also a tensor, but let us prove this statement. It comes down to a simple application of the transformation matrices of tensors; it suffices to show that the contracted tensor transforms in exactly the same way a tensor of the same type would.

Theorem 32. *The contraction of a tensor $T^{i_1 \dots i_m}_{j_1 \dots j_m}$ is a tensor.*

Proof.

$$T^{i'_1 \dots i'_{p-1} i'_{p+1} \dots i'_m}_{j'_1 \dots j'_{q-1} j'_{q+1} \dots j'_n} = T^{i'_1 \dots i'_p \dots i'_m}_{j'_1 \dots i'_p \dots j'_n} \quad (3.40)$$

$$= \Lambda^{i'_1}_{i_1} \dots \Lambda^{i'_p}_{i_p} \dots \Lambda^{i'_m}_{i_m} \Lambda^{j_1}_{j'_1} \dots \Lambda^{j_q}_{i'_p} \dots \Lambda^{j_n}_{j'_n} T^{i_1 \dots i_p \dots i_m}_{j_1 \dots j_q \dots j_n} \quad (3.41)$$

$$= (\Lambda^{i'_p}_{i_p} \Lambda^{j_q}_{i'_p}) \Lambda^{i'_1}_{i_1} \dots \Lambda^{i'_m}_{i_m} \Lambda^{j_1}_{j'_1} \dots \Lambda^{j_n}_{j'_n} T^{i_1 \dots i_p \dots i_m}_{j_1 \dots j_q \dots j_n} \quad (3.42)$$

$$= \delta^{j_q}_{i_p} \Lambda^{i'_1}_{i_1} \dots \Lambda^{i'_m}_{i_m} \Lambda^{j_1}_{j'_1} \dots \Lambda^{j_n}_{j'_n} T^{i_1 \dots i_p \dots i_m}_{j_1 \dots j_q \dots j_n} \quad (3.43)$$

Note that when $i_p = j_q$, then $\delta^{j_q}_{i_p} = 1$, so we have

$$\Lambda^{i'_1}_{i_1} \dots \Lambda^{i'_m}_{i_m} \Lambda^{j_1}_{j'_1} \dots \Lambda^{j_n}_{j'_n} T^{i_1 \dots i_p \dots i_m}_{j_1 \dots i_p \dots j_n} \quad (3.44)$$

$$= \Lambda^{i'_1}_{i_1} \dots \Lambda^{i'_m}_{i_m} \Lambda^{j_1}_{j'_1} \dots \Lambda^{j_n}_{j'_n} T^{i_1 \dots i_{p-1} i_{p+1} \dots i_m}_{j_1 \dots j_{q-1} j_{q+1} \dots j_n} \quad (3.45)$$

It is clear then that the contraction of a tensor transforms like a tensor. $\square$

We consider a mixed tensor $T : V \times V^* \rightarrow \mathbb{F}$. Then, we note a specific type of contraction known as the trace. Those who are familiar with matrices might recall the trace of a matrix as the sum of all of its diagonal elements; that is, $\sum_{i=1}^n a_{ii}$, where a_{ij} is an element of the matrix of row i and column j . Then, we define the trace from a tensor perspective as follows:

Definition 85 (Trace). *Given a mixed tensor T^i_j , the trace $tr_{(i,j)} T^i_j$ is the contraction of the tensor $T^i_i = T$.*

Let us have a tensor $T^{i_1 \dots i_p \dots i_m}_{j_1 \dots j_n}$ and another tensor $S^{i_{m+1} \dots i_{m+k}}_{j_{n+1} \dots j_q \dots j_{n+l}}$ where $i_p = j_q$. We take the tensor

$$\begin{aligned} & U^{i_1 \dots i_{p-1} i_{p+1} \dots i_{m+k}}_{j_1 \dots j_{q-1} j_{q+1} \dots j_{n+l}} \\ &= T^{i_1 \dots i_p \dots i_m}_{j_1 \dots j_n} S^{i_{m+1} \dots i_{m+k}}_{j_{n+1} \dots j_{q-1} i_p j_{q+1} \dots j_{n+l}} \end{aligned} \quad (3.46)$$

$$= \text{tr}_{(i_p, j_q)}(T^{i_1 \dots i_p \dots i_m}_{j_1 \dots j_n} \otimes S^{i_{m+1} \dots i_{m+k}}_{j_{n+1} \dots j_q \dots j_{n+l}}) \quad (3.47)$$

Therefore, we can rigorously define the tensor acting on another tensor in such a way as the trace of the tensor product of the two tensors. We exploit the equivalence of a mixed tensor of order two and a linear operator, and it suffices to show that the trace of a tensor is simply the trace of its corresponding matrix as defined in matrix algebra.

Theorem 33. *The contraction of a tensor T_j^i is equivalent to taking the trace of the corresponding linear operator (matrix) $S : V \rightarrow V$.*

Proof. We are given that $T = T_i^i$ (recall the Einstein summation convention). We define

$$T(\varphi^i, e_j) = \varphi^i(S(e_j)) \quad (3.48)$$

where S is a matrix composed of elements a_{ij} . Then

$$T_j^i = T(\varphi^i, e_j) = \varphi^i(S(e_j)) = a_{ij} \quad (3.49)$$

Therefore, taking the trace is simply a matter of summing up all elements a_{ii} . $\square$

Note that one cannot contract two contravariant indices or two covariant indices; it always has to be one contravariant and one covariant. The reader should verify this.

Now we reach a theorem that once again may seem obvious, but ought to be proved rigorously once again. That is, the contraction of different indices results in different tensors. As a simple example of this, we must have that $T^i_{ij} \neq T^i_{ji}$. We outline the proof like so: we compare the sums of the tensor with its specific basis vectors and dual basis vectors as inputs, and consider how there exists an apparent difference between the two.

Theorem 34. *The contraction of different indices results in different tensors.*

Proof. Consider the tensor $T^{i_1 \dots i_p \dots i_m}_{j_1 \dots j_q \dots j_r \dots j_n}$. Consider the two contractions where $i_p = j_q$ and $i_p = j_r$. Then we must have, for the first contraction,

$$\sum_{i_p} T(\varphi^{i_1}, \dots, \varphi^{i_p}, \dots, \varphi^{i_m}, e_{j_1}, \dots, e_{i_p}, \dots, e_{j_r}, \dots, e_{j_n}) \quad (3.50)$$

and for the second contraction,

$$\sum_{i_p} T(\varphi^{i_1}, \dots, \varphi^{i_p}, \dots, \varphi^{i_m}, e_{j_1}, \dots, e_{j_q}, \dots, e_{i_p}, \dots, e_{j_n}) \quad (3.51)$$

These are, upon close inspection, not at all necessarily the same. Therefore, the contraction of different indices must result in different tensors. $\square$

By this point, the Einstein summation convention might make more sense now. The operation of contraction is an often used operation, and therefore, we will be taking many sums of tensors like so. It is an incredible economy of notation that we omit the sigma notation for all contractions, and instead note that whenever there is the same index as both a contravariant and covariant index, then we must sum the tensor over that index.

3.3 Symmetrization

We continue to gather more terminology with regard to tensors by discussing the concept of symmetric tensors. A symmetric matrix is one where the transpose of the matrix is equal to the matrix itself. The symmetric tensor is, as many concepts of tensors are, a natural generalization of the concept relating to the matrix. Note that a matrix is symmetric if $a_{ij} = a_{ji}$, where a_{ij} is an element of the matrix in row i and column j . Therefore, we can define symmetry of a tensor to be the following:

Definition 86 (Symmetric). *A tensor is considered symmetric with respect to the indices i_m, i_n , if $T^{i_1 \dots i_m \dots i_n \dots} = T^{i_1 \dots i_n \dots i_m \dots}$. A similar definition can be stated for covariant tensors and mixed*

tensors (so long as the indices are either both contravariant or both covariant). A tensor is called symmetric if we have

$$T^{i_1 \dots i_m \dots i_n \dots} = T^{i_1 \dots i_n \dots i_m \dots} \quad (3.52)$$

for any $m \geq 1$ and $n \geq 1$.

We remark that the symmetric tensor has fewer independent components when regarded as an array of numbers than a tensor that is not considered to be symmetric; this is the case for matrices, and it will end up being the case for tensors as well.

Remark 13. *If we can represent the tensor as a matrix, we find that symmetry reduces the number of independent components of the matrix in question.*

For instance, if we have a tensor that can be represented by the matrix A , where

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad (3.53)$$

This matrix clearly has 9 independent components. For a symmetric tensor represented by the matrix B , where

$$B = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{12} & b_{22} & b_{23} \\ b_{13} & b_{23} & b_{33} \end{pmatrix} \quad (3.54)$$

We count 6 independent components for this matrix.

Now we will specify some of our terms a little bit. We understand what it means for a matrix to be symmetric with regard to two indices, and we understand what it means for a matrix to be symmetric for all indices. Now, we examine the following definition of complete symmetry:

Definition 87 (Completely Symmetric). *A tensor $T^{i_1 \dots i_p \dots i_q \dots}$ is considered completely symmetric in the indices $\{i_p, \dots, i_q\}$ if it is symmetric for any i_m, i_n where $p \leq m \leq q$ and $p \leq n \leq q$. Note that this definition is stronger than symmetry with respect to a pair of indices, but weaker than symmetry in general.*

Note that the symmetric tensor is also considered to be completely symmetric in all indices. Now we reach an important caveat when it comes to the symmetry of tensors. As it turns out, it does not serve us well to define symmetry with respect to a contravariant and a covariant index. This point is illustrated below:

Remark 14. *The tensor can be defined to be symmetric with regard to contravariant indices or symmetric with regard to covariant indices, but cannot be symmetric with regard to a pair of indices, one being contravariant and one being covariant.*

This is because, if we consider a simple case, that would imply

$$T^i_{jk} = T_j^i{}_k \quad (3.55)$$

which does not carry any meaning whatsoever. This expression implies that we have $T(\varphi^i, e_j, e_k) = T(e_j, \varphi^i, e_k)$, which is obviously not an equation of interest; we are instead interested in those equations where we are able to swap basis vectors or dual basis vectors in the argument and obtain equality. Therefore, the definition of symmetry applies only to pairs of contravariant indices and pairs of covariant indices.

It quickly follows that a completely symmetric tensor cannot be a mixed tensor:

Corollary 1. *A tensor that is considered to be symmetric must either be contravariant or covariant, but not mixed.*

The proof for this theorem follows swiftly from many of the claims above, and therefore will be left to the reader.

It is also useful for us to “symmetrize” tensors; that is, we want to take a tensor that is not symmetric inherently and perform such an operation that the resulting tensor will be symmetric. We can define the following operation below which accomplishes this feat:

Definition 88 (Symmetrize). *We symmetrize a tensor with respect to certain indices (these indices must be either all contravariant or all covariant, as we can see) via the following*

process: we take the sum of all tensors where the desired indices are arranged in different permutations and divide by the factorial of the number of desired indices (the number of terms in our sum).

It may help to illustrate this process with an example. Let us symmetrize the tensor T_{ij} . We take the sum of all tensors with the indices permuted in different manners: $T_{ij} + T_{ji}$, and divide by the number of terms in this sum: 2. Therefore,

$$T_{(ij)} = \frac{1}{2}(T_{ij} + T_{ji}) \quad (3.56)$$

We use parentheses to denote the symmetrization of a tensor.

It will be left to the reader to verify that the symmetrization of a tensor is really symmetric.

3.4 Antisymmetrization

Now we move from the concept of symmetry to that of antisymmetry. We begin by illustrating the concept of antisymmetry in a matrix. An example of an antisymmetric matrix would be as follows:

$$A = \begin{pmatrix} 0 & 2 & 3 & 4 \\ -2 & 0 & 4 & 5 \\ -3 & -4 & 0 & 6 \\ -4 & -5 & -6 & 0 \end{pmatrix} \quad (3.57)$$

We consider the definition of symmetry of a matrix, and understand how it can be generalized to that of a tensor. Likewise, antisymmetry, or skew-symmetry, can also be generalized to tensors. We establish the following definition.

Definition 89 (Antisymmetric). *A tensor is considered antisymmetric with respect to the indices i_m, i_n , if $T^{i_1 \dots i_m \dots i_n \dots} = -T^{i_1 \dots i_n \dots i_m \dots}$. A similar definition can be stated for covariant tensors and mixed tensors (so long as the indices are either both contravariant or both covariant). A tensor is called antisymmetric if we have*

$$T^{i_1 \dots i_m \dots i_n} = -T^{i_1 \dots i_n \dots i_m} \quad (3.58)$$

for any $m \geq 1$ and $n \geq 1$.

Just as in symmetry, it is easy to ascertain the following remark that antisymmetry reduces the number of independent components of a matrix.

Remark 15. *If we can represent the tensor as a matrix, then antisymmetry reduces, just as symmetry, the number of independent components of the matrix in question.*

We can understand this by simply noting that, for a 3×3 matrix, we have 9 independent components, but as is the case for symmetric matrices, we only have 3 independent components for an antisymmetric 3×3 matrix. The reader should verify this remark more rigorously.

We now give the following definition of complete antisymmetry, which is analogous to that of complete symmetry in the expected manner.

Definition 90 (Completely Antisymmetric). *A tensor $T^{i_1 \dots i_p \dots i_q}$ is called completely antisymmetric in the indices $\{i_p, \dots, i_q\}$ if it is antisymmetric for any i_m, i_n where $p \leq m \leq q$ and $p \leq n \leq q$. Note again that this definition is stronger than symmetry with respect to a pair of indices, but weaker than symmetry in general.*

Just as with symmetric tensors, we find that antisymmetric tensors cannot be defined with regard to a mix of contravariant and covariant indices; that is to say, you cannot “swap” a contravariant and a covariant index and have it be the same tensor. The motivation for this is very similar to the motivation for the analogous claim in the section on symmetric tensors.

Remark 16. *The tensor can be defined to be antisymmetric with regard to contravariant indices or antisymmetric with regard to covariant indices, but cannot be antisymmetric with regard to a pair of indices, one being contravariant and one being covariant.*

It follows shortly that an antisymmetric tensor in general must either be contravariant or covariant. It cannot be mixed.

Corollary 2. *A tensor that is considered to be antisymmetric must either be contravariant or covariant, but not mixed.*

Now, it follows that we have an operation that allows us to symmetrize a tensor, so it only makes sense that we shall develop an operation that allows us to antisymmetrize a tensor. The reader should verify that the described operation results in an antisymmetric tensor.

Definition 91 (Antisymmetrize). *We can antisymmetrize a tensor with respect to certain indices (these indices must be either all contravariant or all covariant) via the following process: we take the **alternating** sum of all tensors where the desired indices are arranged in different permutations and divide once again by the factorial of the number of desired indices (the number of terms in our sum).*

This definition might be confusing at first. Here are a couple examples. We shall antisymmetrize the tensor T_{ij} . We take the alternating sum of all tensors with the indices permuted in different manners: T_{ij} stays the same sign since it requires zero swaps between adjacent indices, but T_{ji} must have a negative sign since it requires one swap between adjacent indices, between i and j . The alternating sum thus becomes $T_{ij} - T_{ji}$. We divide all of this by the number of terms in the alternating sum, which is clearly 2. Therefore, we must have

$$T_{[ij]} = \frac{1}{2}(T_{ij} - T_{ji}) \quad (3.59)$$

Unlike symmetrization, we use brackets to denote the antisymmetrization of a tensor.

Now let us compute the antisymmetrization of the tensor T_{ijk} . The alternating sum is as follows:

$$T_{ijk} - T_{ikj} - T_{jik} + T_{jki} + T_{kji} - T_{kji} \quad (3.60)$$

Notice how the sign reflects how many adjacent indices are swapped; if there is an odd number of swaps, then there is a negative sign, whereas if there is an even number of swaps, then there is a positive sign. The number of terms in this alternating sum, as one may count, is 6. Therefore, we have

$$T_{[ijk]} = \frac{1}{6}(T_{ijk} - T_{ikj} - T_{jik} + T_{jki} + T_{kji} - T_{kji}) \quad (3.61)$$

3.5 The Metric Tensor

The metric tensor is one of the most fundamental parts of differential geometry and tensor calculus, and it primarily stems from the concept of distance. When we examine a manifold, we want to have a way to quantify distance on that manifold. As it turns out, we have a way to formalize this concept, and it comes in the form of the metric tensor, which is also known, in a surface, as the first fundamental form.

Definition 92 (First Fundamental Form). *Given an inner product $\langle, \rangle$ and tangent vectors to a surface x_u, x_v , we have that the first fundamental form can be thought of as*

$$ds^2 = Edu^2 + 2Fdu dv + Gdv^2 \quad (3.62)$$

where $E = \langle x_u, x_u \rangle$, $F = \langle x_u, x_v \rangle = \langle x_v, x_u \rangle$, and $G = \langle x_v, x_v \rangle$.

We can find that we are able to represent the first fundamental form as a matrix, like so:

$$I = \begin{pmatrix} E & F \\ F & G \end{pmatrix} \quad (3.63)$$

Note that the first fundamental form is essentially equivalent to the inner product in that it takes in two tangent vectors $ax_u + bx_v$ and $cx_u + dx_v$ and outputs the following:

$$I(ax_u + bx_v, cx_u + dx_v) = acE + adF + bcF + bdG \quad (3.64)$$

Observe how, by the definition of the first fundamental form, we have defined distance officially on a surface in $\mathbb{R}^3$; indeed, if we consider the Euclidean first fundamental form of a plane,

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (3.65)$$

the distance formula becomes $ds^2 = du^2 + dv^2$, which is precisely the distance formula taught in elementary mathematics. We can

also derive the concept of infinitesimal area from that of the first fundamental form, like so:

$$dA = \sqrt{EG - F^2} du dv \quad (3.66)$$

The proof of this will be left to the advanced reader.

We generalize the first fundamental form in its matrix form to that of the metric tensor, which is applicable to spaces of any dimension, which is defined like so:

Definition 93 (Metric Tensor). *The metric tensor is a bilinear map, where, given a manifold M and a point x , $g : T_x M \times T_x M \rightarrow \mathbb{F}$, where $g_{\mu\nu} u^\mu v^\nu = u \cdot v$.*

We can arrange the components of the metric tensor in the form of a matrix:

$$g_{\mu\nu} = \begin{pmatrix} g_{11} & \cdots & g_{1n} \\ \vdots & \ddots & \vdots \\ g_{n1} & \cdots & g_{nn} \end{pmatrix} \quad (3.67)$$

Definition 94 (Orthogonal). *Given two vectors u and v , if we have that $u \cdot v = 0$, then the two vectors are considered to be orthogonal.*

We also define the norm of a vector through the metric tensor:

Definition 95 (Norm). *The norm of a vector v is simply $g_{\mu\nu} v^\mu v^\nu$.*

We also define an inverse metric $g^{\mu\nu}$ as follows:

Definition 96 (Inverse Metric). *The inverse metric $g^{\mu\nu}$ is a tensor of order two that is defined in such a way that $g_{\mu\nu} g^{\nu\rho} = \delta_\mu^\rho$.*

Let us examine some examples of the metric tensor as well as the inverse metric. In n -dimensional Euclidean geometry, we have that

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \quad (3.68)$$

The inverse metric is simply the matrix that, when multiplied by the metric tensor, gives us the identity matrix (or the Kronecker delta). Thus,

$$g^{\mu\nu} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = g_{\mu\nu} \quad (3.69)$$

as

$$\begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \quad (3.70)$$

Now we consider the metric tensor in Minkowski geometry:

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.71)$$

The inverse metric is clearly

$$g^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.72)$$

due to the fact that

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.73)$$

The metric tensor allows us to define the distance formula in a similar manner as the first fundamental form:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (3.74)$$

So, for a metric tensor represented by a 2×2 matrix, we have

$$ds^2 = g_{11}dx^1dx^1 + g_{12}dx^1dx^2 + g_{21}dx^1dx^2 + g_{22}dx^2dx^2 \quad (3.75)$$

Given a manifold M which is equipped with a metric tensor $g_{\mu\nu}$ at every point $p \in M$, we say that the set of all metric tensors at every point is called a tensor field. The tensor field is a very similar concept to that of the vector field, but with tensors instead of vectors.

3.6 Raising and Lowering Indices

The metric tensor can also be used to raise and lower indices. Let us develop the intuition behind this statement; we recall from the definition of a metric tensor that, given two vectors $u^\mu, v^\nu \in V$, we have

$$g_{\mu\nu}u^\mu v^\nu = u \cdot v \quad (3.76)$$

Note that we also have that

$$u \cdot v = u_\nu v^\nu \quad (3.77)$$

Therefore,

$$g_{\mu\nu}u^\mu v^\nu = u_\nu v^\nu \implies g_{\mu\nu}u^\mu = u_\nu \quad (3.78)$$

Similarly, we find that

$$g^{\mu\nu}g_{\nu\rho}v^\rho = \delta^\mu_\rho v^\rho = v^\mu \quad (3.79)$$

We also have

$$g_{\nu\rho}v^\rho = v_\nu \quad (3.80)$$

by 3.78. Therefore,

$$g^{\mu\nu}v_\nu = v^\mu \quad (3.81)$$

We have observed how the metric tensor can raise or lower the index of a vector; that is to say, the metric tensor can transform a vector into a dual vector, and the inverse metric can transform a dual vector into a vector. It will be left to the reader to extend this property to that of tensor, but it essentially follows that the final result is as follows:

Remark 17 (Raising and Lowering Indices). *The metric tensor can be used to lower the index of a tensor:*

$$g_{ij_n} T^{j_1 \dots j_{n-1} j_n j_{n+1} \dots j_p}_{k_1 \dots k_m} = T^{j_1 \dots j_{n-1} \quad i \quad j_{n+1} \dots j_p}_{k_1 \dots k_m} \quad (3.82)$$

The inverse metric can also be used to raise the index of a tensor:

$$g^{ij_n} T_{j_1 \dots j_{n-1} j_n j_{n+1} \dots j_p}^{k_1 \dots k_m} = T_{j_1 \dots j_{n-1} \quad i \quad j_{n+1} \dots j_p}^{k_1 \dots k_m} \quad (3.83)$$

Further cases can be derived in an analogous way.

We now consider several examples of this property in action. We are given the Minkowski metric

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.84)$$

Then we consider the tensor $T^{\rho\sigma}$:

$$T^{\rho\sigma} = \begin{pmatrix} 2 & 4 & 3 & 1 \\ -3 & 0 & 2 & 7 \\ 1 & 1 & 2 & -1 \\ 0 & 4 & -2 & 5 \end{pmatrix} \quad (3.85)$$

Can we find $T^\mu{}_\nu$? We have

$$T^\mu{}_\nu = g_{\rho\nu} T^{\mu\rho} = T^{\mu\rho} g_{\rho\nu} \quad (3.86)$$

$$= \begin{pmatrix} 2 & 4 & 3 & 1 \\ -3 & 0 & 2 & 7 \\ 1 & 1 & 2 & -1 \\ 0 & 4 & -2 & 5 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.87)$$

$$= \begin{pmatrix} -2 & 4 & 3 & 1 \\ 3 & 0 & 2 & 7 \\ -1 & 1 & 2 & -1 \\ 0 & 4 & -2 & 5 \end{pmatrix} \quad (3.88)$$

One particular point of interest is that given two tensors represented by matrices, you have to ensure that the contracted

indices are adjacent to each other in order to multiply the matrices. We will see more of this in the next example. Let us find the tensor $T_\mu{}^\nu$.

$$T_\mu{}^\nu = g_{\mu\rho} T^{\rho\nu} \quad (3.89)$$

The contracted indices are already next to each other.

$$= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 3 & 1 \\ -3 & 0 & 2 & 7 \\ 1 & 1 & 2 & -1 \\ 0 & 4 & -2 & 5 \end{pmatrix} \quad (3.90)$$

$$= \begin{pmatrix} -2 & -4 & -3 & -1 \\ -3 & 0 & 2 & 7 \\ 1 & 1 & 2 & -1 \\ 0 & 4 & -2 & 5 \end{pmatrix} \quad (3.91)$$

Can we find $T_{\mu\nu}$? We can! Observe:

$$T_{\mu\nu} = g_{\mu\rho} T^\rho{}_\nu \quad (3.92)$$

$$= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 4 & 3 & 1 \\ 3 & 0 & 2 & 7 \\ -1 & 1 & 2 & -1 \\ 0 & 4 & -2 & 5 \end{pmatrix} \quad (3.93)$$

$$= \begin{pmatrix} 2 & -4 & -3 & -1 \\ 3 & 0 & 2 & 7 \\ -1 & 1 & 2 & -1 \\ 0 & 4 & -2 & 5 \end{pmatrix} \quad (3.94)$$

These matrices are all very similar, but for more interesting metric tensors, we may obtain more diverse results. Now, given $T_\mu{}^\nu$, can we obtain $T^{\mu\nu}$? We will have to use the inverse metric.

$$T^{\mu\nu} = g^{\mu\rho} T_\rho{}^\nu \quad (3.95)$$

$$= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & -4 & -3 & -1 \\ -3 & 0 & 2 & 7 \\ 1 & 1 & 2 & -1 \\ 0 & 4 & -2 & 5 \end{pmatrix} \quad (3.96)$$

$$= \begin{pmatrix} 2 & 4 & 3 & 1 \\ -3 & 0 & 2 & 7 \\ 1 & 1 & 2 & -1 \\ 0 & 4 & -2 & 5 \end{pmatrix} \quad (3.97)$$

3.7 Exterior Algebra

We now turn our attention to that field of mathematics known as exterior algebra, which concerns itself especially with covariant antisymmetric tensors. The reason why will become clearer later, but the short answer is that these tensors are especially useful for integrating on manifolds. We begin this section by restating the definition of a covariant antisymmetric tensor for the reader's convenience.

Definition 97 (Covariant Antisymmetric Tensor). *A tensor is considered covariant and antisymmetric if it is a multilinear map $T_{i_1 \dots i_n} : V \times \dots \times V \rightarrow \mathbb{F}$, where $T_{i_1 \dots i_n}(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = -T_{i_1 \dots i_n}(v_1, \dots, v_j, \dots, v_i, \dots, v_n)$ for any $1 \leq i, j \leq n$.*

As it turns out, the covariant antisymmetric tensors of the same type form a set known as the exterior product space. This space is a vector subspace of the set of all tensors of the same type. The proof for this is relatively straightforward and will be outlined shortly after.

Definition 98 (Antisymmetric Tensor Space, Exterior Product Space). *We have that the set of all antisymmetric $(0, n)$ tensors forms a vector subspace $\Lambda^n(V^*)$, also denoted $AT^n(V)$ of the set of all $(0, n)$ tensors. These two notations represent the same subspace; the only difference is that someone may prefer one instead of the other.*

Theorem 35. *The exterior product space $\Lambda^n(V^*)$ is a vector subspace of the set of all $(0, n)$ tensors.*

Proof. It is straightforward that $\Lambda^n(V^*)$ is a subset of the set of all $(0, n)$ tensors, by the definition of the exterior product space. Note that there exists a “zero” tensor $\mathbf{0}_{i_1 \dots i_n} : V \times$

$\dots \times V \rightarrow \mathbb{F}$ that maps the Cartesian product of all vectors to zero. It is left to the reader to verify that this tensor fulfills the necessary properties. Then, we consider that we are able to add antisymmetric tensors of the same type and we find that the resulting tensor is also antisymmetric and of the same type. Similarly, if we multiply an antisymmetric tensor by a scalar, we find that the resulting tensor is also antisymmetric and of the same type. The machinery behind this proof will be relegated to the reader as an exercise. $\square$

Now that we have established a definition for the antisymmetric tensor space, it naturally follows that we can define a symmetric tensor space as well consisting of symmetric tensors of the same type.

Definition 99 (Symmetric Tensor Space). *The set of all symmetric $(0, n)$ tensors forms a vector subspace $ST^n(V)$ of the set of all $(0, n)$ tensors.*

The reader may also come across the idea of the exterior product space within the context of a manifold. Recall the notation for the tangent space $T_p(M)$ and its dual $T_p^*(M)$. We can define an antisymmetric tensor space on a manifold as well:

Remark 18. *When $V^* = T_p^*M$, where M is a manifold and $p \in M$, we have that the set of all antisymmetric $(0, n)$ tensors can also be represented by $\Lambda^n T_p^*(M)$ or $\Omega_p^r(M)$.*

Now we come across one of the most important operations in exterior algebra: the wedge product. The wedge product is an operation that takes two tensors and returns a tensor. It is the antisymmetrization of the tensor product of two covariant antisymmetric tensors. The definition is fairly straightforward:

Definition 100 (Wedge Product). *The wedge product $\wedge$ is an operation on two tensors $T_{i_1 \dots i_m} \in \Lambda^m(V^*)$ and $S_{i_1 \dots i_n} \in \Lambda^n(V^*)$ such that*

$$T \wedge S = (T \otimes S)_{[i_1 \dots i_{m+n}]} \quad (3.98)$$

The wedge product is also known as the exterior product. Now, it is time for us to define a basis for each antisymmetric

tensor space using the dual basis and the wedge product. The first statement will be presented without proof, as if it were a definition. The second and third statements will be left to the reader to verify.

Theorem 36. *If we have a dual basis $\{\varphi_1, \dots, \varphi_k\}$, then $\{\varphi_I = \varphi_{i_1} \wedge \dots \wedge \varphi_{i_p}\}$ where $1 \leq i_1, \dots, i_p \leq k$ is a basis for $\Lambda^p(V^*)$. It follows that*

$$\dim \Lambda^p(V^*) = \binom{k}{p} \quad (3.99)$$

It follows that the dimension of $\Lambda^n(V^)$ where $n > k = 0$.*

At last, we come across the definition of the exterior algebra. We define the exterior algebra as the direct sum of all the possible antisymmetric tensor spaces, up to $\dim V^* = k$, like so:

Definition 101 (Exterior Algebra). *The exterior algebra $\Lambda(V^*)$ or $AT(V)$ is defined as*

$$\Lambda(V^*) = \bigoplus_{i=0}^k \Lambda^i(V^*) = \Lambda^0(V^*) \oplus \Lambda^1(V^*) \oplus \dots \oplus \Lambda^k(V^*) \quad (3.100)$$

Similarly, we are able to define a symmetric tensor algebra, which, unlike its antisymmetric counterpart, is an infinite direct sum. The reader should ponder why this is the case.

Definition 102 (Symmetric Tensor Algebra). *The symmetric tensor algebra $ST(V)$ is defined as*

$$ST(V) = \bigoplus_{i=0}^{\infty} ST^i(V) = ST^0(V) \oplus ST^1(V) \oplus \dots \quad (3.101)$$

3.8 Wedge Product

Before we dive deeper into our study of the wedge product, we must first adjust our notation slightly so that we are not burdened by indices to such a degree. We first introduce the concept of the permutation, which should be familiar to most readers. However, the notation may be novel. Observe the definition below.

Definition 103 (Permutation). *We call $\pi \in S_p$ a permutation of the numbers 1 to p . When we denote a permuted tensor T as T^π , we say that*

$$T^\pi(v_1, \dots, v_p) = T(v_{\pi(1)}, \dots, v_{\pi(p)}) \quad (3.102)$$

where $\{\pi(n)\}$ is a permutation of 1 to p . We have that $(-1)^\pi$ must be either 1 or -1 depending on if π is even or odd.

Additionally, we add new notation for the concept of antisymmetrization. Since antisymmetric tensors are also known as alternating tensors, we shall use the notation Alt for antisymmetrization. This saves us from using all those indices for tensors.

Remark 19 (Antisymmetrization Alternate Notation). *The antisymmetrization of a tensor T can also be written as $\text{Alt}(T)$.*

Now we define the wedge product. Although the wedge product has already been defined in the previous section, we define it using the novel notation below.

Definition 104 (Wedge Product). *The wedge product $T \wedge S \in \Lambda^{m+n}(V^*)$ where $T \in T^m(V)$ and $S \in T^n(V)$, is defined as follows:*

$$T \wedge S = \text{Alt}(T \otimes S) \quad (3.103)$$

Note that $T \otimes S$ may not always be antisymmetric if both T and S are antisymmetric. Now we observe a lemma and a theorem from Guillemin and Pollack, both of which concern fundamental properties of the wedge product. The lemma, as always, is necessary for the proof of the theorem. Therefore, we begin with the lemma, that the wedge product of a tensor whose antisymmetrization is zero must be zero as well.

Lemma 3. *If $\text{Alt}(T) = 0$, then $T \wedge S = S \wedge T = 0$.*

Proof. Consider the set A of possible permutations of $(1, \dots, m+n)$ where $m+1, \dots, m+n$ is fixed. This is a copy of the set

S_m . We have a map $A \rightarrow S_m$ where $\pi \mapsto \pi'$ where $\pi \in A$ and $\pi' \in S_m$ and π' is the restriction of π to $(1, \dots, m)$. We have that

$$(T \otimes S)^\pi = T^{\pi'} \otimes S \quad (3.104)$$

as $(T \otimes S)(v_{\pi(1)}, \dots, v_{\pi(m)}, v_{m+1}, \dots, v_{m+n}) = T(v_{\pi(1)}, \dots, v_{\pi(m)}) \cdot S(v_{m+1}, \dots, v_{m+n})$. It is also clear that $(-1)^\pi = (-1)^{\pi'}$. Therefore, we must have

$$\sum_{\pi \in A} (-1)^\pi (T \otimes S)^\pi = \left[\sum_{\pi' \in S_m} (-1)^{\pi'} T^{\pi'} \right] \otimes S \quad (3.105)$$

$$= p! \text{Alt}(T) \otimes S = 0 \quad (3.106)$$

We assert the fact that for any element in S_{m+n} , we can represent it as a union of all the cosets (a coset is a set created by multiplying each element of A by a fixed element of S_n like so) $A \circ \sigma = \{\pi \circ \sigma : \pi \in A\}$. The reader who is well-versed in group theory may be able to verify this claim. We find that for each coset,

$$\sum_{\pi \in A} (-1)^{\pi \circ \sigma} (T \otimes S)^{\pi \circ \sigma} = (-1)^\sigma \left[\sum_{\pi \in A} (-1)^\pi (T \otimes S)^\pi \right]^\sigma \quad (3.107)$$

$$= (-1)^\sigma (0)^\sigma = 0 \quad (3.108)$$

It follows that $T \wedge S$ must be zero, as well as $S \wedge T$. $\square$

Note that the antisymmetrization is a linear operation; that is, $\text{Alt}(S + T) = \text{Alt}(S) + \text{Alt}(T)$, and $\text{Alt}(\lambda T) = \lambda \text{Alt}(T)$.

At last we come to the ultimate theorem of this section, which simply states that the wedge product is associative. Unlike the proof that the tensor product is associative, which is relatively straightforward, this proof is not necessarily as straightforward. Some of it will be left to the reader as an exercise, so pay attention closely.

Theorem 37. *The wedge product is associative.*

Proof. By the definition of the wedge product, we know that

$$(T \wedge S) \wedge R = \text{Alt}((T \wedge S) \otimes R) \quad (3.109)$$

Therefore, we have that

$$(T \wedge S) \wedge R - \text{Alt}(T \otimes S \otimes R) \quad (3.110)$$

$$= \text{Alt}((T \wedge S) \otimes R) - \text{Alt}((T \otimes S) \otimes R) \quad (3.111)$$

$$= \text{Alt}((T \wedge S - T \otimes S) \otimes R) \quad (3.112)$$

Because $T \wedge S$ is already alternating, then $\text{Alt}(T \wedge S) = T \wedge S$, so

$$\text{Alt}(T \wedge S - T \otimes S) = \text{Alt}(T \wedge S) - \text{Alt}(T \otimes S) \quad (3.113)$$

$$= T \wedge S - T \wedge S = 0 \quad (3.114)$$

Therefore, by Lemma 3,

$$\text{Alt}((T \wedge S - T \otimes S) \otimes R) = 0 \quad (3.115)$$

Therefore, $(T \wedge S) \wedge R = \text{Alt}(T \otimes S \otimes R)$. A similar argument, which will be left to the reader, claims that $T \wedge (S \wedge R) = \text{Alt}(T \otimes S \otimes R)$, and from there, the theorem is proven. $\square$

3.9 The Determinant

Before we explore the first definition of a determinant that is presented in this book, the reader will be first reminded of the Levi-Civita symbol, in n dimensions. This symbol is not necessarily a tensor, but it is important, and, as we will see soon, can become a tensor as well with some modifications.

Definition 105 (Levi-Civita Symbol). *The Levi-Civita symbol $\epsilon_{i_1 \dots i_n}$ is*

$$\epsilon_{i_1 \dots i_n} = \begin{cases} 1 & i_1, \dots, i_n \text{ is an even permutation of } 1, \dots, n \\ -1 & i_1, \dots, i_n \text{ is an odd permutation of } 1, \dots, n \\ 0 & \text{otherwise} \end{cases} \quad (3.116)$$

With this in mind, the following definition of a determinant is presented, using the Levi-Civita symbol for notational clarity.

Definition 106 (Determinant). *The determinant of a matrix A composed of elements a_{ij} is denoted $|A|$ and is equivalent to*

$$|A| = \sum_{i_1, \dots, i_n} \epsilon_{i_1 \dots i_n} a_{1i_1} \dots a_{ni_n} \quad (3.117)$$

where $\epsilon_{i_1 \dots i_n}$ is the Levi-Civita symbol.

At this point, we explore some elementary properties of the determinant. We also begin to observe the concept of an orthogonal matrix, which represents an orthogonal linear transformation. A matrix is considered orthogonal if its transpose times the matrix itself equals the identity matrix. Besides the properties concerning an orthogonal matrix, we find that the determinant preserves multiplication, transforming matrix multiplication into scalar multiplication, like so:

Theorem 38. *We have that $|AB| = |A| \cdot |B|$.*

The definition of the orthogonal matrix is presented below.

Definition 107 (Orthogonal Matrix). *A matrix A is considered to be orthogonal if its transpose times the matrix itself equals the identity matrix, that is, $A^T A = I$.*

We now consider some of the fundamental properties of orthogonal matrices. Our task is to ultimately prove that the determinant of an orthogonal matrix must either be 1 or -1, but in order to achieve this goal, we must consider the following lemma, presented here without proof:

Lemma 4. *We have that, for an orthogonal matrix A , $|A^T| \cdot |A| = |I| = 1$.*

Considering this lemma, we are now able to reach the following theorem regarding orthogonal matrices; namely, that the determinant of an orthogonal matrix must either be 1 or -1. This theorem follows shortly from Lemma 4, and the reader should verify this theorem easily.

Theorem 39. *For an orthogonal matrix A , it follows that $|A|^2 = 1$, and therefore the determinant of an orthogonal matrix can either be 1 or -1 .*

Finally, we reach yet another fundamental property of determinants; if we consider a matrix A and its transformation $A' = BAB^{-1}$, we reach the following conclusion: the determinants of these matrices are the same. In that way, we are able to conclude that the determinant remains invariant under a coordinate change. Perhaps that description should ring a bell to the reader who has been paying attention, but we will reach the obvious conclusion later in the section.

Theorem 40. *Given a matrix A and a transformed matrix $A' = BAB^{-1}$, we have that $|A'| = |A|$.*

Proof. Consider the expression $A' = BAB^{-1}$. Right-multiplying both sides by B , we obtain $A'B = BA$. It follows that $|A'| \cdot |B| = |B| \cdot |A|$. Since the determinant of a matrix is a scalar, we divide by $|B|$, giving the desired expression. $\square$

One of the most pivotal determinants in mathematics is that of the Jacobian determinant. The Jacobian determinant is what its name simply entails; that is, it is the determinant of the Jacobian matrix.

Definition 108 (Jacobian Determinant). *The Jacobian determinant is written as*

$$\left| \frac{\partial(x_1, \dots, x_n)}{\partial(u_1, \dots, u_n)} \right| = \begin{vmatrix} \partial_1 x_1 & \partial_1 x_2 & \dots & \partial_1 x_n \\ \partial_2 x_1 & \partial_2 x_2 & \dots & \partial_2 x_n \\ \vdots & \vdots & \ddots & \vdots \\ \partial_n x_1 & \partial_n x_2 & \dots & \partial_n x_n \end{vmatrix} \quad (3.118)$$

One of the most important properties of the Jacobian determinant is that we are able to use the aforementioned determinant to apply a change of variables to an integral. That is, as those readers who have taken multivariable calculus are aware, we are able to change the integration variables based on the relationship between these variables and the Jacobian determinant based on such

relationships. Since we have $dx_1 \dots dx_n = \left| \frac{\partial(x_1, \dots, x_n)}{\partial(u_1, \dots, u_n)} \right| du_1 \dots du_n$, we have that

Theorem 41. *Given a region R in the coordinates $u_1, \dots, u_n$ that is being integrated and its corresponding region R' in the coordinates $x_1, \dots, x_n$, we have that*

$$\int_{R'} dx_1 \dots dx_n = \int_R \left| \frac{\partial(x_1, \dots, x_n)}{\partial(u_1, \dots, u_n)} \right| du_1 \dots du_n \quad (3.119)$$

At this point, we shift our attention to a more abstract definition of the determinant. The presentation of this section closely follows the beginning of Axler's section on determinants. We begin by defining a certain antisymmetric covariant tensor a_T based on another antisymmetric covariant tensor a as follows:

Definition 109 (a_T). *Consider an antisymmetric covariant tensor a of type $(0, n)$. a_T is another antisymmetric covariant tensor of the same type that is defined by $a_T(v_1, \dots, v_n) = a(Tv_1, \dots, Tv_n)$ for all vectors $v_1, \dots, v_n \in V$.*

At last we have reached the second, more abstract, definition of the determinant. We obtain the following definition:

Definition 110 (Determinant (2)). *The determinant of a linear operator T , denoted $|T|$, is defined to be the unique scalar such that*

$$a_T = |T|a \quad (3.120)$$

We conclude with a theorem on determinants circling back to the idea that the determinant itself is an antisymmetric covariant tensor in a sort of way; that is, it takes in a certain amount of vectors and produces a scalar in such a way that preserves multilinearity. The proof is rather short and elegant, like so:

Theorem 42. *Consider a matrix where the vectors $v_1, \dots, v_n$ constitute of the columns of the matrix. The map that takes the vectors $v_1, \dots, v_n \in V$ to $|v_1 \ \dots \ v_n|$ is an antisymmetric covariant tensor of type $(0, n)$.*

Proof. We let T be the linear transformation such that $Te_k = v_k$ and let a be an antisymmetric multilinear map such that $a(e_1, \dots, e_n) = 1$. Then we must have

$$|v_1 \quad \dots \quad v_n| = |T| = |T|a(e_1, \dots, e_n) = a(Te_1, \dots, Te_n) \quad (3.121)$$

$$= a(v_1, \dots, v_n) \quad (3.122)$$

We have found that the determinant is a covariant antisymmetric tensor that takes in n vectors and outputs a scalar. $\square$

3.10 The Volume Element

At this point, we will now examine the application of tensor algebra to the act of integrating along a manifold, at a basic level. In order to do this, however, we need to define a couple concepts. The first between them is that of orientation, which allows us to examine the “sides” of a manifold. We consider two bases of the tangent space of the manifold and consider whether they are part of the same orientation or not, like so:

Definition 111 (Orientation). *We classify the arrangement of bases $(e_1, \dots, e_n)$ and $(e'_1, \dots, e'_n)$ into two groups depending on their relationship with each other; if $e'_i = a^{ij}e_j$, then consider the matrix A with elements a^{ij} . If $|A| > 0$, then the two arrangements are in the same group. Each of the two groups is called an orientation μ for the vector space with the bases span.*

Since we understand the definition of orientation, we are now able to define the volume element of a given manifold. This volume element must be a tensor, as it must remain invariant under coordinate transformation. We want the volume element to contain the following conditions, as written below:

Definition 112 (Volume Element). *The volume element ω is a unique antisymmetric covariant tensor given by an orientation μ such that if $(e_1, \dots, e_n) = \mu$, then $\omega(e_1, \dots, e_n) = 1$.*

If we have the typical inner product on Euclidean spaces and an orientation μ , the absolute value of the determinant of

vectors $v_1, \dots, v_n$ is the volume of the parallepiped defined by the vectors $v_1, \dots, v_n$. There are some other remarks that can be made about the volume element on notation and the fact that it can be nowhere zero. The reason why it cannot be zero at any point should be intuitive to the reader.

Remark 20. *The volume element must be nowhere zero, and is sometimes denoted dV . It can also be denoted ds in one dimension and dA in two dimensions. The volume of a manifold M is equal to*

$$V = \int_M dV \quad (3.123)$$

We recall how the determinant can be used to define the volume in certain cases. We can generalize this claim. So long as the typical inner product is used, we are able to define the volume element on a surface in three-dimensional space as the determinant of the matrix with rows consisting of the two vectors as well as the unit normal, as denoted below.

Remark 21. *If ω is a $(0, 2)$ tensor with the typical inner product, we have that*

$$\omega(v, w) = \begin{vmatrix} v \\ w \\ n(x) \end{vmatrix} \quad (3.124)$$

where n is the unit normal of the surface M at a point $x \in M$.

We have that, through some elementary properties of the determinant, assuming ω is defined as in Remark 21,

$$\omega(v, w) = \langle v \times w, n(x) \rangle \quad (3.125)$$

Since $n(x)$ is parallel to $v \times w$ and is of length 1, we have that

$$\langle v \times w, n(x) \rangle = |v \times w| \quad (3.126)$$

However, since $\omega(v, w) = dA$, we conclude that, for a surface with the typical inner product,

$$dA = |v \times w| \quad (3.127)$$

We now present a theorem that the reader has seen before; now, it is time to prove it.

Theorem 43. *Given a surface c ,*

$$\int_c dA = \int_{[0,1]^2} \sqrt{EG - F^2} \quad (3.128)$$

where E, F, G are the coefficients of the first fundamental form.

Proof. To prove this claim, the following lemma must be used. We present it here without proof:

Lemma 5.

$$\int_c dA = \int_{[0,1]^2} c^* dA \quad (3.129)$$

Therefore, we claim that

$$\int_{[0,1]^2} c^* dA(e_1, e_2) = \int_{[0,1]^2} dA(c_* e_1, c_* e_2) \quad (3.130)$$

To proceed forward in the proof, we need to understand conceptually the relationship between the basis vectors and partial derivatives. Recall that the basis vectors of the tangent space of a manifold can be described by partial derivatives: ∂_i . Therefore, we claim that

$$\int_{[0,1]^2} dA(c_* e_1, c_* e_2) = \int_{[0,1]^2} dA(\partial_1 c, \partial_2 c) \quad (3.131)$$

$$= \int_{[0,1]^2} |\partial_1 c \times \partial_2 c| \quad (3.132)$$

$$\begin{aligned} &= \int_{[0,1]^2} |(\partial_1 c^2 \partial_2 c^3 - \partial_1 c^3 \partial_2 c^2, \partial_1 c^3 \partial_2 c^1 \\ &\quad - \partial_1 c^1 \partial_2 c^3, \partial_1 c^1 \partial_2 c^2 - \partial_1 c^2 \partial_2 c^1)| \end{aligned} \quad (3.133)$$

Expanding the expression in such a way that the absolute value is removed (i.e., by squaring and adding up the resulting squares, and taking the square root), we obtain

$$= \int_{[0,1]^2} \sqrt{(\partial_1 c^1)^2 (\partial_2 c^2)^2 + (\partial_1 c^2)^2 (\partial_2 c^1)^2 - 2 \partial_1 c^1 \partial_1 c^2 \partial_2 c^1 \partial_2 c^2}$$

$$\frac{+(\partial_1 c^2)^2 (\partial_2 c^3)^2 + (\partial_1 c^3)^2 (\partial_2 c^2)^2 - 2\partial_1 c^2 \partial_1 c^3 \partial_2 c^2 \partial_2 c^3}{+(\partial_1 c^3)^2 (\partial_2 c^1)^2 + (\partial_1 c^1)^2 (\partial_2 c^3)^2 - 2\partial_1 c^1 \partial_1 c^3 \partial_2 c^1 \partial_2 c^3} \quad (3.134)$$

Let

$$E = (\partial_1 c^1)^2 + (\partial_1 c^2)^2 + (\partial_1 c^3)^2 \quad (3.135)$$

$$F = \partial_1 c^1 \partial_2 c^1 + \partial_1 c^2 \partial_2 c^2 + \partial_1 c^3 \partial_2 c^3 \quad (3.136)$$

$$G = (\partial_2 c^1)^2 + (\partial_2 c^2)^2 + (\partial_2 c^3)^2 \quad (3.137)$$

It follows that our expression becomes

$$\int_{[0,1]^2} \sqrt{EG - F^2} \quad (3.138)$$

Hence proving our claim. $\square$

This formula is useful to compute the surface area of surfaces in 3-dimensional space, but sometimes, we would like to work in different types of spaces and geometries. For that, we need to define some notation. As the integrand dx must transform like a covariant tensor, the wedge product can apply to different dx s, creating an expression of the form $dx^0 \wedge dx^1$. We can repeat this process, using the following definition:

Definition 113 (Integrand). *The integrand dx is a $(0, 1)$ tensor, and $dx^n = dx^0 \wedge \dots \wedge dx^{n-1}$ is an antisymmetric $(0, n)$ tensor.*

Now, it is time to examine a second definition of the volume element, this definition concerning the metric tensor. We consider the notation above regarding integrands, and we make the following claim without a rigorous justification, save for the proposition that the addition of the square root of the determinant of the metric tensor allows the element to transform like a tensor:

Definition 114 (Volume Element (with Metric)). *The volume element transforms as a tensor does, and can be described by $\sqrt{|g|}dx^0 \wedge \dots \wedge dx^{n-1}$, where $|g|$ is the determinant of the metric tensor.*

From there, the following remark becomes clear:

Remark 22. *The integral of a scalar function f over a manifold M of dimension n is*

$$\int_M f(x) \sqrt{|g|} dx^0 \wedge \dots \wedge dx^{n-1} \quad (3.139)$$

3.11 Exercises

Exercise 1: Prove that $(V \otimes W)^* = V^* \otimes W^*$.

Exercise 2: Explain, in your own words, the proof of Theorem 31 in full.

Exercise 3: Find the order and type of the tensor $T_1^{i_1}{}_{i_2} \otimes T_2^{i_2}{}_{i_3} \otimes \dots \otimes T_k^{i_k}{}_{i_{k+1}}$. Justify using a rigorous proof.

Exercise 4: Let π be a permutation of the indices $\{1, \dots, n\}$. Prove that there must exist an isomorphism

$$\sigma : V_1 \otimes \dots \otimes V_n \rightarrow V_{\pi(1)} \otimes \dots \otimes V_{\pi(n)} \quad (3.140)$$

where

$$\sigma(v_1 \otimes \dots \otimes v_n) = v_{\pi(1)} \otimes \dots \otimes v_{\pi(n)} \quad (3.141)$$

Hint: an isomorphism means that there is a bijection (one-to-one and onto) map between the two spaces.

Exercise 5: Verify that all the definitions of the tensor product are consistent with each other.

Exercise 6: Consider the $(2,1)$ tensor. What sort of tensor results from a contraction of its indices? What familiar object is this tensor?

Exercise 7: Explain, in your own words, the point of the Einstein summation convention.

Exercise 8: Prove that the double contraction of a tensor also transforms like a tensor.

Exercise 9: Consider the set of all tensors which are in the tensor algebra of a vector space V , where

$$T(V) = \bigoplus_{i=0}^{\infty} T^i(V) \quad (3.142)$$

Consider all contractions of said tensors. Does this set itself form a tensor algebra? Prove your claim.

Exercise 10: Justify Remark 13. Justify Remark 14.

Exercise 11: Prove Corollary 1.

Exercise 12: Verify that the symmetrization of a tensor is itself symmetric.

Exercise 13: Find the formula for a symmetrization of a tensor as follows: $T_{(\mu\nu\rho)}$.

Exercise 14: Of a tensor with n covariant indices, how many different symmetrizations can this tensor have? Explain your thought process.

Exercise 15: Verify that for an antisymmetric or symmetric $n \times n$ matrix, the number of independent components is reduced compared to that of a regular $n \times n$ matrix. By how much is it reduced?

Exercise 16: Prove that antisymmetrizing a tensor makes it antisymmetric.

Exercise 17: Find the alternating sum with respect to the permutations of indices of $T_{\mu\nu\rho\sigma}$

Exercise 18: Find $T_{i(jkl)m}$. Find $T_{i[jkl]m}$.

Exercise 19: Derive the first fundamental form of the paraboloid $z = x^2 + y^2$.

Exercise 20: Derive the first fundamental form of a surface of revolution. Show all steps.

Exercise 21: Find the inverse metric of the following metric tensor:

$$g_{\mu\nu} = \begin{pmatrix} -\frac{1}{a} & 0 & 0 & 0 \\ 0 & \frac{1}{a} & 0 & 0 \\ 0 & 0 & \frac{1}{a} & 0 \\ 0 & 0 & 0 & \frac{1}{a} \end{pmatrix} \quad (3.143)$$

Exercise 22: Find dA for the paraboloid in Exercise 19.

Exercise 23: Verify that the metric tensor and inverse metric can lower and raise one of the indices of a tensor, respectively. Only a couple cases need to be shown. Conclude that the properties observed for vectors (Eqs. 3.78, 3.81) translate accordingly for tensors.

Exercise 24: Write the expression for $T_{\mu\nu\rho\sigma}$ in terms of $T^{\mu\nu}{}_{\rho}{}^{\sigma}$.

Exercise 25: Given a tensor

$$T^i_j = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ -1 & -2 & -3 \end{pmatrix} \quad (3.144)$$

and a metric

$$g_{ij} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix} \quad (3.145)$$

Find T^{ij} and T_{ij} .

Exercise 26: Compose a rule for what the components of the tensor $T_{\mu\nu}$ will be given a tensor $T^{\mu\nu}$ represented as a matrix and the Minkowski metric.

Exercise 27: Complete the proof of Theorem 35.

Exercise 28: Prove the consequences of Theorem 36.

Exercise 29: Is it true that the union of the symmetric tensor space and the antisymmetric tensor space is the set of all tensors of a certain type? Justify your answer.

Exercise 30: Is either the symmetric tensor algebra or the exterior algebra considered to be a tensor algebra? Justify your answer.

Exercise 31: Why is it that the symmetric tensor algebra is an infinite direct sum, yet the exterior algebra is not? Try to think of a profound answer for this.

Exercise 32: Complete the proof of Theorem 37.

Exercise 33: Given V_μ and W_ν , where

$$V_\mu = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 0 \end{pmatrix}, W_\nu = \begin{pmatrix} 2 \\ 4 \\ -1 \\ 5 \end{pmatrix} \quad (3.146)$$

as well as the outer product $\otimes$, find $V \wedge W$.

Exercise 34: Make up two tensors of order 2 T and S . Is it possible to numerically calculate $T \wedge S$?

Exercise 35: Let $v = ae_1 + be_2$ and $w = ce_1 + de_2$. Follow these steps to find $v \wedge w$:

(a) The wedge product is distributive. Express $v \wedge w$ in terms of a, b, c, d, e_1, e_2 .

(b) Note that the wedge product of a basis vector with itself is zero. Simplify the expression in (a).

(c) Note that $v \wedge w = -w \wedge v$. Considering the case for basis vectors, simplify the expression in (b).

(d) Does the expression of the coefficient look familiar? Find a formula reminiscent of this expression.

Exercise 36: Prove that $|AB| = |A| \cdot |B|$ for any matrices A, B .

Exercise 37: Verify that all definitions of the determinant are equivalent.

Exercise 38: Find the determinant of the following matrix:

$$A = \begin{pmatrix} 1 & 0 & 2 & 3 \\ 0 & 2 & 4 & -1 \\ 2 & 4 & 1 & -2 \\ 0 & 1 & 2 & 3 \end{pmatrix} \quad (3.147)$$

Exercise 39: Why is the determinant not a symmetric multilinear form? Use an example to justify.

Exercise 40: Prove that $dA = n^1 dy \wedge dz + n^2 dz \wedge dx + n^3 dx \wedge dy$, where the unit normal is n^i . Hint: in three dimensions, the wedge product is analogous to the cross product.

Exercise 41: Explain why $\int \sqrt{EG - F^2}$ is a special case of $\int \sqrt{|g|} dx^0 \wedge \dots \wedge dx^{n-1}$ (there is a discrepancy in notation, so be aware of that).

Exercise 42: Find the surface area of a paraboloid on the circle with center at the origin and radius 1.

Exercise 43: Generalize the expression in Exercise 40.

Challenge 1: In this challenge, the reader is asked to prove some advanced properties concerning the tensor product.

(a) Consider two finite-dimensional vector spaces U, V over the field $\mathbb{F}$. We define $\text{Hom}(U, V)$ as the space of all linear maps $U \rightarrow V$. Prove that there exists an isomorphism between the spaces $U^* \otimes V$ and $\text{Hom}(U, V)$.

(b) Describe the inverse of the map that you found in part (a), if you have not already.

(c) We define a ring as follows:

Definition 115 (Ring). *A ring R is a mathematical structure consisting of a set with two defined operations, $+$ and $\cdot$, where, given $x, y, z \in R$, $(x + y) + z = x + (y + z)$, $x + y = y + x$, $\exists 0 \in R$ s.t. $x + 0 = x$, $\forall x \in R \exists y \in R$ s.t. $x + y = 0$, $(x \cdot y) \cdot z = x \cdot (y \cdot z)$, $\exists 1 \in R$ s.t. $x \cdot 1 = 1 \cdot x = x$, $x \cdot (y + z) = x \cdot y + x \cdot z$, and $(x + y) \cdot z = x \cdot z + y \cdot z$.*

Using this definition, we can therefore define a module:

Definition 116 (Module). *A left R -module is an abelian group $(M, +)$ equipped with an operation analogous to scalar multiplication for vector spaces, $\cdot : R \times M \rightarrow M$, such that given $r, s \in R$ and $a, b \in M$, we have $r \cdot (a+b) = r \cdot a + r \cdot b$, $(r+s) \cdot a = r \cdot a + s \cdot a$, $(rs) \cdot a = r \cdot (sa)$, and $1 \cdot a = a$. A right R -module is defined analogously.*

How might a tensor product be defined of modules over a ring?

(d) We define an R -algebra as follows:

Definition 117 (R -Algebra). *An R -algebra is simply a vector space equipped with an inner product that follows the properties of distributivity and compatibility with scalars, though the set of scalars is not a field, but rather a ring.*

Show that the tensor product of two R -algebras A and B , $A \otimes_R B$, is an R -algebra itself.

Challenge 2: For this challenge, we explore the topic of symmetric tensors.

(a) Many tensors in physics tend to be symmetric. Make a qualitative argument for why this might be the case.

(b) How many independent components are in a symmetric tensor of order 3, which can be represented by a $3 \times 3 \times 3$ dimensional array of numbers? What about a tensor of order 4 that is represented by a $3 \times 3 \times 3 \times 3$ dimensional array of numbers? Show all calculations.

(c) Let $T = v_1 \otimes v_2 \otimes \dots \otimes v_n$. We define:

Definition 118 (Symmetric Product). *The symmetric product $\odot$ is defined as*

$$v_1 \odot v_2 \odot \dots \odot v_n = \frac{1}{n!} \sum_{\pi} v_{\pi(1)} \otimes v_{\pi(2)} \otimes \dots \otimes v_{\pi(n)} \quad (3.148)$$

Define $T_1 \odot T_2 = \text{Sym}(T_1 \otimes T_2)$, where Sym denotes the symmetrization of the tensor. Show that this product is commutative and associative.

(d) We know that for second-order tensors, the rank of the tensor corresponds to the rank of a matrix representing such a tensor. Does this hold true for higher-order tensors and its corresponding higher-order arrays of numbers? Justify your answer.

Challenge 3: Create a program simulating the metric tensor of a given three-dimensional space for the following metric tensors (let t, x, y be the three variables):

(a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

(d) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{5} \end{pmatrix}$

(e) $\begin{pmatrix} x & 0 & 0 \\ 0 & t & 1 \\ 0 & 1 & 1 \end{pmatrix}$

(f) $\begin{pmatrix} -1 & 0 & x \\ 0 & x^2 & y \\ x & y & y^2 \end{pmatrix}$

(g) Choose any two points in each of the 6 spaces you have constructed. Find the distance between them using the metric tensor.

(h) Would it be possible for the distance between two points to be zero, and the points are not the same? Justify your answer with several examples.

Challenge 4: We will now examine the exterior algebra.

(a) Prove that $x \wedge y = -y \wedge x$, given that $x \wedge x = 0$. Consider the expression $(x + y) \wedge (x + y)$.

(b) We consider the following definition:

Definition 119 (Decomposable). *A tensor T is said to be decomposable if it may be written as*

$$T = v_1 \wedge v_2 \wedge \dots \wedge v_n \quad (3.149)$$

where $v_1, \dots, v_n \in V$.

Consider $AT(V)$. Is every element of this space decomposable? Why or why not?

(c) Let us define $\mathcal{A}(T)$ as the antisymmetrization of T without the factor of $1/n!$. Find $\mathcal{A}(T)\mathcal{A}(T)$.

(d) Define in detail the relationship between the wedge product and the tensor product. How does this relate to the exterior algebra being a quotient product of the tensor algebra?

CHAPTER 4

Tensor Decompositions

4.1 Eigenvalues and Eigenvectors for Second-Order Tensors

In our exploration of linear algebra throughout this textbook, we have covered many different topics in introductory linear algebra. However, there is one topic that we have yet to cover, which we will present shortly. That is the concept of the eigenvalue. When we observe operators, we are particularly interested in those vectors where the operator applied to a vector returns a scalar multiple of that same vector. We call that scalar an “eigenvalue.” That vector is called an “eigenvector.” As it turns out, all the eigenvectors of a given eigenvalue form a vector space known as, one might guess, an “eigenspace.” Here are the definitions of an eigenvalue and an eigenvector:

Definition 120 (Eigenvalue, Eigenvector). *Let $T^\mu{}_\nu$ be a second-order tensor. An eigenvalue is some scalar $\lambda \in \mathbb{F}$ such that the application of a tensor to a given vector yields λ times such a vector. This vector is called an eigenvector.*

It follows shortly that when we are studying the eigenvalues of a second-order tensor, we want to find solutions λ to the following equation:

Remark 23. *When we are interested in studying the eigenvalues of a given tensor, we consider the following equation:*

$$T^\mu{}_\nu v^\nu = \lambda v^\mu \tag{4.1}$$

We are able to prove many elementary properties about eigenvectors. However, as this text does not concern itself primarily

with linear algebra, we will demonstrate the proof of one property, and leave the rest for the curious reader. We will show that we can represent repeated applications of the operator as the eigenvalue raised to a given power for some eigenvector, worded more precisely below:

Theorem 44. *If v^μ is an eigenvector of the tensor $T^\mu{}_\nu$, then we must have that*

$$T^{i_1}{}_{i_2} T^{i_2}{}_{i_3} \dots T^{i_n}{}_{i_{n+1}} v^{i_{n+1}} = \lambda^n v^{i_1} \quad (4.2)$$

Proof. We know that

$$T^\mu{}_\nu v^\nu = \lambda v^\mu \quad (4.3)$$

Consider the application of the tensor $T^\rho{}_\mu$ to the equation. We obtain

$$T^\rho{}_\mu T^\mu{}_\nu v^\nu = T^\rho{}_\mu \lambda v^\mu = \lambda T^\rho{}_\mu v^\mu = \lambda^2 v^\rho \quad (4.4)$$

Repeating this process, we will ultimately achieve our desired result. $\square$

Definition 121 (Characteristic Polynomial). *In order to find all eigenvalues λ of a given second-order tensor $T^\mu{}_\nu$, we utilize the characteristic polynomial:*

$$|T^\mu{}_\nu - \lambda \delta^\mu{}_\nu| = 0 \quad (4.5)$$

For those who are familiar with the rudiments of linear algebra, this equation should look very similar to the characteristic polynomial you were taught in linear algebra; that is, given an operator A and an identity matrix I , we solve for λ :

$$|A - \lambda I| = 0 \quad (4.6)$$

Definition 122 (Rotation). *A rotation tensor $R^i{}_j$ has the matrix representation as follows:*

$$R^i{}_j = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (4.7)$$

Consider the case when $\theta = \pi/2$. In this case, we are rotating by 90 degrees:

$$R^i_j = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (4.8)$$

To find the eigenvalues of this tensor, we first find $R^i_j - \lambda\delta^i_j$:

$$R^i_j - \lambda\delta^i_j = \begin{pmatrix} -\lambda & -1 & 0 \\ 1 & -\lambda & 0 \\ 0 & 0 & 1 - \lambda \end{pmatrix} \quad (4.9)$$

Therefore, the characteristic polynomial is

$$\begin{vmatrix} -\lambda & -1 & 0 \\ 1 & -\lambda & 0 \\ 0 & 0 & 1 - \lambda \end{vmatrix} = 0 \implies \lambda^2 - \lambda^3 + 1 - \lambda = 0 \quad (4.10)$$

$$\implies (1 - \lambda)(\lambda^2 + 1) = 0 \quad (4.11)$$

Which, it is plain to see, we have $\lambda = 1, \pm i$. We have discovered the three eigenvalues of this rotation tensor. Finding the eigenvectors will be left as an exercise to the reader.

Definition 123 (Shear). *A shear tensor, S^i_j , has the matrix representation as follows:*

$$S^i_j = \begin{pmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (4.12)$$

where γ is some scalar. This matrix can sometimes be used as the deformation gradient tensor, which we will learn about in the applications section of the book.

It is time to find the eigenvalues of this tensor. We first find $S^i_j - \lambda\delta^i_j$:

$$S^i_j - \lambda\delta^i_j = \begin{pmatrix} 1 - \lambda & \gamma & 0 \\ 0 & 1 - \lambda & 0 \\ 0 & 0 & 1 - \lambda \end{pmatrix} \quad (4.13)$$

Then, we use the characteristic polynomial:

$$\begin{vmatrix} 1 - \lambda & \gamma & 0 \\ 0 & 1 - \lambda & 0 \\ 0 & 0 & 1 - \lambda \end{vmatrix} = 0 \implies (1 - \lambda)^3 = 0 \quad (4.14)$$

Therefore, we see that the only eigenvalue that this matrix has is $\lambda = 1$. This does not mean that we have calculated incorrectly. Some tensors have fewer eigenvalues than the dimension of the vector space that the tensor is a tensor product of (in this case, we have that the dimension is 3).

4.2 Spectral Theorem

We approach one of the most foundational theorems in linear algebra: that of the spectral theorem. Before we observe the spectral theorem in full, a couple different concepts must be explained first. Among them is the definition of a Hermitian matrix, which has been explored briefly before, but for the reader's convenience, will be presented again in its full and complete definition.

Definition 124 (Hermitian Matrix). *A Hermitian matrix H is a matrix such that it is equal to its transpose conjugate; that is, $H = H^\dagger = (H^T)^*$.*

It follows that, for Hermitian matrices, we have

$$\langle v, Hw \rangle = \langle Hv, w \rangle \quad (4.15)$$

It will be left to the reader to prove this claim.

Now, we consider the concept of the spectrum. The spectrum is where the spectral theorem gets its name from, and hence pertains to the theorem. We define a spectrum as a set of eigenvalues of a given tensor; one can imagine this spectrum as being a “spectrum” of points among the real line or complex plane.

Definition 125 (Spectrum). *A spectrum S is a set of points λ_i , which are eigenvalues of a given tensor T .*

Now, we examine a lemma that will be used to assist in the proof of the spectral theorem. We assert that the characteristic equation of a tensor must give us at least one complex eigenvalue and a nonzero eigenvector for any tensor. The proof of this claim is beyond the focus of this book, but the curious reader may choose to delve further into this particular lemma.

Lemma 6. *If we examine the characteristic equation of a given tensor T^i_j , we find that there must be at least one complex eigenvalue λ and a nonzero eigenvector v^i .*

We now define the orthogonal complement, which is another concept instrumental to proving the spectral theorem. The orthogonal complement of a vector is simply the set of all vectors orthogonal to that vector. The orthogonal complement of two vectors is the set of all vectors orthogonal to both vectors, and so on. We generalize this definition like so:

Definition 126 (Orthogonal Complement). *The orthogonal complement $O = \text{span}(v_1^i, \dots, v_n^i)^\perp$ of a subspace $U = \text{span}(v_1^i, \dots, v_n^i)$ is the set of all vectors that is orthogonal to any of the vectors in the subspace $\text{span}(v_1^i, \dots, v_n^i)$; that is to say, given $o \in O$ and $u \in U$, we have that $\langle o, u \rangle = \langle u, o \rangle = 0$.*

We now observe the spectral theorem. The spectral theorem is one of the pivotal results in linear algebra, and it states that we can find an orthonormal basis of a vector space consisting of eigenvectors of a Hermitian tensor. Additionally, every eigenvalue of the Hermitian tensor is real. This statement is so profound because it allows us to guarantee real eigenvalues for certain tensors, as well as the understanding that the eigenvectors of the tensor span the vector space (this is not always the case for any tensor). The proof is rather straightforward, relying on an iterative process. We present the spectral theorem with proof below:

Theorem 45 (Spectral Theorem). *Consider the vector space V . If we have a Hermitian tensor H^i_j , then there must be an orthonormal basis of V consisting of eigenvectors of H^i_j . We can also conclude that every eigenvalue of H^i_j is real.*

Proof. Let $\mathbb{F} = \mathbb{C}$. By lemma 6, letting λ be that given eigenvalue and v^i be the corresponding eigenvector, we may write

$$\lambda \langle v^i, v^i \rangle = \langle H^i_j v^j, v^i \rangle \quad (4.16)$$

And, since H^i_j is Hermitian,

$$= \langle v^i, H^i_j v^j \rangle = \lambda^* \langle v^i, v^i \rangle \quad (4.17)$$

Therefore, it follows that $\lambda = \lambda^*$, which obviously entails that λ is real. Now we consider the orthogonal complement of v^i , $O = \text{span}(v^i)^\perp$. Let $o^i \in O$. Then $\langle o^i, v^i \rangle = 0$ by the definition of an orthogonal complement. Now, since

$$\langle H^i_j o^j, v^i \rangle = \langle o^i, H^i_j v^j \rangle \quad (4.18)$$

as H^i_j is Hermitian, and

$$\langle o^i, H^i_j v^j \rangle = \langle o^i, \lambda v^i \rangle = \lambda \langle o^i, v^i \rangle = 0 \quad (4.19)$$

it follows that

$$\langle H^i_j o^j, v^i \rangle = 0 \quad (4.20)$$

so $H^i_j o^j$ is orthogonal to v^i as well; therefore $H^i_j o^j \in O$. Now we consider the space O . Since H^i_j is an operator on O , there exists an eigenvalue λ' and an associated eigenvector $v'^i \in O$. We are able to continue creating orthogonal complements $O', O'', \dots$ in a repeated manner until we have completely spanned V . We therefore have an orthonormal basis of V consisting of eigenvectors of H^i_j , and every eigenvalue of the second-order tensor is real. $\square$

We observe some more specific results, regarding that of symmetric second order tensors. This theorem states that given an eigenvalue of a second order symmetric tensor, we have that its so called “algebraic multiplicity” is equal to that of its “geometric multiplicity.” The theorem statement below clarifies what is meant by these terms. The proof is slightly more complicated than that of the spectral theorem.

Theorem 46. *If we have an eigenvalue λ_0 of a symmetric second order tensor T , then the number of linearly independent eigenvectors related to that eigenvalue is equal to the number of times that the eigenvalue appears as a root of the characteristic polynomial.*

Proof. If we have a set of linearly independent eigenvectors e_i such that $i = 1, 2, \dots, n$, where n is the number of linearly independent eigenvectors related to λ_0 , then it follows that every linear combination $v = a^i e_i$ is an eigenvector associated with λ_0 . We have that

$$T \sum_{i=1}^n a^i e_i = \sum_{i=1}^n a^i T e_i = \sum_{i=1}^n a^i \lambda_0 e_i = \lambda_0 \sum_{i=1}^n a^i e_i \quad (4.21)$$

Let m be the dimension of the vector space. Then we have, by writing T as a tensor product of basis vectors and considering that we were able to find a basis such that the first $n \times n$ block in the matrix is λ_0 multiplied by the identity,

$$T = \sum_{i,j=1}^m T^{ij} e_i \otimes e_j = \lambda_0 \sum_{k=1}^n e_k \otimes e_k + \sum_{i,j=n+1}^m T^{ij} e_i \otimes e_j \quad (4.22)$$

It follows that, since T can be thought of as

$$T = \begin{pmatrix} \lambda_0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\ 0 & \lambda_0 & \dots & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_0 & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & T^{n+1,n+1} & T^{n+1,n+2} & \dots & T^{n+1,m} \\ 0 & 0 & \dots & 0 & T^{n+2,n+1} & T^{n+2,n+2} & \dots & T^{n+2,m} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & T^{m,n+1} & T^{m,n+2} & \dots & T^{m,m} \end{pmatrix}$$

$$= \begin{pmatrix} \lambda_0 \sum_{k=1}^n e_k \otimes e_k & \mathbf{0} \\ \mathbf{0} & \sum_{i,j=n+1}^m T^{ij} e_i \otimes e_j \end{pmatrix}$$

it follows that the determinant must be a product of the two outlined nontrivial matrices,

$$|T| = \left| \lambda_0 \sum_{k=1}^n e_k \otimes e_k \right| \left| \sum_{i,j=n+1}^m T^{ij} e_i \otimes e_j \right|$$

$$\implies |T - \lambda \delta^{ij}| = |T^{ij} - \lambda \delta^{ij}| (\lambda_0 - \lambda)^n \quad (4.23)$$

Therefore, we find that the number of times that λ_0 appears as a root in the characteristic polynomial must be greater than or equal to n . Now, since

$$T' = \sum_{i,j=n+1}^m T^{ij} e_i \otimes e_j \quad (4.24)$$

is a map from this space to itself, the eigenvectors must be different than that of e_k in Equation 4.22. Therefore, λ_0 cannot be an eigenvalue of T' , and it follows shortly that

$$|T^{ij} - \lambda \delta^{ij}| \quad (4.25)$$

does not have a root $\lambda = \lambda_0$. Therefore, we have that the number of linearly independent eigenvectors related to that eigenvalue is equal to the number of times that the eigenvalue appears as a root of the characteristic polynomial. $\square$

4.3 Principal Axes

We now concern ourselves with the idea of diagonalization. This idea is particularly important in linear algebra, and has its use with tensors as well. The idea of diagonalization is to find a basis in which a given matrix becomes a diagonal matrix. As it turns out, this can be accomplished when the eigenvectors span the vector space. It is left as an exercise to the reader to prove this claim.

It may be useful to acquaint ourselves with a different nomenclature as well. We define principal axes as the eigenvectors by which we are able to diagonalize the matrix, or tensor, for that matter, as follows:

Definition 127 (Principal Axes). *The principal axes of a given second-order tensor are the basis vectors by which the matrix corresponding to the tensor becomes a diagonal matrix; that is, they are the basis vectors where the components of the second-order tensor are zero when not along the diagonal.*

Similarly, we redefine the elements of the diagonal matrix, or the components of a tensor, as principal values.

Definition 128 (Principal Values). *The principal values of a second-order tensor are the values that the components take in the tensor's diagonal form.*

The idea of introducing different nomenclatures for this core concept stems from its applicability to a wide variety of areas. Diagonalization is particularly important in areas of quantum mechanics, in which one may often diagonalize matrices, or “Hamiltonians,” in order to solve relevant problems. Diagonalization is also known as such in the area of linear algebra, where, depending on the level of computation required, may be a more abstract endeavor. This is not to say that the principal axes do not have use. Anywhere tensors are found in engineering or physics, one considers the principal axes of a tensor. We will consider some examples later.

Now, we ought to ask the question: how can we find the principal axes of a tensor? The answer, by this point, should be obvious: it pertains to eigenvectors and eigenvalues. We make this statement more clear below.

Theorem 47. *The eigenvectors of a second-order tensor T are the principal axes of the tensor.*

Proof. Let $e_1, \dots, e_n$ be the eigenvectors of a tensor T^i_j . Then we have that

$$T^i_j e_j = \lambda_j e_j \quad (4.26)$$

where λ_j is the corresponding eigenvalue of e_j . Therefore, by

utilizing the linearity of T^i_j , we can construct a matrix as follows:

$$T^i_j v = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix} \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix} \quad (4.27)$$

where the basis is in $\{e_1, \dots, e_n\}$. We have found that the eigenvectors of a second-order tensor T are therefore the principal axes of the tensor. $\square$

We now consider an application of principal axes to that of physics and engineering. We recall the moment of inertia tensor I_{ij} . We make the following remark:

Remark 24. *The inertia tensor I_{ij} is a real and symmetric matrix. This is because*

$$I_{ij} = \int \rho(r) \left(\delta_j^i \sum_k x_k^2 - x_i x_j \right) dV \quad (4.28)$$

$$= \int \rho(r) \left(\delta_i^j \sum_k x_k^2 - x_j x_i \right) dV = I_{ji} \quad (4.29)$$

The inertia tensor, of course, has a principal axis system, alongside its principal values, which are commonly referred to as I_j , since the matrix is diagonal, and there exists no need for two indices anymore. In this case, we call these principal values the principal moments of inertia.

It is also a fact that if the angular velocity vector ω points alongside one of the principal axes like so:

$$\omega = v^1 e_1 \quad (4.30)$$

Then we have that the angular momentum L is parallel to the angular velocity ω , and we can discern the vector's magnitude by the following equation:

$$L_1 e_1 = I_1 \omega_1 e_1 \quad (4.31)$$

The idea of a principal axis system belonging to a given second-order tensor is a powerful one, because it allows us to simplify calculations greatly and reduces the number of indices that we must consider when considering a given tensor.

4.4 Invariants of Second-Order Tensors

When it comes to the study of tensors in linear algebra and their eigenvalues and eigenvectors, we are also concerned with those functions that remain invariant under similar tensors (i.e. a transformation of the type $T \rightarrow ATA^{-1}$, where A is invertible). We call these functions invariants.

Definition 129 (Invariant). *We consider the function $f : V \otimes V^* \rightarrow \mathbb{F}$. This function f is called an invariant of a second-order tensor T^i_j if for all invertible matrices A^i_j , we have that*

$$f(T) = f(ATA^{-1}) \quad (4.32)$$

Now we examine a similar topic in that of the principal invariants. These are the coefficients of the characteristic polynomial of the given tensor. One property of note is that principal invariants do not change under rotation.

Definition 130 (Principal Invariant). *Given a tensor T^i_j , we have that the principal invariants I_n of T are the coefficients of the characteristic polynomial.*

More specifically, for a tensor of three dimensions, we have that the characteristic polynomial can be written as

$$-\lambda^3 + I_1\lambda^2 - I_2\lambda + I_3 = 0 \quad (4.33)$$

where I_1, I_2, I_3 are the principal invariants.

One may note that we may also represent the principal invariants in the form of the eigenvalues of the second-order tensor.

Theorem 48. *Given eigenvalues $\lambda_1, \lambda_2, \lambda_3$ for a three dimensional second-order tensor, we have that*

$$I_1 = \lambda_1 + \lambda_2 + \lambda_3 \quad (4.34)$$

$$I_2 = \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3 \quad (4.35)$$

$$I_3 = \lambda_1\lambda_2\lambda_3 \quad (4.36)$$

The proof of this is a straightforward application of Vieta's formulas, which will be left to the reader as an exercise. Let us now calculate I_n using the coefficients of T^i_j . We observe the following:

$$|T^i_j - \lambda\delta^i_j| = \begin{vmatrix} T^1_1 - \lambda & T^1_2 & T^1_3 \\ T^2_1 & T^2_2 - \lambda & T^2_3 \\ T^3_1 & T^3_2 & T^3_3 - \lambda \end{vmatrix} \quad (4.37)$$

$$= (T^1_1 - \lambda)(T^2_2 - \lambda)(T^3_3 - \lambda) + T^1_2T^2_3T^3_1 + T^1_3T^2_1T^3_2 - T^1_3T^3_1(T^2_2 - \lambda) - T^2_3T^3_2(T^1_1 - \lambda) - T^1_2T^2_1(T^3_3 - \lambda) \quad (4.38)$$

Expanding the factored expressions, we obtain

$$\begin{aligned} &= -\lambda^3 + (T^1_1 + T^2_2 + T^3_3)\lambda^2 - (T^1_1T^2_2 + T^1_1T^3_3 + T^2_2T^3_3)\lambda \\ &\quad + T^1_1T^2_2T^3_3 + T^1_2T^2_3T^3_1 + T^1_3T^2_1T^3_2 \\ &\quad + (T^1_2T^2_1 + T^1_3T^3_1 + T^2_3T^3_2)\lambda \\ &\quad - T^1_1T^2_3T^3_2 - T^2_2T^1_3T^3_1 - T^3_3T^1_2T^2_1 \end{aligned} \quad (4.39)$$

Furthermore, we simplify by combining like terms, resulting in the following:

$$\begin{aligned} &= -\lambda^3 + (T^1_1 + T^2_2 + T^3_3)\lambda^2 - (T^1_1T^2_2 \\ &\quad + T^1_1T^3_3 + T^2_2T^3_3 - T^1_2T^2_1 - T^1_3T^3_1 - T^2_3T^3_2)\lambda \\ &\quad + T^1_1T^2_2T^3_3 + T^1_2T^2_3T^3_1 + T^1_3T^2_1T^3_2 \\ &\quad - T^1_1T^2_3T^3_2 - T^2_2T^1_3T^3_1 - T^3_3T^1_2T^2_1 \end{aligned} \quad (4.40)$$

Therefore, we conclude that

$$I_1 = T^1_1 + T^2_2 + T^3_3 \quad (4.41)$$

$$I_2 = T^1_1 T^2_2 + T^1_1 T^3_3 + T^2_2 T^3_3 - T^1_2 T^2_1 - T^1_3 T^3_1 - T^2_3 T^3_2 \quad (4.42)$$

$$I_3 = T^1_1 T^2_2 T^3_3 + T^1_2 T^2_3 T^3_1 + T^1_3 T^2_1 T^3_2 \\ - T^1_1 T^2_3 T^3_2 - T^2_2 T^1_3 T^3_1 - T^3_3 T^1_2 T^2_1 \quad (4.43)$$

Furthermore, through Theorem 48, we have that

$$T^1_1 + T^2_2 + T^3_3 = \lambda_1 + \lambda_2 + \lambda_3 \quad (4.44)$$

$$T^1_1 T^2_2 + T^1_1 T^3_3 + T^2_2 T^3_3 - T^1_2 T^2_1 - T^1_3 T^3_1 - T^2_3 T^3_2 \\ = \lambda_1 \lambda_2 + \lambda_1 \lambda_3 + \lambda_2 \lambda_3 \quad (4.45)$$

$$T^1_1 T^2_2 T^3_3 + T^1_2 T^2_3 T^3_1 + T^1_3 T^2_1 T^3_2 \\ - T^1_1 T^2_3 T^3_2 - T^2_2 T^1_3 T^3_1 - T^3_3 T^1_2 T^2_1 \\ = \lambda_1 \lambda_2 \lambda_3 \quad (4.46)$$

There is one more topic that should be discussed in this section: that of main invariants. These are scalars that can be constructed as sums of powers of eigenvalues.

Definition 131 (Main Invariants). *We define main invariants as the sum of powers of the eigenvalues; that is to say,*

$$J_1 = \lambda_1 + \lambda_2 + \lambda_3 \quad (4.47)$$

$$J_2 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2 \quad (4.48)$$

$$J_3 = \lambda_1^3 + \lambda_2^3 + \lambda_3^3 \quad (4.49)$$

4.5 Higher-Order Tensor Decompositions

At this point, we consider ways to decompose tensors, specifically of higher order than two. As it turns out, this topic is full of nuance and complexity. The reader will be introduced to some of the most pertinent tensor decompositions, as well as associated topics related to tensors.

It will be worth noting that this particular section will not be of much use to mathematicians, but rather may be much more applicable to computer scientists or any other profession which principally regards tensors not as a multilinear map, but simply as an n -dimensional array of numbers. Keep this in mind as we continue throughout the section.

Before we discuss tensor decompositions, we must introduce a couple of key concepts necessary to understand tensor decompositions. First is that of the mode. The reader is already familiar with the order of a tensor; the mode is very much related to the order. We define the mode as follows:

Definition 132 (Mode). *A tensor of order n is considered to have n modes, where mode- i corresponds to the i th index.*

As an example, a tensor of order 3 has 3 modes, mode-1, mode-2, and mode-3. These are specifically the dimensions of the tensor.

Additionally, we are also interested in types of “cross-sections” of a tensor which are created by fixing certain numbers of indices. There are two of these mathematical objects that are primarily used; first between them is that of the fiber.

Definition 133 (Fiber). *A fiber of a tensor is a vector that is constructed by fixing every index of the tensor but one to a constant. For a tensor of order 3, that means fixing 2 indices to a constant, and for a tensor of order 4, that means fixing 3 indices to a constant.*

It is easy to imagine why these mathematical objects are called fibers, if we imagine a 3-tensor as a cubic array of numbers. The second mathematical object of this variety is that of the slice:

Definition 134 (Slice). *A slice of a tensor is a tensor of order two that is constructed by fixing every index of the tensor but two to a constant. For a tensor of order 3, that means fixing 1 index to a constant, and for a tensor of order 4, that means fixing 2 indices to a constant.*

Now that we have covered some of the basic concepts and mathematical objects related to tensor decompositions, we introduce two fundamental operations of tensors that we have not covered before. The first is known as unfolding, and transforms a tensor into a matrix like so:

Definition 135 (Mode- n Unfolding). *The mode- n unfolding of a tensor T , denoted*

$$T_{(n)} \quad (4.50)$$

is the matrix obtained by taking every mode- n fiber and stacking them as columns.

It is easy to verify that there are n different unfoldings of an n -tensor.

The second operation of relevance is that of mode- n multiplication, which is slightly more complicated. It is defined as follows:

Definition 136 (Mode- n Multiplication). *If T is a tensor with dimensions $i_1 \times \dots \times i_m \times \dots \times i_n$, and A is a matrix with dimensions $j \times i_m$, then the mode- n product $T \times_n A$ is the tensor whose mode- n fibers are multiplied by A . The result is a tensor of size*

$$i_1 \times \dots \times i_{m-1} \times j \times i_{m+1} \times \dots \times i_n \quad (4.51)$$

This operation can be essentially thought of as a replacement of the mode- n coordinate system with a new basis given by the rows of A .

Before we discuss tensor decompositions, we must discuss the primary method of matrix decomposition, which is known as singular value decomposition. This takes a matrix and reduces it into a product of simpler matrices, specifically two rotation matrices and a diagonal matrix. This decomposition is notable for its simplicity and its applicability to all matrices.

Definition 137 (Singular Value Decomposition). *Singular Value Decomposition, or SVD for short, is a method of decomposing a matrix through factorizing a matrix M into two rotation matrices U and $V^\dagger$ and a diagonal matrix Σ in between the two, like so:*

$$M = U \Sigma V^\dagger \quad (4.52)$$

Our goal is to generalize this idea to higher-order tensors. However, unlike matrices, tensors admit multiple non-equivalent analogues of SVD, each emphasizing different structure.

We are already familiar with the rank of a tensor. We understand that a rank-1 tensor can be expressed as the tensor product of vectors and dual vectors

$$T_{j_1 \dots j_n}^{i_1 \dots i_m} = v^{i_1} \otimes \dots \otimes v^{i_m} \otimes \varphi_{j_1} \otimes \dots \otimes \varphi_{j_n} \quad (4.53)$$

We must keep this in mind as we discuss our first method of tensor decomposition, the canonical polyadic decomposition, which takes a tensor and reduces it to a linear combination of rank-1 tensors.

Definition 138 (Canonical Polyadic Decomposition). *Canonical polyadic decomposition expresses a tensor as a linear combination of rank-1 terms. For example, for a tensor of order three, we have*

$$T = \sum_{i=1}^R \lambda_i a_i \otimes b_i \otimes c_i \quad (4.54)$$

The smallest R possible is known as the CP rank.

Canonical polyadic decomposition generalizes the idea of the matrix rank to tensors. The decomposition of a tensor is not unique. However, the main drawback to this decomposition method is that computing CP rank is NP-hard; in other words, that means that it takes a lot more time to calculate as complexity increases.

The second notable method of tensor decomposition is known as Tucker decomposition, named after American mathematician Ledyard Tucker. This decomposition aims to be a higher-order analogue of singular value decomposition, as we see below.

Definition 139 (Tucker Decomposition). *A tensor T of dimension $i_1 \times \dots \times i_n$ admits a Tucker decomposition of the form*

$$T = \mathcal{G} \times_1 U^{(1)} \times_2 \dots \times_n U^{(n)} \quad (4.55)$$

where $U^{(p)}$ is a matrix with dimension $i_p \times r_p$, and $\mathcal{G}$ is called the core tensor, with dimension $r_1 \times \dots \times r_n$.

This is closely analogous to SVD, where $\mathcal{G}$ is analogous to Σ , and $U^{(p)}$ is analogous to $U, V^\dagger$. The factor matrices give a separate subspace/basis for each mode. Additionally, the core tensor encodes the interactions between these mode-wise coordinates.

These are the two primary methods of tensor decomposition. They may not be of great interest to pure mathematicians, but they are crucial to the applied mathematician or scientist's understanding of tensors.

4.6 Exercises

Exercise 1: Find the eigenvectors of the rotation matrix

$$R^i_j = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (4.56)$$

This exercise is best for those who have not taken linear algebra yet, as this would be routine for the student who has already taken linear algebra.

Exercise 2: Find the eigenvectors of the shear matrices

$$S^i_j = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (4.57)$$

and

$$S^i_j = \begin{pmatrix} 1 & -5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (4.58)$$

Can you find a rule for the eigenvectors of any shear matrix?

Exercise 3: Is it true that all the eigenvectors of a given second-order tensor span the relevant vector space? Why or why not?

(You will need to examine the wording of this statement very closely.)

Exercise 4: Find the eigenspace of the rotation matrix examined in Section 4.1, for each eigenvalue: $\lambda = 1$, $\lambda = i$, and $\lambda = -i$.

Exercise 5: Prove that

$$\langle Hv, w \rangle = \langle v, Hw \rangle \quad (4.59)$$

for a Hermitian matrix H and two vectors $v, w \in V$.

Exercise 6: We define a normal operator N as an operator that commutes with itself. Let $W \subset V$ be a subspace of V such that $w \in W \implies Nw \in W$. Prove that $W^\perp$ is also invariant under N ; that is, $u \in W^\perp \implies Nu \in W^\perp$.

Exercise 7: Based on Theorem 46, find the spectral decomposition of a symmetric second order tensor T based on its eigenvalues and orthonormal eigenvectors.

Exercise 8: The Cayley-Hamilton theorem states that $p_A(A) = 0$, where A is an operator and p_A is its characteristic polynomial. Prove this theorem.

Exercise 9: Diagonalize the following matrix:

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{pmatrix} \quad (4.60)$$

Exercise 10: Verify that one can diagonalize a matrix if and only if the eigenvectors span the vector space.

Exercise 11: Does any symmetric matrix have a principal axis system? How about an anti-symmetric matrix? Prove your claim.

Exercise 12: For those who are acquainted with multivariable calculus, find the moment of inertia tensor for a solid sphere with radius R and mass M .

Exercise 13: Prove Theorem 48.

Exercise 14: Find J_1 , J_2 , and J_3 in terms of T^i_j .

Exercise 15: Show that principal invariants do not change under rotation of coordinates.

Exercise 16: How would you find the invariants of higher-order tensors? (That is, tensors of order three or more.)

Exercise 17: For a 3-tensor of dimensions $i \times j \times k$, what will be the dimensions of the matrix induced by mode-1 unfolding? What about mode-2 and mode-3 unfolding?

Exercise 18: Show that

$$(T \times_n A)_{(n)} = AT_{(n)} \quad (4.61)$$

Exercise 19: Find the SVD of the following matrix:

$$M = \begin{pmatrix} 2 & 3 \\ 1 & 0 \end{pmatrix} \quad (4.62)$$

Exercise 20: Explain what happens if one or more of the $U^{(p)}$

matrices is the identity matrix. How does the formula for Tucker decomposition simplify?

Challenge 1: If we consider a symmetric tensor $T_{i_1 \dots i_d}$ of dimension d , consisting of an array of numbers specified to be $n \times n \times \dots \times n$, the following theorem applies.

Theorem 49. *The number of complex eigenvectors of $T_{i_1 \dots i_d}$ is*

$$\frac{(d-1)^n - 1}{d-2} = \sum_{i=0}^{n-1} (d-1)^i \quad (4.63)$$

(a) Prove this theorem.

(b) Is it possible to identify a symmetric tensor defined above with the space of homogeneous polynomials of degree d and in n variables? How so? Consider the relation of quadratic forms to bilinear forms.

(c) Describe the dimension and characteristics of the tensors corresponding to the following polynomials:

$$f(x, y, z) = xyz(x + y + z) \quad (4.64)$$

$$f(x, y) = x^2 + 2xy + y^2 \quad (4.65)$$

$$f(x, y, z, w) = xyzw + x^4 + x^5yz + w^3 \quad (4.66)$$

$$f(x_1, x_2, \dots, x_n) = (x_1 + x_2 + \dots + x_n)^p \quad (4.67)$$

(d) Using the cubic function $f(x, y, z) = x^3 + y^3 + z^3$, determine the number of eigenvectors that a $3 \times 3 \times 3$ tensor can have.

Challenge 2: The Faddeev-LeVerrier algorithm is defined as follows:

Definition 140 (Faddeev-LeVerrier Algorithm). *The Faddeev-LeVerrier algorithm is an algorithm used for calculating the coefficients c_i of the characteristic polynomial $|\lambda \delta_j^i - A_j^i|$. It is defined recursively:*

$$M_0 = 0, \quad c_n = 1 \quad (4.68)$$

$$M_k = AM_{k-1} + c_{n-k+1}I, \quad c_{n-k} = -\frac{1}{k} \text{tr}(AM_k) \quad (4.69)$$

- (a) Rederive this algorithm.
- (b) Can you construct another algorithm that calculates the coefficients of the characteristic polynomial?

Challenge 3: Write a paper on potential novel applications of tensors to a field of your interest, particularly concerning the use of modes, fibers, and slices.

Challenge 4: Construct a novel algorithm for higher-order tensor decomposition, either CP, Tucker, or a different kind. How does it compare to the current standard?

CHAPTER 5

Coordinate Systems and Change of Basis

5.1 Contravariant and Covariant Components

Now that we have developed much of the foundations for understanding tensors, it is time to switch gears into a more geometric point of view from our algebraic fundamentals. Even so, much of the content may be familiar to the reader, but it will be written again here for the sake of clarity before the coming chapters on differential geometry. We begin with a simple definition of the position vector:

Definition 141 (Position Vector). *The position vector is some vector r that represents the position of a given point in $\mathbb{R}^n$, relative to another point called the origin, which is labeled O .*

Every time we have represented a point in Cartesian coordinates by the ordered pair (x, y) , or perhaps the ordered triplet (x, y, z) , we have been describing a point using the position vector. This definition is simply a more general equivalent to the aforementioned examples.

Recall the definitions of contravariance and covariance. Some objects are contravariant; that is, they transform in a certain way, and some objects are covariant; that is, they transform in the opposite way. An example of an object that transforms contravariantly is the vector. It transforms against the basis. That is to say, if the basis rotates by a quarter of a full turn, from the basis' "point of view," it will be as if the components of the vector are rotating by a quarter of a full turn the other

way. It is in this sense that we mean that a vector transforms contravariantly.

Let us consider an example of an object that transforms covariantly: the gradient vector of a function ∇f . If the coordinate basis is scaled by a factor n , so too is the gradient.

We have seen much of this content before. Now, let us define a coordinate system in Euclidean space, so we may begin to see the geometric properties of vectors, bases, and the metric.

Definition 142 (Coordinate System). *A coordinate system, for now, is a bijective mapping $x^i = x^i(r)$ that takes a vector in $\mathbb{R}^n$ and assigns to it n numbers in $\mathbb{R}$, each one called coordinates of the vector.*

We will see later on in the book how to define a coordinate system on a manifold, but for now, it is perfectly fine to only consider coordinate systems in $\mathbb{R}^n$, as that is where we will be working for the majority of the chapter. There also exist tangent vectors in $\mathbb{R}^n$, which we define below:

Definition 143 (Tangent Vector). *A vector given by*

$$e_i = \frac{\partial r}{\partial x^i} \quad (5.1)$$

is called a tangent vector.

These vectors are called tangent vectors because, when we apply the partial derivative operation in a certain way, we are able to create a tangent line, plane, or space of some kind. Note that we also label tangent vectors as e_i . This is because they constitute a basis of $\mathbb{R}^n$, as we will see below.

Theorem 50. *Tangent vectors are linearly independent and span $\mathbb{R}^n$.*

Proof. We begin with the claim that tangent vectors are linearly independent. We show this by taking $r = x_j$. We have that

$$\left(\frac{\partial}{\partial x^i} \right) (x^j) = \delta_i^j \quad (5.2)$$

That is to say, the partial derivative of x^j with respect to x^i is equal to 1 if $i = j$, and 0 if $i \neq j$. Furthermore, let us show that we can construct $\mathbb{R}^n$ from all e_i . We can think of the tangent vector as the n -tuple

$$\frac{\partial r}{\partial x^i} \sim (0, \dots, 0, 1, 0, \dots, 0) \quad (5.3)$$

where 1 is in the i th position. We are able to construct a linear combination of tangent vectors

$$v = a^1 \frac{\partial r}{\partial x^1} + \dots + a^n \frac{\partial r}{\partial x^n} \sim (a^1, \dots, a^n) \quad (5.4)$$

which we see is equivalent to the ordered n -tuple on the right-hand side, where $a^1, \dots, a^n \in \mathbb{R}$. Furthermore, we see that the set of all n -tuples of the above form describes $\mathbb{R}^n$. Therefore, as the set of tangent vectors span $\mathbb{R}^n$ and are linearly independent, they constitute a basis of $\mathbb{R}^n$. $\square$

We have seen that the covariant basis e_i is recovered from the position vector by differentiation with respect to each of its coordinates:

$$e_i = \frac{\partial r}{\partial x^i} \quad (5.5)$$

Now we will see some basic examples of coordinate transformations, before we delve deeper into transformation rules in the second section of the chapter.

Let us consider a coordinate system $x^i = x^i(r)$ and another coordinate system $\bar{x}^i = \bar{x}^i(r)$. These describe $\mathbb{R}^n$, yet assign different coordinates to each point. Let us observe the covariant law of transformation:

Theorem 51. *Given two bases for coordinate systems e_i and $\bar{e}_i$, we have that these two are related by the following equation:*

$$\bar{e}_i = e_j \frac{\partial x^j}{\partial \bar{x}^i} \quad (5.6)$$

Proof. Consider $\bar{e}_i$. We have that

$$\bar{e}_i = \frac{\partial r}{\partial \bar{x}^i} = \frac{\partial r}{\partial x^j} \frac{\partial x^j}{\partial \bar{x}^i} \quad (5.7)$$

by the multivariable chain rule (do not forget the Einstein summation convention here, although it is tricky to spot). By the definition of a tangent vector, we have that this is equal to

$$e_j \frac{\partial x^j}{\partial \bar{x}^i} \quad (5.8)$$

giving us the desired result. $\square$

Similarly, there also exists a contravariant law of transformation:

$$\bar{\varphi}^i = \varphi^j \frac{\partial \bar{x}^i}{\partial x^j} \quad (5.9)$$

It will be left to the reader as an exercise to verify this law.

Now, let us understand vector and tensor transformations. We have seen some of this before, so hopefully this should not be completely new to you. We consider the vector $v = v_i \varphi^i = \bar{v}^i e_i$ and the second-order tensor $T = T^{ij} e_i \otimes e_j = T_{ij} \varphi^i \otimes \varphi^j = \dots$. We can apply these transformation laws as well. We are able to do so below:

$$v^i = v \cdot \varphi^i = v \cdot \bar{\varphi}^j \frac{\partial x^i}{\partial \bar{x}^j} = (v \cdot \bar{\varphi}^j) \frac{\partial x^i}{\partial \bar{x}^j} = \bar{v}^j \frac{\partial x^i}{\partial \bar{x}^j} \quad (5.10)$$

Therefore, we have that

$$v^i = \bar{v}^j \frac{\partial x^i}{\partial \bar{x}^j} \quad (5.11)$$

We have seen this before as

$$v^i = v^{j'} \Lambda_{j'}^i \quad (5.12)$$

Similarly, we have

$$v_i = v \cdot e_i = \bar{v}_j \frac{\partial \bar{x}^j}{\partial x^i} \quad (5.13)$$

according to the covariant transformation law. Tensors of order two transform in the expected manner:

$$T^{ij} = \varphi^i T \varphi^j \quad (5.14)$$

$$= \frac{\partial x^i}{\partial \bar{x}^k} \frac{\partial x^j}{\partial \bar{x}^l} \bar{T}^{kl} \quad (5.15)$$

Tensors of order three and higher transform similarly. The derivation of the transformation laws for tensors of order three will be left as an exercise for the reader.

5.2 Transformation Rules

In this section of the chapter we shall provide a more detailed explanation of tensor transformation rules, as well as a particularly important example in the identity tensor and its transformation. Tensor transformations can be easily found simply by following certain reasonable rules concerning indices, but we have been able to derive some of the transformation rules before (and hopefully by this point you have worked on some exercises similarly deriving more). We will continue our excursion into tensor transformations.

We consider the vector. The vector itself, being an invariant object, has no indices, yet the components of the vector have indices, like so:

$$v = v^i e_i = v_i \varphi^i \quad (5.16)$$

Recall that the Einstein summation convention applies here. Similarly, the tensor itself being invariant has no indices, yet its components may have any number of indices:

$$T = T^{ij} e_i \otimes e_j = T_{ij} \varphi^i \otimes \varphi^j = T^i{}_j e_i \otimes \varphi^j = T_i{}^j \varphi^i \otimes e_j \quad (5.17)$$

Notice that for every contravariant index in the component of the tensor, there exists a basis vector in the tensor product, whereas for every covariant index in the component of the tensor, there exists a dual basis vector in the tensor product. When working with tensors, we commonly consider just the components, just as when we work with vectors, we usually work with the components of the vector instead of the abstract invariant object, since we want the numbers to transform in the appropriate ways.

Before we move onto more on transformations, we briefly review the metric tensor and the inverse metric. The metric

tensor can be defined as

$$g_{ij} = \langle e_i, e_j \rangle \quad (5.18)$$

and the inverse metric can be defined as

$$g^{ij} = \langle \varphi^i, \varphi^j \rangle \quad (5.19)$$

The metric tensor can be used to “lower” the index of a given tensor; that is, we can transform one of the indices of a tensor from contravariant to covariant, whereas the inverse metric does precisely the opposite.

Why is this the case? Let us derive this phenomenon. We understand that

$$v^i = \langle v, \varphi^i \rangle \quad (5.20)$$

This is because

$$\langle v, \varphi^i \rangle = \langle v^j e_j, \varphi^i \rangle = v^j \langle e_j, \varphi^i \rangle = v^j \delta_j^i = v^i(1) = v^i \quad (5.21)$$

Then, we are able to say that

$$\langle v, \varphi^i \rangle = \langle v, g^{ij} e_j \rangle = \langle v_j \varphi^j, g^{ij} e_j \rangle \quad (5.22)$$

$$= v_j g^{ij} \langle \varphi^j, e_j \rangle = v_j g^{ij} \delta_j^j = v_j g^{ij} \quad (5.23)$$

In this way, we have demonstrated another way to prove that the metric tensor lowers the index of a vector. We can utilize a similar proof to show how the metric transforms a covariant index into a contravariant index:

$$v_i = \langle v, e_i \rangle = \langle v, g_{ij} \varphi^j \rangle \quad (5.24)$$

$$= \langle v^j e_j, g_{ij} \varphi^j \rangle = v^j g_{ij} \langle e_j, \varphi^j \rangle = v^j g_{ij} \quad (5.25)$$

To examine how the metric tensor and the inverse metric affects tensors, we need to understand one simple relation:

$$T^{ij} = \varphi^i T \varphi^j \quad (5.26)$$

as well as other, analogous relations for tensors of various types. The derivation of this specific identity will be left as an exercise

to the reader. Moving on, using the properties of the metric on basis vectors,

$$T^{ij} = \varphi^i T(g^{jk} e_k) \quad (5.27)$$

$$= (g^{il} e_l) T(g^{jk} e_k) \quad (5.28)$$

$$= T^i{}_k g^{kj} \quad (5.29)$$

$$= g^{il} T_{lk} g^{kj} \quad (5.30)$$

Therefore, the formula for taking a tensor with two covariant indices and transforming it into two contravariant indices using the metric tensor and its inverse can be easily found above. The reader should be able to generalize the general concept of raising and lowering indices to higher dimensions.

We consider a special case of tensor transformation now. We define the identity tensor as follows:

Definition 144. *The identity tensor $I^i{}_j$ is defined as the tensor such that*

$$I^i{}_j x^j = x^i \quad (5.31)$$

In fact, this is another name for a type of tensor that we have seen before. Do you know what tensor this identity tensor is equivalent to?

Interestingly, when we raise or lower the indices of the identity tensor, we recover the inverse metric and the metric tensor, respectively. We derive this claim as follows:

$$I^{ij} = \varphi^i I \varphi^j = \langle \varphi^i, \varphi^j \rangle = g^{ij} \quad (5.32)$$

Similarly,

$$I_{ij} = e_i I e_j = \langle e_i, e_j \rangle = g_{ij} \quad (5.33)$$

In this way, we are able to represent the identity tensor I as

$$I = g_{ij} \varphi^i \otimes \varphi^j = g^{ij} e_i \otimes e_j \quad (5.34)$$

If the basis is orthonormal, which means that $\langle e_i, e_j \rangle = \delta_j^i$, then

$$I = \sum_{i=1}^n e_i \otimes e_i \quad (5.35)$$

It will be left to the reader to verify this statement.

We have seen a manner of different transformations in the past two sections. We have seen how tensors and vectors react to coordinate transformations according to the contravariant and covariant law, as well as combinations of the two. We have also seen how the metric tensor is able to raise and lower indices on a vector or tensor. Now, we will examine some examples of coordinate transformations and change of bases in Euclidean space.

5.3 Examples in Euclidean $\mathbb{R}^n$

To supplement our understanding of coordinate systems and how they transform according to a change of basis, we will see below several simple examples that will serve to enhance the reader's conception of how we are able to mathematically analyze the properties of coordinate transformations. We will examine these coordinate transformations not yet on any manifold; that will be reserved for the second part of this book and beyond. Instead, we will be working in $\mathbb{R}^n$, or n -dimensional Euclidean space. For these particular examples, we will be looking at $\mathbb{R}^2$ and $\mathbb{R}^3$.

We are familiar with linear coordinates; that is, the coordinates $(x, y, z) \in \mathbb{R}^3$ such that the basis $\{e_1, e_2, e_3\}$ is orthonormal, positively oriented, and unchanging. That is to say, e_1 runs along the x axis, e_2 along the y axis, and e_3 along the z axis. Let us consider, for a moment, two-dimensional linear coordinates, $(x, y) \in \mathbb{R}^2$, and its transformation to polar coordinates. Polar coordinates, instead of considering x and y as if the Cartesian plane were to be subdivided into squares of length 1, considers r and θ , where r is the radius of the point from the origin, and θ is the angle made between the x axis and the line containing (r, θ) , passing through the origin. This angle is taken counterclockwise, from the x axis to the line the point sits on. It is easy to see that

$$r = \sqrt{x^2 + y^2} \tag{5.36}$$

and

$$\theta = \arctan \frac{y}{x} \quad (5.37)$$

This is a simple exercise in trigonometry, which the reader may wish to perform. Meanwhile, the equations converting polar to linear coordinates are

$$x = r \cos \theta \quad (5.38)$$

and

$$y = r \sin \theta \quad (5.39)$$

We begin by taking the derivatives of x and y with respect to r and θ . For x ,

$$\frac{\partial x}{\partial r} = \cos \theta \quad (5.40)$$

and

$$\frac{\partial x}{\partial \theta} = -r \sin \theta \quad (5.41)$$

For y ,

$$\frac{\partial y}{\partial r} = \sin \theta \quad (5.42)$$

and

$$\frac{\partial y}{\partial \theta} = r \cos \theta \quad (5.43)$$

We can construct a matrix from these partial derivatives called the Jacobian matrix, which we have seen before in the book. It consists of the matrix

$$\begin{pmatrix} \frac{\partial x^i}{\partial r} & \frac{\partial x^i}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = \begin{pmatrix} \frac{x}{r} & -\frac{y}{r} \\ \frac{y}{r} & x \end{pmatrix} \quad (5.44)$$

By taking the inverse of this matrix, we will find the reciprocal derivatives:

$$\begin{pmatrix} \frac{\partial x^i}{\partial r} & \frac{\partial x^i}{\partial \theta} \end{pmatrix}^{-1} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\frac{\sin \theta}{r} & \frac{\cos \theta}{r} \end{pmatrix} = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix} \quad (5.45)$$

By performing this exercise, we have found the transformation matrix of basis vectors from the linear or Cartesian coordinate system to the polar coordinate system, and back again, in $\mathbb{R}^2$.

Now, let us perform this exercise in $\mathbb{R}^3$, using cylindrical coordinates. We define cylindrical coordinates using three variables:

1. r , the radius of the projection of the point onto the xy -plane,
2. θ , the angle with respect to the x axis that the projection of the point onto the xy -plane makes, as in polar coordinates,
3. z , equivalent to the z coordinate in Cartesian coordinates.

We can derive three equations converting linear coordinates to cylindrical coordinates as follows:

$$r = \sqrt{x^2 + y^2} \quad (5.46)$$

$$\theta = \arctan \frac{y}{x} \quad (5.47)$$

$$z = z \quad (5.48)$$

Similarly, we can derive three equations converting cylindrical back to linear:

$$x = r \cos \theta \quad (5.49)$$

$$y = r \sin \theta \quad (5.50)$$

$$z = z \quad (5.51)$$

Therefore, we can construct the Jacobian matrix as follows:

$$J = \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (5.52)$$

We can easily find the inverse of this matrix, which is presented as follows:

$$J^{-1} = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} & 0 \\ -\frac{y}{r^2} & \frac{x}{r^2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (5.53)$$

The observant reader will notice something odd, however. If you try to apply the Jacobian matrix to a vector, it will not give the correct components for the transformed vector. Why is that? The answer will be left as an exercise for the reader to think about, to ensure that the reader understands these coordinate transformations on a deep level.

Let us consider the rotation of the xy -plane counterclockwise by $\pi/2$. We can represent two coordinate systems, (x, y) and (x', y') using these equations:

$$x' = y \quad (5.54)$$

$$y' = -x \quad (5.55)$$

The Jacobian is therefore

$$\begin{pmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (5.56)$$

And the inverse Jacobian is

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (5.57)$$

Interestingly, we can use the Jacobian matrix to calculate integrals with regard to a change of variables. The proof to this theorem can be found in any multivariable calculus book:

Theorem 52. *Consider two regions R in the xy -plane and S in the uv -plane such that $x = g(u, v)$ and $y = h(u, v)$, and g, h are bijective. If f is continuous on R , then, when the theorem is applicable, we have the following:*

$$\iint_R f(x, y) dx dy = \iint_S f(g(u, v), h(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv \quad (5.58)$$

where $\left| \frac{\partial(x, y)}{\partial(u, v)} \right|$ is the Jacobian matrix

$$\left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \quad (5.59)$$

5.4 Curvilinear Coordinates

In particular simplistic applications, we prefer to use linear coordinates. However, there exist a second type of coordinates

that may often be used called curvilinear coordinates. These are coordinate systems where the coordinate lines may not be completely straight. We have seen cylindrical coordinates in the previous section; as these contain curved coordinate lines, this coordinate system is a curvilinear coordinate system. Similarly, we will see spherical coordinates later on in the section; these are also curvilinear.

Our primary goal in this section will be to see how certain operations transform from linear to curvilinear coordinates; in this case, we will be considering the gradient, divergence, and curl. We begin by expressing the basis vectors for cylindrical coordinates e_r, e_θ, e_z in terms of the Cartesian basis vectors e_x, e_y, e_z :

$$e_r = \cos(\theta)e_x + \sin(\theta)e_y \quad (5.60)$$

$$e_\theta = -\sin(\theta)e_x + \cos(\theta)e_y \quad (5.61)$$

$$e_z = e_z \quad (5.62)$$

Inverting our formulas, we obtain

$$e_x = \cos(\theta)e_r - \sin(\theta)e_\theta \quad (5.63)$$

$$e_y = \sin(\theta)e_r + \cos(\theta)e_\theta \quad (5.64)$$

And, of course,

$$e_z = e_z \quad (5.65)$$

We recall the formula for the gradient:

$$\nabla f(x, y, z) = \frac{\partial f}{\partial x}e_x + \frac{\partial f}{\partial y}e_y + \frac{\partial f}{\partial z}e_z \quad (5.66)$$

Through the chain rule, we expand this expression:

$$\begin{aligned} &= \left(\frac{\partial f}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial x} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial x} \right) e_x \\ &+ \left(\frac{\partial f}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial y} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial y} \right) e_y \\ &+ \left(\frac{\partial f}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial z} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial z} \right) e_z \end{aligned} \quad (5.67)$$

Recalling the Jacobian of the linear-to-cylindrical coordinate transformation, we simplify:

$$\begin{aligned}
 &= \left(\frac{\partial f}{\partial r} \cos \theta - \frac{\partial f}{\partial \theta} \frac{\sin \theta}{r} \right) e_x \\
 &+ \left(\frac{\partial f}{\partial r} \sin \theta + \frac{\partial f}{\partial \theta} \frac{\cos \theta}{r} \right) e_y + \frac{\partial f}{\partial z} e_z \quad (5.68)
 \end{aligned}$$

However, recall the equations for e_x, e_y, e_z in e_r, e_θ, e_z . We write:

$$\begin{aligned}
 &= \left(\frac{\partial f}{\partial r} \cos \theta - \frac{\partial f}{\partial \theta} \frac{\sin \theta}{r} \right) (\cos(\theta)e_r - \sin(\theta)e_\theta) \\
 &+ \left(\frac{\partial f}{\partial r} \sin \theta + \frac{\partial f}{\partial \theta} \frac{\cos \theta}{r} \right) (\sin(\theta)e_r + \cos(\theta)e_\theta) + \frac{\partial f}{\partial z} e_z \quad (5.69)
 \end{aligned}$$

Simplifying, we obtain

$$\begin{aligned}
 &\frac{\partial f}{\partial r} \cos^2(\theta)e_r + \frac{\partial f}{\partial \theta} \frac{\sin^2 \theta}{r} e_\theta + \frac{\partial f}{\partial r} \sin^2(\theta)e_r \\
 &+ \frac{\partial f}{\partial \theta} \frac{\cos^2 \theta}{r} e_\theta + \frac{\partial f}{\partial z} e_z \quad (5.70)
 \end{aligned}$$

$$= \frac{\partial f}{\partial r} e_r + \frac{\partial f}{\partial \theta} \frac{1}{r} e_\theta + \frac{\partial f}{\partial z} e_z \quad (5.71)$$

We have thus derived the formula for the gradient in cylindrical coordinates. Now, we consider the divergence:

$$\nabla \cdot f = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z} \quad (5.72)$$

But $f_x = f_r \cos \theta - f_\theta \sin \theta$ and $f_y = f_r \sin \theta + f_\theta \cos \theta$, as we recall from the way we express Cartesian basis vectors in terms of cylindrical basis vectors. Therefore, we have

$$\begin{aligned}
 &= \frac{\partial(f_r \cos \theta - f_\theta \sin \theta)}{\partial x} + \frac{\partial(f_r \sin \theta + f_\theta \cos \theta)}{\partial y} + \frac{\partial f_z}{\partial z} \quad (5.73) \\
 &= \frac{\partial(f_r \cos \theta - f_\theta \sin \theta)}{\partial r} \cos \theta - \frac{\partial(f_r \cos \theta - f_\theta \sin \theta)}{\partial \theta} \frac{\sin \theta}{r}
 \end{aligned}$$

$$\begin{aligned} & \frac{\partial(f_r \sin \theta + f_\theta \cos \theta)}{\partial r} \sin \theta + \frac{\partial(f_r \sin \theta + f_\theta \cos \theta)}{\partial \theta} \frac{\cos \theta}{r} \\ & \quad + \frac{\partial f_z}{\partial z} \end{aligned} \quad (5.74)$$

$$\begin{aligned} &= \frac{\partial f_r}{\partial r} \cos^2 \theta - \frac{\partial f_\theta}{\partial r} \sin \theta \cos \theta + \frac{\partial f_r}{\partial r} \sin^2 \theta + \frac{\partial f_\theta}{\partial r} \sin \theta \cos \theta \\ & \quad - \frac{\partial f_r}{\partial \theta} \frac{\sin \theta \cos \theta}{r} + f_r \frac{\sin^2 \theta}{r} + \frac{\partial f_\theta}{\partial \theta} \frac{\sin^2 \theta}{r} + f_\theta \frac{\sin \theta \cos \theta}{r} \\ & \quad + \frac{\partial f_r}{\partial \theta} \frac{\sin \theta \cos \theta}{r} + f_r \frac{\cos^2 \theta}{r} + \frac{\partial f_\theta}{\partial \theta} \frac{\cos^2 \theta}{r} \\ & \quad - f_\theta \frac{\sin \theta \cos \theta}{r} + \frac{\partial f_z}{\partial z} \end{aligned} \quad (5.75)$$

$$= \frac{\partial f_r}{\partial r} + \frac{f_r}{r} + \frac{\partial f_\theta}{\partial \theta} \frac{1}{r} + \frac{\partial f_z}{\partial z} \quad (5.76)$$

$$= \frac{\partial(r f_r)}{\partial r} \frac{1}{r} + \frac{\partial f_\theta}{\partial \theta} \frac{1}{r} + \frac{\partial f_z}{\partial z} \quad (5.77)$$

Let us examine the curl:

$$\nabla \times f = \left(\frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z} \right) e_x + \left(\frac{\partial f_x}{\partial z} - \frac{\partial f_z}{\partial x} \right) e_y + \left(\frac{\partial f_y}{\partial x} - \frac{\partial f_x}{\partial y} \right) e_z \quad (5.78)$$

We will go term-by-term:

$$\frac{\partial f_z}{\partial y} = \frac{\partial f_z}{\partial r} \sin \theta + \frac{\partial f_z}{\partial \theta} \frac{\cos \theta}{r} \quad (5.79)$$

$$\frac{\partial f_y}{\partial z} = \frac{\partial(f_r \sin \theta + f_\theta \cos \theta)}{\partial z} = \frac{\partial f_r}{\partial z} \sin \theta + \frac{\partial f_\theta}{\partial z} \cos \theta \quad (5.80)$$

$$\frac{\partial f_x}{\partial z} = \frac{\partial(f_r \cos \theta - f_\theta \sin \theta)}{\partial z} = \frac{\partial f_r}{\partial z} \cos \theta - \frac{\partial f_\theta}{\partial z} \sin \theta \quad (5.81)$$

$$\frac{\partial f_z}{\partial x} = \frac{\partial f_z}{\partial r} \cos \theta - \frac{\partial f_z}{\partial \theta} \frac{\sin \theta}{r} \quad (5.82)$$

$$\begin{aligned} \frac{\partial f_y}{\partial x} &= \frac{\partial(f_r \sin \theta + f_\theta \cos \theta)}{\partial r} \cos \theta - \frac{\partial(f_r \sin \theta + f_\theta \cos \theta)}{\partial \theta} \frac{\sin \theta}{r} \\ &= \frac{\partial f_r}{\partial r} \sin \theta \cos \theta + \frac{\partial f_\theta}{\partial r} \cos^2 \theta - \frac{\partial f_r}{\partial \theta} \frac{\sin^2 \theta}{r} \end{aligned}$$

$$-f_r \frac{\sin \theta \cos \theta}{r} - \frac{\partial f_\theta}{\partial \theta} \frac{\sin \theta \cos \theta}{r} + f_\theta \frac{\sin^2 \theta}{r} \quad (5.83)$$

$$\begin{aligned} \frac{\partial f_x}{\partial y} &= \frac{\partial(f_r \cos \theta - f_\theta \sin \theta)}{\partial r} \sin \theta + \frac{\partial(f_r \cos \theta - f_\theta \sin \theta)}{\partial \theta} \frac{\cos \theta}{r} \\ &= \frac{\partial f_r}{\partial r} \sin \theta \cos \theta - \frac{\partial f_\theta}{\partial r} \sin^2 \theta + \frac{\partial f_r}{\partial \theta} \frac{\cos^2 \theta}{r} \\ &\quad - f_r \frac{\sin \theta \cos \theta}{r} - \frac{\partial f_\theta}{\partial \theta} \frac{\sin \theta \cos \theta}{r} - f_\theta \frac{\cos^2 \theta}{r} \end{aligned} \quad (5.84)$$

Considering e_x, e_y, e_z , we calculate (with great suffering):

$$\begin{aligned} &\left(\frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z} \right) e_x + \left(\frac{\partial f_x}{\partial z} - \frac{\partial f_z}{\partial x} \right) e_y \\ &= \left(\frac{\partial f_z}{\partial r} \sin \theta + \frac{\partial f_z}{\partial \theta} \frac{\cos \theta}{r} - \frac{\partial f_r}{\partial z} \sin \theta - \frac{\partial f_\theta}{\partial z} \cos \theta \right) \\ &\quad (\cos(\theta) e_r - \sin(\theta) e_\theta) \\ &\quad + \left(\frac{\partial f_r}{\partial z} \cos \theta - \frac{\partial f_\theta}{\partial z} \sin \theta - \frac{\partial f_z}{\partial r} \cos \theta + \frac{\partial f_z}{\partial \theta} \frac{\sin \theta}{r} \right) \\ &\quad (\sin(\theta) e_r + \cos(\theta) e_\theta) \end{aligned} \quad (5.85)$$

$$= \left(\frac{\partial f_z}{\partial \theta} \frac{1}{r} - \frac{\partial f_\theta}{\partial z} \right) e_r + \left(-\frac{\partial f_z}{\partial r} + \frac{\partial f_r}{\partial z} \right) e_\theta \quad (5.86)$$

$$\left(\frac{\partial f_y}{\partial x} - \frac{\partial f_x}{\partial y} \right) e_z = \left(\frac{\partial f_\theta}{\partial r} - \frac{\partial f_r}{\partial \theta} \frac{1}{r} + f_\theta \frac{1}{r} \right) e_z \quad (5.87)$$

Therefore, at last, we have finally computed the curl in cylindrical coordinates:

$$\begin{aligned} \nabla \times f &= \left(\frac{\partial f_z}{\partial \theta} \frac{1}{r} - \frac{\partial f_\theta}{\partial z} \right) e_r \\ &\quad + \left(-\frac{\partial f_z}{\partial r} + \frac{\partial f_r}{\partial z} \right) e_\theta + \frac{1}{r} \left(\frac{\partial(r f_\theta)}{\partial r} - \frac{\partial f_r}{\partial \theta} \right) e_z \end{aligned} \quad (5.88)$$

Before I end this section, the reader should also be acquainted with spherical coordinates, which consist of r , the total distance from the origin, θ , the angle made between the projection of the point onto the xy -plane and the x axis, and ϕ , the angle made

between the point and the xy -plane. In Cartesian coordinates, we have:

$$x = r \sin \phi \cos \theta \quad (5.89)$$

$$y = r \sin \phi \sin \theta \quad (5.90)$$

and

$$z = r \cos \phi \quad (5.91)$$

Furthermore, for the basis vectors, we have

$$e_r = \sin \phi \cos \theta e_x + \sin \phi \sin \theta e_y + \cos \phi e_z \quad (5.92)$$

$$e_\theta = -\sin \theta e_x + \cos \theta e_y \quad (5.93)$$

and

$$e_\phi = \cos \phi \cos \theta e_x + \cos \phi \sin \theta e_y - \sin \phi e_z \quad (5.94)$$

I will not derive the gradient, divergence, and curl in spherical coordinates, as it would take up too much space; the reader should derive them for practice. However, the formulas are presented below:

$$\nabla f = \frac{\partial f}{\partial r} e_r + \frac{1}{r} \frac{\partial f}{\partial \phi} e_\phi + \frac{1}{r \sin \phi} \frac{\partial f}{\partial \theta} e_\theta \quad (5.95)$$

$$\nabla \cdot f = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 f_r) + \frac{1}{r \sin \phi} \frac{\partial v_\theta}{\partial \theta} + \frac{1}{r \sin \phi} \frac{\partial}{\partial \phi} (\sin \phi v_\phi) \quad (5.96)$$

$$\begin{aligned} \nabla \times f &= \frac{1}{r \sin \phi} \left(\frac{\partial}{\partial \phi} (\sin \phi v_\theta) - \frac{\partial v_\phi}{\partial \theta} \right) e_r \\ &+ \frac{1}{r} \left(\frac{\partial}{\partial r} (r v_\phi) - \frac{\partial v_r}{\partial \phi} \right) e_\theta + \frac{1}{r} \left(\frac{1}{\sin \phi} \frac{\partial v_r}{\partial \theta} - \frac{\partial}{\partial r} (r v_\theta) \right) e_\phi \end{aligned} \quad (5.97)$$

One last note is that physicists and mathematicians use different conventions for spherical coordinates. We have used the mathematical convention: r, θ, ϕ , in the order described above. The physical convention is r, ϕ, θ , where ϕ is swapped with θ in terms of function. In the case that you are a physicist or will read physics literature, bear this one caveat in mind.

5.5 Metric and Levi-Civita Symbols

We now cover certain symbols and tensors that we have seen before in slightly more detail. We have seen both the metric tensor and the Levi-Civita symbol before, in various applications. Both of these mathematical objects are crucial to tensor calculus, and their properties may be used in both calculation and proof, regardless of whether the reader is concerned with pure mathematics, physics, or engineering.

We begin with the metric tensor. The metric tensor is hardly unfamiliar; we define it as the inner product of two basis vectors, as such:

$$g_{ij} = \langle e_i, e_j \rangle \quad (5.98)$$

In that way, the metric tensor depends on the coordinate system that we are using. We are able to calculate the inverse metric tensor in a rather straightforward manner, as well:

$$g^{ij} g_{jk} = \delta_k^i \quad (5.99)$$

As we have discussed before, the inverse metric multiplied by the metric tensor gives us the Kronecker delta, or the identity matrix (in the matrix formulation).

The dual basis, which we have seen all the way back in Chapter 1, may also be known as the contravariant basis; we define it as follows:

Definition 145 (Contravariant Basis). *The contravariant basis is defined as the set φ^i such that*

$$\varphi^i = g^{ij} e_j \quad (5.100)$$

Recall that we are taking the sum of the terms over all the j indices, by the Einstein summation convention.

The metric tensor allows us to calculate distance on a given surface or manifold. We consider the line element, which is defined as follows:

Definition 146 (Line Element). *The line element ds is defined as*

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (5.101)$$

where μ and ν are summed over, as in the Einstein summation convention.

Recall that dx^μ is contravariant, and transforms contravariantly. In Cartesian space, the line element should look very familiar; that is, given the metric for the Cartesian coordinate system

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (5.102)$$

We have that

$$ds^2 = 1(dx)(dx) + 0(dx)(dy) + 0(dx)(dz) + \dots \quad (5.103)$$

$$= dx^2 + dy^2 + dz^2 \quad (5.104)$$

Now, given this line element, we are thus able to convert to different coordinates, provided we are given a correspondence between one set of coordinates and the other. For example, we recall that the equation converting spherical coordinates to Cartesian coordinates looks like the following:

$$x = r \sin \phi \cos \theta \quad (5.105)$$

$$y = r \sin \phi \sin \theta \quad (5.106)$$

and

$$z = r \cos \phi \quad (5.107)$$

We derive the line element for spherical coordinates like so:

$$\begin{aligned} ds^2 = & \left(\frac{\partial x}{\partial r} dr + \frac{\partial x}{\partial \theta} d\theta + \frac{\partial x}{\partial \phi} d\phi \right)^2 + \left(\frac{\partial y}{\partial r} dr + \frac{\partial y}{\partial \theta} d\theta + \frac{\partial y}{\partial \phi} d\phi \right)^2 \\ & + \left(\frac{\partial z}{\partial r} dr + \frac{\partial z}{\partial \theta} d\theta + \frac{\partial z}{\partial \phi} d\phi \right)^2 \end{aligned} \quad (5.108)$$

$$\begin{aligned}
&= (\sin \phi \cos \theta dr - r \sin \phi \sin \theta d\theta + r \cos \phi \cos \theta d\phi)^2 \\
&+ (\sin \phi \sin \theta dr + r \sin \phi \cos \theta d\theta + r \cos \phi \sin \theta d\phi)^2 \\
&+ (\cos \phi dr - r \sin \phi d\phi)^2 \tag{5.109}
\end{aligned}$$

$$\begin{aligned}
&= \sin^2 \phi \cos^2 \theta dr^2 - 2r \sin^2 \phi \sin \theta \cos \theta dr d\theta \\
&+ 2r \sin \phi \cos \phi \cos^2 \theta dr d\phi + r^2 \sin^2 \phi \sin^2 \theta d\theta^2 \\
&- 2r^2 \sin \phi \cos \phi \sin \theta \cos \theta d\theta d\phi + r^2 \cos^2 \phi \cos^2 \theta d\phi^2 \\
&+ \sin^2 \phi \sin^2 \theta dr^2 + 2r \sin^2 \phi \sin \theta \cos \theta dr d\theta \\
&+ 2r \sin \phi \cos \phi \sin^2 \theta dr d\phi + r^2 \sin^2 \phi \cos^2 \theta d\theta^2 \\
&+ 2r^2 \sin \phi \cos \phi \sin \theta \cos \theta d\theta d\phi + r^2 \cos^2 \phi \sin^2 \theta d\phi^2 \\
&+ \cos^2 \phi dr^2 - 2r \sin \phi \cos \phi dr d\phi + r^2 \sin^2 \phi d\phi^2 \tag{5.110}
\end{aligned}$$

$$= dr^2 + r^2 \sin^2 \phi d\theta^2 + r^2 d\phi^2 \tag{5.111}$$

As we can see, it is fairly simple yet somewhat tedious work when it comes to deriving the line element for another set of coordinates based on a set of equations denoting the coordinate transformation.

Notice that, then, we are able to represent the arc length of some curve S in terms of the metric:

$$L = \int_S ds = \int_S \sqrt{g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}} dt \tag{5.112}$$

This allows us to calculate certain simple shapes on manifolds rather easily without relying on Cartesian coordinates.

A few more key terms regarding the metric: we may speak of putting the metric into its canonical form, which is simply defined as follows:

Definition 147 (Canonical Form). *The canonical form of a given metric is the resulting matrix when the metric is arranged*

in this specific order:

$$g_{\mu\nu} = \begin{pmatrix} -1 & \dots & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \ddots & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \dots & -1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \dots & 0 & 1 & \dots & 0 & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \ddots & 0 & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \dots & 1 & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 & 0 & \ddots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix} \quad (5.113)$$

In other words, we put the -1 s on the diagonal first, then the $+1$ s, and then any remaining 0 s. We define the signature of a metric below:

Definition 148 (Signature). *The signature of a metric is the number of its positive and negative eigenvalues. For example, Minkowski space, with one negative eigenvalue and three positive eigenvalues, having the diagonal matrix $\text{diag}(-1, 1, 1, 1)$, has signature minus-plus-plus.*

Furthermore, we are able to categorize the metric like so:

Definition 149 (Types of Metrics). *The metric tensor can have the following descriptions:*

- **Riemannian:** *If the signature is positive at every point, the metric is Riemannian.*
- **Lorentzian/Pseudo-Riemannian:** *If the signature has a single negative, but all other eigenvalues/entries on the diagonal are positive, then the metric is Lorentzian, or pseudo-Riemannian.*
- **Indefinite:** *If the metric is neither of the above, then the metric is indefinite.*

We continue onward with the Levi-Civita symbol. However, before we reintroduce this character in tensor calculus to the reader, we must consider the definition of the object which it is; for the Levi-Civita symbol does not quite transform like a tensor, but rather transforms like something called a tensor density.

Definition 150 (Tensor Density). *A tensor density is a mathematical object D which may have indices that follow this property during coordinate transformation:*

$$D' = \left| \frac{\partial x^{i'}}{\partial x^i} \right|^n D \quad (5.114)$$

for some n .

We call this n the weight of the tensor density:

Definition 151 (Weight). *The weight n of a tensor density is a number that denotes the power to which the Jacobian is raised in the transformation equation of the tensor density.*

Note that a tensor is essentially a tensor density of weight 0. Now, we redefine the Levi-Civita symbol, not exactly a tensor but a tensor density:

Definition 152 (Levi-Civita Symbol). *The Levi-Civita symbol is*

$$\epsilon_{i_1 \dots i_n} = \begin{cases} 1 & i_1, \dots, i_n \text{ is an even permutation of } 1, \dots, n \\ -1 & i_1, \dots, i_n \text{ is an odd permutation of } 1, \dots, n \\ 0 & \text{otherwise} \end{cases} \quad (5.115)$$

The curious reader wonders, why is this not a tensor? We present the following formula without proof, though the reader may verify this statement for lower-dimensional matrices if they wish:

$$\epsilon_{i'_1 \dots i'_n} |A| = \epsilon_{i_1 \dots i_n} A^{i_1}_{i'_1} \dots A^{i_n}_{i'_n} \quad (5.116)$$

Let $A^{i_n}_{i'_n} = \partial x^{i_n} / \partial x^{i'_n}$. We obtain

$$\epsilon_{i'_1 \dots i'_n} \left| \frac{\partial x^{i_n}}{\partial x^{i'_n}} \right| = \epsilon_{i_1 \dots i_n} \frac{\partial x^{i_1}}{\partial x^{i'_1}} \dots \frac{\partial x^{i_n}}{\partial x^{i'_n}} \quad (5.117)$$

Multiplying by the inverse determinant, we obtain

$$\epsilon_{i'_1 \dots i'_n} = \left| \frac{\partial x^{i'_n}}{\partial x^{i_n}} \right| \epsilon_{i_1 \dots i_n} \frac{\partial x^{i_1}}{\partial x^{i'_1}} \dots \frac{\partial x^{i_n}}{\partial x^{i'_n}} \quad (5.118)$$

It doesn't take a rocket scientist to determine that this symbol does not transform like a tensor. In fact, one observes that it appears to have weight 1.

5.6 Epsilon Tensor

However, one may note that it is actually possible to turn a “modified” Levi-Civita symbol into a tensor. We simply apply the following process. Let us multiply the Levi-Civita symbol $\epsilon_{i_1 \dots i_n}$ by the square root determinant of the metric $\sqrt{|g|}$. Consider how this new object transforms:

$$\sqrt{|g(x^{i'})|} \epsilon_{i'_1 \dots i'_n} = \left| \frac{\partial x^{i'}}{\partial x^i} \right|^{-1} \sqrt{|g(x^i)|} \left| \frac{\partial x^{i'}}{\partial x^i} \right| \epsilon_{i_1 \dots i_n} \frac{\partial x^{i_1}}{\partial x^{i'_1}} \dots \frac{\partial x^{i_n}}{\partial x^{i'_n}} \quad (5.119)$$

$$= \sqrt{|g(x^i)|} \epsilon_{i_1 \dots i_n} \frac{\partial x^{i_1}}{\partial x^{i'_1}} \dots \frac{\partial x^{i_n}}{\partial x^{i'_n}} \quad (5.120)$$

This mathematical object transforms like a tensor! We call this the epsilon tensor:

Definition 153 (Epsilon Tensor). *The epsilon tensor is defined as*

$$\tilde{\epsilon}_{i_1 \dots i_n} = \sqrt{|g|} \epsilon_{i_1 \dots i_n} \quad (5.121)$$

One important thing to note is that this tensor is also known as the Levi-Civita tensor. In certain literature, the notation is flipped; ϵ denotes the tensor whereas $\tilde{\epsilon}$ denotes the density.

However, because ϵ is often used to denote the symbol, I have elected to use $\tilde{\epsilon}$ for the tensor instead. Additionally, the epsilon tensor, or Levi-Civita tensor, may also be represented by E .

Note that, in two dimensions, the square root of the determinant of the metric tensor is also denoted as $\sqrt{EG - F^2}$, where E, F, G are the coefficients of the first fundamental form. Therefore, the epsilon tensor in two dimensions can be written as

$$\tilde{\epsilon}_{ij} = \sqrt{EG - F^2} \epsilon_{ij} \quad (5.122)$$

Since the epsilon tensor is a tensor, it is possible to define the epsilon tensor with raised indices rather than lower indices. We define the Levi-Civita *symbol* with upper indices as follows:

Definition 154 (Levi-Civita Symbol (Upper Indices)). *The Levi-Civita symbol with upper indices is denoted by*

$$\epsilon^{i_1 \dots i_n} = \text{sgn}(|g|) \epsilon_{i_1 \dots i_n} \quad (5.123)$$

where $\text{sgn}(|g|)$ represents the sign of the determinant of the metric tensor.

By doing so, we are able to construct a tensor like so:

Definition 155 (Epsilon Tensor (Upper Indices)). *The epsilon tensor with upper indices is denoted by*

$$\tilde{\epsilon}^{i_1 \dots i_n} = \frac{1}{\sqrt{|g|}} \epsilon^{i_1 \dots i_n} \quad (5.124)$$

$$= \frac{\text{sgn}(|g|)}{\sqrt{|g|}} \epsilon_{i_1 \dots i_n} \quad (5.125)$$

If we recall the operation of contraction of tensors, we may contract the epsilon tensor; in fact, it turns out that if we contract m indices of the epsilon tensor, we have the following:

$$\tilde{\epsilon}^{i_1 \dots i_m j_{m+1} \dots j_n} \tilde{\epsilon}_{i_1 \dots i_m k_{m+1} \dots k_n} = (-1)^p m! (n - m)! \delta_{k_{m+1}}^{[j_{m+1}} \dots \delta_{k_n}^{j_n]} \quad (5.126)$$

where p is the number of negative eigenvalues of the metric, or, in canonical form, the number of -1 s. The reason for this will become clear. First, we present a lemma without proof:

Lemma 7. *We have that*

$$\epsilon^{i_1 \dots i_n} \epsilon_{j_1 \dots j_n} = n! \delta_{j_1}^{[i_1} \dots \delta_{j_n}^{i_n]} \quad (5.127)$$

Consider how the Levi-Civita symbol consists of both 1s and -1 s, so one can assume that there is an element of antisymmetry in the resulting product. Additionally, consider the arrangements of the top indices in the Kronecker delta symbols. There are n indices, so the number of possible permutations of the indices is $n!$ (we do not antisymmetrize the bottom indices, because we are already antisymmetrizing the top indices; we do not need to antisymmetrize the bottom indices to produce the combinations of indices we are looking for).

Now, notice that when we are contracting m out of n indices, there are $m!$ possible permutations of the contracted indices, and therefore $(n - m)!$ possible permutations of the non-contracted indices on the top (we do not see the contracted indices on the right-hand side because they have “disappeared,” because the top and the bottom canceled out with each other). However, consider the $(-1)^p$ at the very front of the right-hand side, where p is the number of negative eigenvalues of the metric. This turns out to be equivalent to the sign of the metric determinant. Therefore, because the Levi-Civita tensor with upper indices implies a multiplication by a factor of $(-1)^p$, we obtain the desired result.

Unfortunately, there are no elementary properties that we can build off of to prove this identity, but we can reason through with some intuition and understand it that way.

Now, we return to the topic of the volume element, with a broader understanding of the epsilon tensor and Levi-Civita symbol. Let us define the expression $d^n x$ as the following:

$$d^n x = \frac{1}{n!} \epsilon_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n} \quad (5.128)$$

We consider the way this mathematical object transforms with regard to a coordinate change, hoping that it will transform as a tensor does. We consider first the Levi-Civita symbol:

$$= \frac{1}{n!} \left| \frac{\partial x^\mu}{\partial x^{\mu'}} \right| \epsilon_{\mu'_1 \dots \mu'_n} \frac{\partial x^{\mu'_1}}{\partial x^{\mu_1}} \dots \frac{\partial x^{\mu'_n}}{\partial x^{\mu_n}} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n} \quad (5.129)$$

Then we consider the wedge product of the dx^μ , where each differential transforms exactly as we have seen:

$$= \frac{1}{n!} \left| \frac{\partial x^\mu}{\partial x^{\mu'}} \right| \epsilon_{\mu'_1 \dots \mu'_n} \frac{\partial x^{\mu'_1}}{\partial x^{\mu_1}} \dots \frac{\partial x^{\mu'_n}}{\partial x^{\mu_n}} \frac{\partial x^{\mu_1}}{\partial x^{\mu'_1}} \dots \frac{\partial x^{\mu_n}}{\partial x^{\mu'_n}} dx^{\mu'_1} \wedge \dots \wedge dx^{\mu'_n} \quad (5.130)$$

$$= \frac{1}{n!} \left| \frac{\partial x^\mu}{\partial x^{\mu'}} \right| \epsilon_{\mu'_1 \dots \mu'_n} dx^{\mu'_1} \wedge \dots \wedge dx^{\mu'_n} \quad (5.131)$$

$$= \left| \frac{\partial x^\mu}{\partial x^{\mu'}} \right| \frac{1}{n!} \epsilon_{\mu'_1 \dots \mu'_n} dx^{\mu'_1} \wedge \dots \wedge dx^{\mu'_n} = \left| \frac{\partial x^\mu}{\partial x^{\mu'}} \right| d^n x' \quad (5.132)$$

Therefore, we may conclude that

$$d^n x' = \left| \frac{\partial x^{\mu'}}{\partial x^\mu} \right| d^n x \quad (5.133)$$

This is precisely why, when we change coordinates in multivariable calculus, we multiply by the Jacobian determinant:

$$\iint_R f(x, y) dx dy = \iint_S f(g(u, v), h(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv \quad (5.134)$$

This object therefore appears to transform like a tensor density. However, by multiplying the object by $\sqrt{|g|}$, we create an invariant object that we call the volume element:

$$\sqrt{|g'|} d^n x' = \sqrt{|g|} d^n x \quad (5.135)$$

This allows the integral when we calculate a volume of sorts to remain the same no matter which coordinate system we choose, which is extremely helpful for performing calculus on manifolds, which we will explore in greater detail later. We are therefore able to integrate a scalar function $f(x)$ over a manifold of dimension n , by the following formula:

Theorem 53.

$$V = \int f(x) \sqrt{|g|} d^n x \quad (5.136)$$

This equation may leave you with more questions than answers, to be honest. What is a manifold? Why does it have dimension n ? How are we able to actually perform calculus on these mathematical objects? The answers to all of these questions lie in Part II of this book.

5.7 Exercises

Exercise 1: Verify that the two examples of contravariant and covariant objects really are contravariant and covariant, respectively. Can you think of other contravariant and covariant mathematical objects?

Exercise 2: Verify the contravariant law of transformation, using the proof for the covariant law of transformation as a guide.

Exercise 3: Show that $\bar{\varphi}^i \cdot \bar{e}_j = \delta_j^i$ using the transformation laws from before, and given that $\varphi^i \cdot e_j = \delta_j^i$ in a different coordinate system.

Exercise 4: Verify that differentials of coordinates also transform according to contravariant law. Accomplish this by taking some dx^i and attempting to derive a $d\bar{x}^j$.

Exercise 5: Derive at least three transformation laws for a tensor of order three.

Exercise 6: Verify that $T^{ij} = \varphi^i T \varphi^j$.

Exercise 7: Generalize tensor transformation laws using both partial derivatives and metric tensors to tensors of order n .

Exercise 8: What is another name for the identity tensor?

Exercise 9: Verify that

$$I = \sum_{i=1}^n e_i \otimes e_i \quad (5.137)$$

if the basis is orthonormal.

Exercise 10: Verify that the equations converting linear (Cartesian) coordinates to polar coordinates and backwards are the ones mentioned in the text.

Exercise 11: Verify that the Jacobian which transforms polar coordinates to Cartesian coordinates is as follows:

$$\begin{pmatrix} \frac{x}{r} & \frac{y}{r} & 0 \\ -\frac{y}{r^2} & \frac{x}{r^2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (5.138)$$

Exercise 12: Why is it that the Jacobian matrix does not transform components of a vector correctly? [*Hint:* Think about how vectors and basis vectors transform.]

Exercise 13: If R is the region bounded by the square with vertices $(0, 1)$, $(1, 2)$, $(2, 1)$, $(1, 0)$, find the integral

$$\iint_R (x + y)^5 e^{x-y} dA \quad (5.139)$$

Exercise 14: Find some mathematical object that you are able to correctly apply the Jacobian operator to. What object is it?

Exercise 15: Derive the formula for the gradient ∇f in spherical coordinates.

Exercise 16: Derive the formula for the divergence $\nabla \cdot f$ in spherical coordinates.

Exercise 17: Derive the formula for the curl $\nabla \times f$ in spherical coordinates.

Exercise 18: The Laplacian $\nabla^2 f = \Delta f$ is equal to $\nabla \cdot (\nabla f)$, where $f(x, y, z)$ is a scalar function. Find the formula for the Laplacian in Cartesian, cylindrical, and spherical coordinates.

There is another, more elegant way to derive the gradient, divergence, and curl for any coordinate system. The following exercises will walk you through this process. Consider the infinitesimal displacement vector,

$$dl = f(du)e_u + g(dv)e_v + h(dw)e_w \quad (5.140)$$

where u, v, w are the variables in the coordinate system, and f, g, h are functions specifically dedicated for that coordinate system. For instance, in Cartesian coordinates, $f = g = h = 1$, in cylindrical coordinates, $f = h = 1, g = r$, and in spherical coordinates $f = 1, g = r, h = r \sin \phi$. We begin:

Exercise 19 (The Gradient): We consider the following expression:

$$dt = \frac{\partial t}{\partial u} du + \frac{\partial t}{\partial v} dv + \frac{\partial t}{\partial w} dw \quad (5.141)$$

We can express this in terms of the inner product,

$$dt = \nabla t \cdot dl \quad (5.142)$$

Use these formulas to derive the gradient.

Exercise 20 (The Divergence): To calculate the divergence, we calculate the integral $\oint t \cdot da$ of a vector function t over an infinitesimal surface starting at (u, v, w) and ending at $(u + du, v + dv, w + dw)$. Let $d\tau = dl_u dl_v dl_w = (fgh) du dv dw$. Then, for one of the six surfaces in this infinitesimal surface, we have $da = -(gh) dv dw e_u$ (the opposite side has its sign switched). Therefore,

$$t \cdot da = -(ght_u) dv dw \quad (5.143)$$

Consider the relation

$$F(u + du) - F(u) = \frac{dF}{du} du \quad (5.144)$$

Using all of this, we are able to calculate the divergence:

$$\oint t \cdot da = (\nabla \cdot f) d\tau \quad (5.145)$$

Find the divergence using the coordinate system u, v, w .

Exercise 21 (The Curl): When we wish to calculate the curl, we consider the line integral of the vector-valued function t ,

$$\oint t \cdot dl \quad (5.146)$$

around a loop starting at (u, v, w) and going to $(u + du, v + dv, w)$, before returning back.

a) Verify that the bottom segment has the line element $dl = t du e_u$. Find $t \cdot dl$. What is the line element for the top segment? What about the right and left sides?

b) Verify that the integral to be taken is

$$\oint t \cdot dl = \left(\frac{\partial}{\partial u}(gt_v) - \frac{\partial}{\partial v}(ft_u) \right) du dv \quad (5.147)$$

Represent this in terms of $da = (fg) du dv e_w$. The resulting expression on the right-hand side (besides the $e_w \cdot da$) becomes the part of the curl multiplied by e_w . You can derive the parts multiplied by e_u and e_v . In a similar manner. Derive it.

Exercise 22: Find the line element in cylindrical coordinates.

Exercise 23: Consider the sphere, which can be thought of in spherical coordinates by setting $r = 1$ and letting θ, ϕ remain free variables. Can you derive the metric tensor of the sphere using the line element in spherical coordinates?

Exercise 24: Calculate the arc length of a circle with constant angle $\phi = a$ and constant radius $r = 1$ using spherical coordinates.

Exercise 25: Using the Lorentzian metric, is it possible for the distance between two points to be less than zero? In what context could this possibly be useful?

Exercise 26: Write out the equations for tensor transformation and tensor density transformation. In what way do they differ?

Exercise 27: Consider the determinant of the metric: $|g_{\mu\nu}|$. Using the formula

$$g_{\mu'\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} g_{\mu\nu} \quad (5.148)$$

assess whether the determinant of the metric is a tensor, a tensor density, or neither. In case of the second option, identify its weight.

Exercise 28: Provide a general method for converting a tensor density into a tensor.

Exercise 29: What is the weight of the Levi-Civita symbol with upper indices? How does it compare to the weight of the Levi-Civita symbol with lower indices?

Exercise 30: Construct the identity for the contraction of the epsilon tensor where $m = n - 1$.

Exercise 31: Verify that the volume element $\sqrt{|g|}d^n x$ is indeed invariant under coordinate change.

Challenge 1: In this challenge, we will consider multiple fascinating coordinate systems. Remark on the properties of each coordinate system, performing the necessary calculations when possible.

(a) We consider the ingoing Eddington-Finkelstein coordinates, which are used to describe the metric near a spherically symmetric black hole.

$$ds^2 = \left(1 - \frac{2GM}{r}\right) dv^2 - 2dvdr - r^2 d\Omega^2 \quad (5.149)$$

where $v = t + r + 2GM \ln \left| \frac{r}{2GM} - 1 \right|$. We also consider the outgoing Eddington-Finkelstein coordinates, the line element of which is written as follows:

$$ds^2 = \left(1 - \frac{2GM}{r}\right) du^2 + 2dudr - r^2 d\Omega^2 \quad (5.150)$$

where $u = t - r - 2GM \ln \left| \frac{r}{2GM} - 1 \right|$.

(b) We now consider Gaussian polar coordinates, which are a family of coordinate systems whose line elements are of the form

$$ds^2 = -a(r)^2 dt^2 + dr^2 + b(r)^2 (d\phi^2 + \sin^2(\phi) d\theta^2) \quad (5.151)$$

Of course, the dt term implies that this coordinate system describes spacetime. What sorts of coordinate systems can you create that fall under this coordinate system? Do they have any remarkable properties?

(c) The Gullstrand-Painleve coordinates also describe a black hole, but from the perspective of a free-falling observer. The line

element is as follows:

$$ds^2 = \left(1 - \frac{2M}{r}\right) dt_r^2 - 2\sqrt{\frac{2M}{r}} dt_r dr - dr^2 - r^2 d\phi^2 - r^2 \sin^2(\phi) d\theta^2 \quad (5.152)$$

where $t_r = t + 2M \left(-2\sqrt{\frac{r}{2M}} + \ln \left(\frac{\sqrt{\frac{r}{2M}} + 1}{\sqrt{\frac{r}{2M}} - 1} \right) \right)$. What are some of the characteristics of this coordinate system?

The Kruskal-Szekeres coordinate system is yet another coordinate system that describes black holes. There are two sets of coordinates, one for the space outside the event horizon, and one for the space inside the event horizon. The line element is as follows:

$$ds^2 = \frac{32G^3 M^3}{r} e^{-r/2GM} (-dT^2 + dX^2) + r^2 d\Omega^2 \quad (5.153)$$

where, outside the event horizon,

$$T = \left(\frac{r}{2GM} - 1 \right)^{1/2} e^{r/4GM} \sinh \left(\frac{t}{4GM} \right) \quad (5.154)$$

$$X = \left(\frac{r}{2GM} - 1 \right)^{1/2} e^{r/4GM} \cosh \left(\frac{t}{4GM} \right) \quad (5.155)$$

and inside the event horizon,

$$T = \left(1 - \frac{r}{2GM} \right)^{1/2} e^{r/4GM} \cosh \left(\frac{t}{4GM} \right) \quad (5.156)$$

$$X = \left(1 - \frac{r}{2GM} \right)^{1/2} e^{r/4GM} \sinh \left(\frac{t}{4GM} \right) \quad (5.157)$$

Challenge 2: In this challenge, we explore the Levi-Civita symbol.

(a) Calculate the following expressions. Justify your answer.

$$\epsilon_{ij} \epsilon^{mn} \quad (5.158)$$

$$\epsilon_{ij} \epsilon^{in} \quad (5.159)$$

$$\epsilon_{ij}\epsilon^{ij} \quad (5.160)$$

$$\epsilon_{ijk}\epsilon^{lmn} \quad (5.161)$$

$$\epsilon_{ijk}\epsilon^{lmk} \quad (5.162)$$

$$\epsilon_{ijk}\epsilon^{ljk} \quad (5.163)$$

$$\epsilon_{ijk}\epsilon^{ijk} \quad (5.164)$$

Connect your answers to Equation 5.137.

(b) Show that

$$\epsilon_{ijk}\epsilon_{lmn} = \begin{vmatrix} \delta_{il} & \delta_{im} & \delta_{in} \\ \delta_{jl} & \delta_{jm} & \delta_{jn} \\ \delta_{kl} & \delta_{km} & \delta_{kn} \end{vmatrix} \quad (5.165)$$

(c) Represent the curl $\nabla \times f$ using the Levi-Civita symbol.

(d) Write the contravariant epsilon tensor $\tilde{\epsilon}^{i_1 \dots i_n}$ in terms of the covariant epsilon tensor $\tilde{\epsilon}_{i_1 \dots i_n}$.

Challenge 3: Consider a twenty-dimensional Cartesian coordinate system, $(x^1, \dots, x^{20})$. Find three nontrivial curvilinear coordinate systems that might be interesting to remark on. Calculate the basis vectors, the metric tensor, the line element, and the epsilon tensor, using computer assistance if need be. Then, consider a twenty-dimensional submanifold of this coordinate system with the induced metric. What would be the volume elements of the integral over the manifold?

Challenge 4: This one is an open problem. Suppose that we have a metric that is continuous, yet its derivatives do not always behave nicely. Is it always possible to find a change of coordinates such that the metric becomes “smoother,” that is, the metric is smooth up to at least four derivatives (the curvature of spacetime, in this case, depends on the second derivatives of the metric)? Or, phrased another way, does there exist a spacetime manifold that cannot be made smooth in this manner no matter the coordinates?

Part II

Tensor Calculus

CHAPTER 6

Bundles, Fields, and Manifolds

6.1 Smooth Manifolds

We have learned, in Part I, about the definition and basic properties of tensors. Tensors allow us to perform calculus in spaces beyond Euclidean $\mathbb{R}^n$; indeed, they allow us to perform the operations of calculus on any manifold. However, in order to learn tensor calculus, we must first define and outline a few key properties of manifolds. A manifold is a sort of special kind of topological space. The reader is not expected to know what a topological space is, so the definition of such follows shortly:

Definition 156 (Topological Space). *A topological space is a set X equipped with a topology $\mathcal{T}$. This topology $\mathcal{T}$ is a collection (set) of subsets of X , with the following conditions:*

1. $X \in \mathcal{T}$ and $\emptyset \in \mathcal{T}$ (X itself is in the topology and the empty set is in the topology).

2. If we have any number of elements $U_1, U_2, \dots \in \mathcal{T}$, then

$$\bigcup_{\alpha} U_{\alpha} \in \mathcal{T} \quad (6.1)$$

(this says that the union of all these sets is also part of the topology).

3. If we have a finite number of elements $U_1, \dots, U_n \in \mathcal{T}$, then

$$\bigcap_{\alpha} U_{\alpha} \in \mathcal{T} \quad (6.2)$$

(this says that the intersection of those sets is also part of the topology).

Any set $U \in \mathcal{T}$ is called an open set. As we will see, the concept of a topological space is crucial to defining a manifold, because the definition of manifolds rely on the concept of open sets, which topological spaces allow us to generalize beyond that of $\mathbb{R}^n$.

Before we define a manifold in particular, though, we need to define the concept of a topological manifold, which is a special kind of topological space. Below we define several categories of topological spaces that are necessary to define the topological manifold.

Definition 157 (Hausdorff Space). *A topological space X is considered a Hausdorff space such that, for any points $x, y \in X$, we can find two open sets $U, V \subset X$ such that $x \in U$ and $y \in V$ but $U \cap V = \emptyset$.*

That is to say, we can consider a topological space to be Hausdorff if any two points can be “separated” by two disjoint open sets. Furthermore, we define:

Definition 158 (Locally Euclidean). *A topological space X is locally Euclidean if, for every point $x \in X$, we can define an open set $U \subset X$ such that $x \in U$ and there exists a homeomorphism*

$$\varphi : U \rightarrow \mathbb{R}^n \quad (6.3)$$

A homeomorphism, for those who do not know, is a mapping that is both bijective and continuous. That is to say, if a mapping φ has an inverse φ^{-1} and an open set $V \subset \mathbb{R}^n$ maps to an open set $\varphi^{-1}(V) \subset X$, then φ is a homeomorphism. If such a homeomorphism exists for all $x \in X$, then X is locally Euclidean. At last, we come to the definition of a topological manifold:

Definition 159 (Topological Manifold). *A topological manifold X is a topological space that is both Hausdorff and locally Euclidean.*

Note that some authors may label additional specifications and constraints for some set to be a topological manifold, but for the sake of simplicity, as the reader may not have enough

topological background to know what paracompactness is, for example, we will keep the definition of a topological manifold as is.

Now we come to the idea of a chart. These allow us to set up coordinates on a given topological manifold. We define it as follows:

Definition 160 (Chart). *A chart, or a coordinate map, at a point p in a topological space X is a function $\varphi_\alpha : U \rightarrow \mathbb{R}^n$, where U is an open set in X containing p which is known as the coordinate neighborhood. This chart must be bijective (it must have an inverse) and continuous.*

Note that we say that the dimension of the chart is n , if the function maps to $\mathbb{R}^n$. We can also create coordinates by projecting φ_α onto some $\mathbb{R}$ using the projection function π^i :

$$x_\alpha^i : U \rightarrow \mathbb{R}, \quad x_\alpha^i = \pi^i \circ \varphi_\alpha \quad (6.4)$$

Definition 161 (Atlas). *An atlas $(U_\alpha, \varphi_\alpha)$ is a collection of charts φ_α and their respective open set domains that cover a topological space; that is to say, any point on a topological space X is in some U_α can be mapped to a point in $\mathbb{R}^n$ by some φ_α in the atlas.*

Before we at last come to the definition of a smooth manifold, we need to talk about smoothness, or the class C^∞ of functions. We say that some function $f : \mathbb{R}^m \supset V \rightarrow \mathbb{R}^n$ is C^∞ if all functions of the form $x^i \circ f$ are continuous and differentiable infinitely. As an example, we consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ such that

$$(x, y, z) \mapsto (x^2 y z^3, e^{xyz}) \quad (6.5)$$

We consider $x^1 \circ f$:

$$(x, y, z) \mapsto x^2 y z^3 \quad (6.6)$$

As you can see, we have projected the resulting function onto the first coordinate, so that we are only considering the first part.

It is infinitely differentiable on all partial derivatives, and so is C^∞ . We then consider the second $x^2 \circ f$:

$$(x, y, z) \mapsto e^{xyz} \quad (6.7)$$

This function is also easily C^∞ , as you can check for yourself. Furthermore, we can extend this concept by calling a map f analytic if $x^i \circ f$ is analytic. The reader may not be familiar with what an analytic function is. Here is the definition:

Definition 162 (Analytic). *A function f is analytic if for all points in the domain, some power series centered at any point converges to the function in some neighborhood (open set) of that point.*

This is not too important to know, but it may come in handy sometimes. Now, knowing what a topological manifold, chart, and C^∞ maps are, we can finally define the smooth manifold, or simply, manifold:

Definition 163 (Smooth Manifold). *A smooth manifold, differentiable manifold, or manifold M is a topological manifold equipped with an atlas $(U_\alpha, \varphi_\alpha)$ such that*

1. *We have that*

$$\bigcup_{\alpha} U_{\alpha} = M \quad (6.8)$$

2. *For every $U_\alpha, U_\beta \subset M$, we have that the function*

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \mathbb{R}^m \rightarrow \mathbb{R}^m \quad (6.9)$$

defined on $\varphi_\alpha^{-1}(U_\alpha \cap U_\beta)$ is C^∞ .

3. *The atlas $(U_\alpha, \varphi_\alpha)$ is maximal, which means that it is the union of all other atlases belonging to M ; it is the biggest one of them all.*

We call the m in the expression $\varphi_\alpha : M \rightarrow \mathbb{R}^m$ the dimension of M . Now we arrive at a certain useful theorem concerning smooth manifolds.

Theorem 54. *A map from a manifold M to a manifold N is C^∞ if its coordinate expression is C^∞ for some smooth C^∞ atlas $(U_\alpha, \varphi_\alpha)$ of M and some smooth atlas (V_β, ψ_β) of N .*

Proof. Let us define the coordinate expression of a map $f : M \rightarrow N$:

$$g = \psi_\beta \circ f \circ \varphi_\alpha^{-1} \quad (6.10)$$

It is clear that, if $\varphi_\alpha : M \rightarrow \mathbb{R}^m$, and $\psi_\beta : N \rightarrow \mathbb{R}^n$, then $g : \mathbb{R}^m \rightarrow \mathbb{R}^n$. It is given that g is C^∞ . Since φ_α and ψ_β are defined to be C^∞ , we understand that

$$\psi_\beta^{-1} \circ g \circ \varphi_\alpha = \psi_\beta^{-1} \circ \psi_\beta \circ f \circ \varphi_\alpha^{-1} \circ \varphi_\alpha = f \quad (6.11)$$

is C^∞ as well. Hence, we have that $f : M \rightarrow N$ is C^∞ . $\square$

Furthermore, it would serve us well to define the concept of a diffeomorphism, which is like a differentiable homeomorphism:

Definition 164 (Diffeomorphism). *A diffeomorphism f from a manifold M to a manifold N is a bijective and C^∞ map such that the inverse map $f^{-1} : N \rightarrow M$ is also C^∞ .*

If such a diffeomorphism exists between two manifolds, the manifolds are said to be diffeomorphic.

We have discussed at length the theory and definition of a smooth manifold. Now it is time for us to study certain objects that exist on manifolds that are of particular importance to us.

6.2 Tangent and Cotangent Spaces

Now that we have defined the concept of a smooth manifold, it is time to shift our attention to that of tangent spaces. Now, in order to understand the concept of a tangent space, we need to get a good grasp of the concept of the derivative beyond that of single-variable calculus. We provide a definition for the derivative for functions in $\mathbb{R}^n$ below:

Definition 165 (Derivative). *Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, for any point $v \in \mathbb{R}^n$, we are able to take the derivative $df_x(v)$ at a point $x \in \mathbb{R}^n$ in the direction described by v as follows:*

$$df_x(v) = \lim_{h \rightarrow 0} \frac{f(x + vh) - f(x)}{h} \quad (6.12)$$

This df_x is a map $\mathbb{R}^n \rightarrow \mathbb{R}^m$ such that $\mathbb{R}^n \ni v \mapsto df_x(v) \in \mathbb{R}^m$. We call this map the derivative of f at x .

You have hopefully learned some very elementary results regarding derivatives in single-variable calculus, such as the product rule or the quotient rule. The chain rule, in fact, applies to derivatives of higher dimensions in a rather elegant way. We state the chain rule like so:

Theorem 55 (Chain Rule). *If we have some open sets $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$, as well as two functions, $f : U \rightarrow V$ and $g : V \rightarrow \mathbb{R}^p$, we are able to state that the map $d(g \circ f)_x : U \rightarrow \mathbb{R}^p$, is equivalent to*

$$d(g \circ f)_x = dg_{f(x)} \circ df_x \quad (6.13)$$

The idea of a tangent vector stems from that of a tangent curve in single-variable calculus. We are looking for a mathematical object that follows generally the same properties as that of the derivative. As it turns out, there are multiple different ways to define the tangent vector in differential geometry, all of which essentially hint at the same geometric object. We begin with the more abstract version, defined independently of coordinates:

Definition 166 (Tangent Vector). *A tangent vector, vector, or contravariant vector, at a point $p \in M$, where M is a smooth manifold, is a function $v : C^\infty(p) \rightarrow \mathbb{R}$ such that, for every $f, g \in C^\infty(p)$ and $a, b \in \mathbb{R}$, the tangent vector v follows these properties:*

1. *We have that $v(f + g) = v(f) + v(g)$.*
2. *We have that $v(af) = av(f)$.*
3. *We have that $v(fg) = fv(g) + gv(f)$. This condition is known as the Leibniz rule, or the product rule.*

Furthermore, we can define the idea of a “tangent space,” which is simply the set of all tangent vectors at a given point. Very soon, we will see that we can even define the idea of a set consisting of all tangent spaces on a manifold, called the tangent bundle, but this idea will be explored mainly in the next section.

Definition 167 (Tangent Space). *The tangent space at a point $p \in M$, where M is a smooth manifold, is the set of all possible tangent vectors at p , denoted as T_pM .*

Now we introduce a more coordinate-dependent definition of the tangent space and tangent vectors. This definition more overtly relies on the derivative mentioned earlier, and is particularly useful for working with defined shapes imbedded in $\mathbb{R}^n$ (the reader should have a colloquial sense of what that means, but we will define it more rigorously later). However, the reader should note that we make use of “parametrizations,” which are simply maps from $\mathbb{R}^n$ to a manifold M itself, or the inverse of a chart.

Remark 25. *The tangent space can be thought of geometrically as the best “flat” approximation to a given manifold such that the space and manifold coincide at a single point; this is analogous to how the tangent line of a curve in single-variable calculus represents the best “flat” approximation to the curve such that the curve and line coincide at a single point. Using this analogy, one may think of the tangent space as the image of the derivative of the parametrization $\varphi : \mathbb{R}^n \rightarrow U \subset M \subset \mathbb{R}^m$ (we take a more coordinate-dependent approach to this definition):*

$$d\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad (6.14)$$

In this case, a tangent vector to a manifold $M \subset \mathbb{R}^m$ at a point $p \in M$ is a point $v \in T_pM \subset \mathbb{R}^m$

At this point, we shall examine some basic properties of tangent vectors. The proofs for this should be relatively straightforward and may even seem rather stupid to reader, but these are foundational results regarding the tangent vector and are important to prove. Note that we use coordinate-independent

notation here. The first proof deals with the constant function, analogous to the fact commonly known that the derivative of a single-variable constant function is zero.

Theorem 56. *For every $v \in T_p M$ and some constant function $c \in C^\infty(M)$, we have that $v(c) = 0$.*

Proof. We utilize the product rule to our advantage. We consider the constant functions c and 1 . We begin by considering $cv(1)$, which is equivalent to $v(c1)$ by the second property of the tangent vector. Then, we have that, by the product rule,

$$cv(1) = v(c1) = cv(1) + 1v(c) \quad (6.15)$$

Subtracting $cv(1)$ from both sides, we obtain

$$0 = 1v(c) \implies v(c) = 0 \quad (6.16)$$

Thus completes the proof. $\square$

The second theorem introduces the idea that the tangent vector of two different functions which coincide in some open set results in the same value for both functions.

Theorem 57. *If we have two functions $f, g \in C^\infty(p)$ such that f and g are equal on some neighborhood, or open set, U , containing a point $p \in M$, then for every $v \in T_p M$, $v(f) = v(g)$.*

Proof. Note that this theorem only states the behavior of the tangent vectors at p , not elsewhere on the manifold. Consider 1 to be the constant function mapping any point on U to 1 , similar to in the proof for the previous theorem. We have that $1f = 1g$. Therefore, using the fundamental properties of the tangent vector, we have

$$v(1f) = 1v(f) + f(p)v(1) = v(f) + f(p)v(1) \quad (6.17)$$

And, since $1f = 1g$ by the given statement, we have that $t(1f) = t(1g)$, which gives us

$$= v(1g) = v(g) + g(p)v(1) = v(g) + f(p)v(1) \quad (6.18)$$

Recall that $f(p) = g(p)$, so we are able to state the equivalence above. This shows us that $f(p) = g(p)$, as given. $\square$

Remark 26. *The above two proofs seem trivial and perhaps even unnecessary, but it is often important to rigorously prove theorems that seem intuitive to us.*

Now that we have adequately defined derivatives on $\mathbb{R}^n$, tangent vectors, and parametrizations, we are able to consider the idea of a derivative on a manifold. Because manifolds are not necessarily flat in the same way Euclidean space is, we need to use parametrizations in order to calculate the derivative of a map between two different manifolds. Observe the following definition and exposition regarding derivatives on manifolds below:

Definition 168 (Derivative). *Let us consider a map f between two smooth manifolds M and N . If we have a point $p \in M$ and $q \in N$ such that $f(p) = q$, then the derivative is the linear transformation described by*

$$df_p : T_p M \rightarrow T_q N \quad (6.19)$$

Let's say that we have a $U \subset \mathbb{R}^m$ such that $\varphi : U \rightarrow M$ parametrizes M , and the same thing goes with $V \subset \mathbb{R}^n$ and N , with a parametrization ψ . We are given a mapping $g : U \rightarrow V$ composed of $g = \psi^{-1} \circ f \circ \varphi$. We are able to take the derivative of f, g, φ , and ψ . Then, it follows that we must have

$$df = d\psi \circ dg \circ d\varphi^{-1} \quad (6.20)$$

This allows us to easily compute the derivative of a mapping from a manifold M to N using coordinates.

As we are able to define the derivative on manifolds, it makes sense that one should be able to find some form of the chain rule for manifolds as well. We are able to find such a rule, and this time, we will prove the chain rule, assuming the basic properties of derivatives on flat space hold true:

Theorem 58 (Chain Rule on Manifolds). *If we have three smooth manifolds X, Y, Z , and $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are smooth maps, then we must have that*

$$d(g \circ f)_p = dg_{f(p)} \circ df_p \quad (6.21)$$

Proof. The proof lies in the properties of derivatives and the use of many different maps. Let us list them all out:

- $f : X \rightarrow Y$
- $g : Y \rightarrow Z$
- $\varphi : \mathbb{R}^m \supset U \rightarrow X$
- $\psi : \mathbb{R}^n \supset V \rightarrow Y$
- $\phi : \mathbb{R}^p \supset W \rightarrow Z$
- $h = \psi^{-1} \circ f \circ \varphi : U \rightarrow V$
- $j = \phi^{-1} \circ g \circ \psi : V \rightarrow W$

From here we are interested in the following compositions of maps:

- $g \circ f : X \rightarrow Z$
- $j \circ h : U \rightarrow W$

We note that $g \circ f = \phi \circ j \circ h \circ \varphi^{-1}$. By the properties of the derivative in coordinates, we have

$$d(g \circ f)_p = d\phi \circ dj \circ dh \circ d\varphi^{-1} \quad (6.22)$$

Then, we have that this is equal to

$$(d\phi \circ dj \circ d\psi^{-1}) \circ (d\psi \circ dh \circ d\varphi^{-1}) = dg_{f(p)} \circ df_p \quad (6.23)$$

We consider that $dg = d\phi \circ dj \circ d\psi^{-1}$, and $df = d\psi \circ dh \circ d\varphi^{-1}$. Therefore, we conclude that

$$d(g \circ f)_p = dg_{f(p)} \circ df_p \quad (6.24)$$

□

We have learned about diffeomorphisms before. However, it is also important for us to consider a concept of local diffeomorphisms, which is defined exactly as how one would think it to be: a map between two manifolds such that at a neighborhood of some point, the restriction of the map to the neighborhood is a diffeomorphism.

Definition 169 (Local Diffeomorphism). *A map $f : M \rightarrow N$ between smooth manifolds such that the restriction of the function f_U to an open set U containing $p \in M$ is a diffeomorphism is called a local diffeomorphism at p .*

The reason why local diffeomorphisms are so important is because there exists one particularly important theorem, known as the inverse function theorem, that tells us if a map is locally diffeomorphic at a point or not. In fact, it is quite easy to tell. All we have to do is examine the derivative of the map itself, as we will see below:

Theorem 59 (Inverse Function Theorem). *If $f : M \rightarrow N$ is a map between smooth manifolds, where the derivative df_p at the point $p \in M$ is an isomorphism (that is, in this case, a bijective linear map), then f is a local diffeomorphism at p .*

Just as how manifolds can be diffeomorphic to each other, we can also relate functions to each other in the expected manner. We call functions equivalent, if the domains are diffeomorphic, the codomains are diffeomorphic, and there exists a certain relation such that we can rederive one function from the above diffeomorphisms and the other function itself. Stated more precisely,

Remark 27. *Two functions $f : M \rightarrow N$ and $g : X \rightarrow Y$ are considered to be equivalent if, given diffeomorphisms $\phi : X \rightarrow M$ and $\psi : Y \rightarrow N$, we have that $f = \psi \circ g \circ \phi^{-1}$. Then, the two functions are considered to be the same up to diffeomorphism.*

We now introduce the concept of the cotangent space. Simply put, the cotangent space is defined as the dual space of the tangent space.

Definition 170 (Cotangent Space). *The cotangent space T_p^*M of a manifold M at a point p is the dual space of the tangent space T_pM .*

That is to say, a cotangent vector, or an element of the cotangent space, is a linear map that takes a tangent vector

$v \in T_p M$ and assigns it to a real number in $\mathbb{F}$. Therefore, we consider $T_p^* M$ to be the set of all linear maps $\omega_p : T_p M \rightarrow \mathbb{F}$. We will see in a few chapters why we use the notation ω , and the properties and intuition regarding this mysterious cotangent space will be revealed in the coming sections and chapters.

6.3 Fiber Bundles

The concept of a bundle is best explained through example first, and definition second. We begin by considering the tangent bundle. The reader has undoubtedly come across the tangent bundle beforehand in the book, where it was defined as the set of all tangent vectors on any point of a smooth manifold M . We will provide the definition again below, and study many definitions regarding the tangent bundle, as well as other types of bundles. (This section in particular will be a very definition-heavy section.)

Definition 171 (Tangent Bundle). *The tangent bundle TM of a smooth manifold M is considered to be the set of tangent vectors on any point $p \in M$; in that case, it can be considered to be*

$$TM = \bigsqcup_{p \in M} T_p M \quad (6.25)$$

where $\bigsqcup$ is called the “disjoint union,” and is the union of disjoint sets.

We have defined the tangent bundle TM , and we have defined a tangent space $T_p M$ for each point $p \in M$. Now, the reader is probably wondering if there exists some sort of classification of $T_p M$ as a subset of TM . As it turns out, $T_p M$ is considered to be a fiber of TM . Let us consider the tangent vector v on point $p \in M$. We have that $(v, p) \in TM$. If we project that given tangent vector onto the point $p \in M$, we have created a surjective mapping. As it turns out, the preimage of p is precisely $T_p M$. Generalizing this example, we consider the fiber to be the preimage of a map π :

Definition 172 (Fiber). *A fiber of a point p under a map π is considered to be the preimage of π , like so:*

$$\pi^{-1}(p) = \{x : \pi(x) = p\} \quad (6.26)$$

With this in mind, we define the fiber bundle. The fiber bundle is an incredibly abstract object, more so than most of the other topics we have covered in the book thus far, so examples will help to clarify. Essentially, we have a space that in small regions (specifically open sets), we have that a space behaves like a product space. In some cases, the fiber bundle actually is a product space! In others, the fiber bundle is not particularly a product space. Anyway, here is the definition, in full mathematical rigor:

Definition 173 (Fiber Bundle). *A fiber bundle is an ordered quadruple (E, B, π, F) such that F, B, E are C^n manifolds, $\pi : E \rightarrow B$ is a continuous surjective map, and for each point $p \in M$ we have a neighborhood $U \subset B$ such that $\phi : \pi^{-1}(U) \rightarrow U \times F$ is a C^n diffeomorphism and $\pi = \text{proj}_U \circ \phi$.*

There is a specific type of fiber bundle that we are especially concerned about called the vector bundle. As it turns out, the tensor bundle is a kind of vector bundle (despite the fact that a vector is a kind of tensor, and not the other way around). The vector bundle is a fiber bundle with the additional stipulation that each and every fiber of the fiber bundle must be a vector space over the field of real numbers. We also have operations defined on the vector bundle in a specific manner. Observe the definition below:

Definition 174 (Vector Bundle). *An n -dimensional vector bundle is an ordered quintuple*

$$\xi = (E, B, \pi, \oplus, \odot) \quad (6.27)$$

where E is called the “total space,” B is called the “base space,” $\pi : E \rightarrow B$ is a continuous surjective map, and $\oplus$ and $\odot$ are operations defined as

$$\oplus : \bigcup_{p \in B} \pi^{-1}(p) \times \pi^{-1}(p) \rightarrow E \quad (6.28)$$

$$\odot : \mathbb{R} \times E \rightarrow E \quad (6.29)$$

such that each fiber $\pi^{-1}(p)$ is closed under both operations, making it a vector space over the field $\mathbb{R}$.

Additionally, we must have that for all $p \in B$, there must exist some open set U such that $p \in U$ and a homeomorphism (a continuous bijective map) $f : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^n$ such that it is an isomorphism (bijective and linear) between the two vector spaces $\pi^{-1}(q)$ and $q \times \mathbb{R}^n$, for all $q \in U$.

Let me add some additional clarification with regard to the operations defined in the definition above. When we “add” things together using the $\oplus$ symbol, the domain is simply two vectors that share the same point. The point can be any point on the manifold (that is the point of the union symbol), but the two vectors must share the same point. The vectors must add to become another vector in the same fiber. Similarly, multiplying a vector using $\odot$ in a given fiber by a real number should produce a vector in that same fiber. The $\oplus$ does NOT mean direct sum, in this case.

We define a simple type of vector bundle, known as the trivial bundle. In this case, the total space is actually a product space containing the base space as well as $\mathbb{R}^n$. The projection π is, of course, easily defined, and it is left to the exercises to show how this admits a vector space structure.

Definition 175 (Trivial Bundle). *The trivial bundle $\epsilon^n(M)$ is considered to be the vector bundle such that $E = M \times \mathbb{R}^n$ and $\pi : M \times \mathbb{R}^n \rightarrow M$ being a simple projection, with the obvious vector space structure defined on each individual fiber.*

Now that we have developed some knowledge regarding vector bundles, it will behoove us to consider the possibility of two vector bundles being equivalent. We call vector bundles equivalent if they have the same base space, but different total spaces, but there is a homeomorphism between the total spaces and the two fibers (one for each total space) generated from a given point are isomorphic. Of course, the homeomorphism in question is considered to be an equivalence.

Definition 176 (Equivalence of Vector Bundles). *Two vector bundles ξ_1 and ξ_2 are considered to be equivalent if there exists a homeomorphism between the total spaces $f : E_1 \rightarrow E_2$ such that $\pi_1^{-1}(p) \simeq \pi_2^{-1}(p)$ for each fiber in the vector bundle. f is called an equivalence.*

Note that any bundle equivalent to the trivial bundle $\epsilon^n(M)$ is simply considered to be trivial as well.

Furthermore, we are able to map between different vector bundles, which do not have the same base space. We wish to add additional stipulations on this map, however. The “bundle map” should actually be a pair of maps, one mapping the total spaces and one mapping the base spaces. These maps and the projections from the total to the base space should commute in the expected manner, as is defined below. Furthermore, we must have that the map from a fiber to the appropriate fiber is linear. More insight regarding these somewhat nebulous terms will be shed in the definition below.

Definition 177 (Bundle Map). *A bundle map between two vector bundles ξ_1 and ξ_2 is a pair of functions $f : E_1 \rightarrow E_2$ and $g : B_1 \rightarrow B_2$ such that $\pi_2 \circ f = g \circ \pi_1 : E_1 \rightarrow B_2$, and $f : \pi_1^{-1}(p) \rightarrow \pi_2^{-1}(g(p))$ is linear.*

Now that we have discovered some basic properties about the vector bundle, let us consider another type of vector bundle. Recall that the cotangent space T_p^*M is the dual space of the tangent space T_pM (one may wish to review section 1.2 of the textbook on dual spaces to further solidify their understanding of this concept). The cotangent bundle is simply the fiber bundle such that each fiber is a cotangent space.

Definition 178 (Cotangent Bundle). *The cotangent bundle T^*M of a smooth manifold M is considered to be the fiber bundle containing all cotangent spaces at all points. Therefore, it is denoted by*

$$T^*M = \bigsqcup_{p \in M} T_p^*M \quad (6.30)$$

For the final topic in this section, let us consider the tensor bundle. This particular object comes in many different forms, but there is one commonality among all of them. Each fiber of a tensor bundle is a product of the tangent space some number of times and the cotangent space some number of times. The definition of a tensor bundle is presented below:

Definition 179 (Tensor Bundle). *The tensor bundle is a fiber bundle such that each fiber is a product of the tangent space and/or cotangent space, like so:*

$$\pi^{-1}(p) = T_p M \otimes \dots \otimes T_p M \otimes T_p^* M \otimes \dots \otimes T_p^* M \quad (6.31)$$

Fiber bundles, vector bundles, and tensor bundles are all important concepts to study in differential geometry, yet this section has barely scratched the surface regarding them. The main takeaway from this section that the reader should obtain is that we can apply our knowledge of vectors, dual vectors, and tensors to objects on manifolds such as bundles, as well as fields, as we will soon see in the next section.

6.4 Vector and Tensor Fields

Now we come to the topic of the vector field. We have that a vector field is simply a map that takes a function on M to a function on M , as if there were a tangent vector on every point on the manifold taking each function to a different function at each point. We will see that there are other definitions for a vector field, but this is the first that we will introduce, due to its similarity to that of the tangent vector in the previous sections.

Definition 180 (Vector Field). *A vector field X on M is a function such that*

$$X : C^\infty(M) \rightarrow C^\infty(M) \quad (6.32)$$

and, for all $f, g \in C^\infty(M)$ and for all $c \in \mathbb{R}$:

1. $v(f + g) = v(f) + v(g)$

$$2. \ v(cf) = cv(f)$$

$$3. \ v(fg) = v(f)g + fv(g)$$

As I have mentioned beforehand, there is more than one way to define a vector field. One can also think of it as a mapping from the manifold to the tangent bundle (taking a point and mapping it to a tangent vector at a given point). Both interpretations are equally valid, and serve their own purposes in various fields of differential geometry and applied mathematics. Furthermore, as it turns out, we are able to define a vector field as a linear combination of partial derivatives, with the coefficients being functions on M .

Remark 28. *A vector field X on M can be thought of as an assignment of a tangent vector $v_p \in T_p M$ to each point $p \in M$, such that v_p is of class C^∞ . In this sense, the vector field can be thought of as a map from M to TM : $X : M \rightarrow TM$. Either definition will be used in differential geometry, so be aware of both. We can represent vector fields as a linear combination of partial derivatives, due to their relation to the tangent space, like so:*

$$X = c^i(p)\partial_i \tag{6.33}$$

Recall that $\partial_i = \partial/\partial x_i$.

We define a coordinate frame field that essentially forms a basis for any other vector field that we create on a given manifold. This will be useful in coordinate-dependent exercises.

Definition 181 (Coordinate Frame Field). *The ordered set of vector fields*

$$(\partial_1, \dots, \partial_n) \tag{6.34}$$

is called a coordinate frame field, or a holonomic frame field.

Since the coordinate frame field forms a basis for the set of vector fields, we have that the coordinate frame field must be linearly independent as well. The reader should ponder why this is the case.

Remark 29. *The vector fields ∂_i on $\mathbb{R}^n$ are linearly independent.*

A helpful hint for understanding this remark: consider the action of two different coordinate frame fields on a function of your choice. Particularly, consider this within the context of the definition of linear independence given all the way back in section 1.1 of the book.

We approach a theorem. If we have that the vector fields coincide when acting upon an arbitrary function, then the vector fields are equal. The given proof of this relies on the previous remark, though it is possible to find alternate proofs. The reader should try to search for one themselves.

Theorem 60. *If M is a smooth manifold and X, Y are vector fields on M , if we have that $Xf = Yf$ for some $f \in C^\infty(M)$, then $X = Y$.*

Proof. All we need to do is prove that $X = Y$ at a single point $p \in M$, since that point is arbitrary. Let

$$X = a^i \partial_i, \quad Y = b^i \partial_i \quad (6.35)$$

Note that ∂_i are linearly independent, so we must have that $a^i \partial_i = b^i \partial_i$ for each individual $i \in \{1, \dots, n\}$. Then, it follows that $a^i = b^i$ for all defined i . Therefore, $Y = a^i \partial_i = X$ on $p \in M$. $\square$

We also consider a useful lemma, that if we have any tangent vector, there exists a vector field containing such a tangent vector in its appropriate position. The proof of this is somewhat complicated despite its relative simplicity, and it is important not to forget the subtlety that we are working on coordinates on a manifold, and beyond the open set the coordinates are defined on, the coordinates mean nothing. Taking this into consideration, we prove the following theorem:

Lemma 8. *For any $v \in T_p M$, there exists a vector field X such that $X(p) = v$.*

Proof. We want to define a vector field such that it is defined only on the chart that p is in, and zero outside of that chart. It

also needs to be continuous. Let's write

$$v = v^i \partial_i \quad (6.36)$$

and let

$$Y = v^i \partial_i \quad (6.37)$$

and v^i are constant functions on the open set U . To accomplish what we mentioned before, we turn to the cut-off function, a specific type of function that is continuous, its value is 1 at a desired point (in this case p), and approaches 0 at a desired place (in this case as we approach the edge of U). Call this function β . Then if

$$X = \beta Y \quad (6.38)$$

and $X = 0$ outside of U , then we have found our desired vector field. $\square$

Furthermore, we will now discuss the existence and uniqueness of a particularly important vector field that can be constructed from two other vector fields. It will be given a name soon, but we consider the following lemma:

Lemma 9. *If we have vector fields X and Y on a smooth manifold M , then there exists a unique vector field $Z = XY - YX$.*

The proof of this lemma will be left as an exercise for the reader. To start the reader off, one can prove uniqueness by finding a single formula describing Z , and to prove existence, consider the clever use of coordinate charts.

This field has a special name. It is called the bracket, or the commutator. It is called the commutator because if $[X, Y] = XY - YX = 0$, then the vector fields X and Y commute.

Definition 182 (Bracket). *The bracket $[X, Y]$ of two vector fields, also known as the commutator, is the vector field defined by $XY - YX$, and is C^∞ (as X and Y are C^∞).*

Now we direct our attention to $C^\infty M$. This particular set is considered to be an algebra over the set of real numbers. That means that it follows certain key properties, which are outlined below.

Remark 30. We say that $C^\infty(M)$ is an algebra over $\mathbb{R}$. This means that, given $f, g \in C^\infty(M)$ and $a, b \in \mathbb{R}$:

- $f + g = g + f$
- $f + (g + h) = (f + g) + h$
- $f(gh) = (fg)h$
- $a(bf) = (ab)f$
- $f(g + h) = fg + fh$
- $(f + g)h = fh + gh$
- $a(f + g) = af + ag$
- $(a + b)f = af + bf$
- $1f = f$

Furthermore, $C^\infty(M)$ is a commutative algebra:

- $fg = gf$

Now that we have introduced the idea of $C^\infty(M)$ being an algebra, we will consider its relationship with a vector field. Since the vector field follows several key properties outlined in its definition, it turns out that the vector field is considered to be a derivation of the algebra.

Remark 31. A vector field X is considered to be a derivation of the algebra $C^\infty(M)$, as it follows the following properties, as stated beforehand:

1. $v(f + g) = v(f) + v(g)$
2. $v(cf) = cv(f)$
3. $v(fg) = v(f)g + fv(g)$

Finally, we consider integral curves. Vector fields may admit “solutions,” meaning that we can find a curve such that its tangent vectors match up directly with that of the vector field. The formal definition for an integral curve is provided below.

Definition 183 (Integral Curve). *If we have a curve on a smooth manifold M , $\gamma : (-\epsilon, \epsilon) \rightarrow M$ such that*

$$\gamma(0) = p \quad (6.39)$$

and

$$\gamma'(t) = X(\gamma(t)) \quad (6.40)$$

where X is some vector field, we call γ an integral curve for X , with an initial condition p .

On the real line, a familiar setting, we examine the idea of Lipschitz continuity. This is particularly important because it pertains to the uniqueness of integral curves for a given vector field and initial condition (if you recall solving ordinary or partial differential equations, you may be familiar with the idea of a vector field admitting a solution, or an integral curve):

Definition 184 (Lipschitz Continuity). *A function f is Lipschitz continuous on an open set U if there exists some c such that for all $x, y \in U$,*

$$|f(x) - f(y)| \leq c|x - y| \quad (6.41)$$

As it turns out, on the real line, if we have an integral curve such that $\gamma'(t) = f(t)$, then given an initial condition, so long as f is Lipschitz continuous, then we will have a unique solution.

We have examined all of these important properties of vector bundles, but let us discuss briefly the idea of a tensor field. The tensor field carries over quite easily from that of the vector field: it is the assignment to each point of $p \in M$ a tensor in some $T_p M \otimes \dots \otimes T_p M \otimes T_p^* M \otimes \dots \otimes T_p^* M$.

Definition 185 (Tensor Field). *A tensor field is a mapping*

$$T : M \rightarrow TM \otimes \dots \otimes TM \otimes T^*M \otimes \dots \otimes T^*M \quad (6.42)$$

such that

$$p \mapsto T^{i_1 \dots i_n}_{j_1 \dots j_n} \in T_p M \otimes \dots \otimes T_p M \otimes T_p^* M \otimes \dots \otimes T_p^* M \quad (6.43)$$

It will be left as an exercise in memory for the reader to try to find an example of a tensor field on an arbitrary manifold, and describe the appropriate mapping.

6.5 Sections of Bundles

Now, we consider sections of bundles. To understand bundles and sections properly, it helps to consider a geometric picture of a bundle inside one's head. The tangent bundle is a suitable example of a bundle, for the purpose of this mental exercise. We can think of a section as essentially a function that takes in a point in the base space of the bundle and assigns it an element of the appropriate fiber. One would note that, if we were to take the section of a point and then take the projection of the resulting element, we would return to the point in question. In fact, this is where the definition of a section arises. We first consider the local section, which deals with points in an open subset of the base space. Observe the definition of the local section below:

Definition 186 (Local Section). *A local section of a bundle is a continuous function $s : U \rightarrow E$, where U is an open subset of B , such that for all $u \in U$, we have that $(\pi \circ s)(u) = u$.*

The global section as defined on a fiber bundle, is simply the same definition, but U is replaced with B :

Definition 187 (Global Section). *A global section of a bundle is a continuous function $s : B \rightarrow E$, such that for all $x \in B$, we have that $(\pi \circ s)(x) = x$.*

We note that if we consider a function ϕ , where $\phi : \pi^{-1}(U) \rightarrow U \times F$ (we map the fibers associated with the open set U to a product space of the open set and F ; this is called a local trivialization of the fiber bundle), and ϕ is a homeomorphism (bijective and continuous), then it naturally follows that there must exist a local section on U .

We can say that the set of all local sections form what is known as a sheaf over B (the base space), which we call the sheaf of sections on E (the total space). The question of what a sheaf is and the study of such mathematical objects are beyond the scope of this textbook.

The question of whether a given fiber bundle has a global section or not is a question that drives part of the field of topology,

and the interested reader may consider studying topology to further pursue the answers to this specific question for different fiber bundles.

6.6 Immersions and Submersions

Now that we have discussed manifolds at length and have concerned ourselves with related topics such as fiber bundles and vector fields, it is time to turn to the topic of an immersion. One should be naturally familiar with the intuition behind an immersion—for instance, many surfaces can be thought of as immersed in $\mathbb{R}^3$, such as the sphere, or the torus. But surprisingly, the sphere, for instance, as represented in 3-dimensional coordinates is not actually a sphere! It is the image of a sphere immersed in $\mathbb{R}^3$ under some map. So, an immersion is some sort of map that takes a manifold of smaller dimension and maps it to a manifold of larger dimension, in a specific way, as we will see below:

Definition 188 (Immersion). *A manifold M is considered to be immersed in a manifold $\overline{M}$ if there exists a C^∞ map $f : M \rightarrow \overline{M}$ such that for all $m \in M$, there exists a neighborhood $U \subset M$ of m and a chart of $\overline{M}$ at $V = f(U)$: $y : V \rightarrow \mathbb{R}^n$, such that $x^i = y^i \circ f$ are coordinates on U for M . The function f is called an immersion.*

It is interesting to note that there are actually multiple equivalent definitions of an immersion. All are equally valid. However, one should also note that the immersion, due to the fact that a manifold of lesser dimension is being mapped onto a manifold of greater dimension, must have that its differential is injective. That is, the linear map df_p , where f is the immersion in question, must be injective. This will prove to be of great importance.

Remark 32. *An immersion $f : M \rightarrow N$, because $\dim M$ is meant to be less than or equal to $\dim N$, implies that*

$$df_p : T_p(M) \rightarrow T_{f(p)}N \quad (6.44)$$

is injective at every point $p \in M$.

Now we come to the first theorem in this section, the local immersion theorem. This theorem merely states that if we have some immersion, then we must have local coordinates such that the immersion is equivalent to what is known as the canonical immersion, which is simply the mapping of coordinates $(x^1, \dots, x^m)$ to $(x^1, \dots, x^m, 0, \dots, 0)$. The theorem is stated below, with proof.

Theorem 61 (Local Immersion Theorem). *If $f : M \rightarrow N$ is an immersion at a point $p \in M$, and $q = f(p)$, there must exist local coordinates around p and q such that*

$$f(x^1, \dots, x^m) = (x^1, \dots, x^m, 0, \dots, 0) \quad (6.45)$$

where $x^1, \dots, x^m$ are coordinates of M around p .

Proof. We already have a function $f : M \rightarrow N$. Let us choose parametrizations $\varphi : U \rightarrow M$ and $\psi : V \rightarrow Y$, such that $U \subset \mathbb{R}^m$, and $V \subset \mathbb{R}^n$. Then we construct a mapping g such that

$$g = \psi^{-1} \circ f \circ \varphi \quad (6.46)$$

We know that $dg_{\varphi^{-1}(p)}$ is injective, so by changing the basis of $\mathbb{R}^n$ we are able to procure a matrix

$$dg_{\varphi^{-1}(p)} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} \quad (6.47)$$

which maps $\mathbb{R}^m$ to $\mathbb{R}^n$, where $m \leq n$. We are thus able to define a function $h : U \times \mathbb{R}^{n-m} \rightarrow \mathbb{R}^n$ such that

$$h(x^1, \dots, x^m, z^{m+1}, \dots, z^n) = g(x^1, \dots, x^m) + \begin{pmatrix} 0 \\ \vdots \\ 0 \\ z^{m+1} \\ \vdots \\ z^n \end{pmatrix} \quad (6.48)$$

It is clear that $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$, and furthermore, we have that

$$dh_{(\varphi^{-1}(p), z^{m+1}, \dots, z^n)} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \quad (6.49)$$

The inverse function theorem therefore shows, since dh is an isomorphism around $(\varphi^{-1}(p), z^{m+1}, \dots, z^n)$, that h is a local diffeomorphism around $(\varphi^{-1}(p), z^{m+1}, \dots, z^n)$. We define the canonical immersion $i : \mathbb{R}^m \rightarrow \mathbb{R}^n$ as

$$i(x^1, \dots, x^m) = (x^1, \dots, x^m, 0, \dots, 0) \quad (6.50)$$

Note that $g = h \circ i$. We know that ψ is also a local diffeomorphism around $\psi^{-1}(q)$, so $\psi \circ h$ is also a local diffeomorphism. It follows therefore that $i : U \rightarrow V$ can be expressed as

$$i = (\psi \circ h)^{-1} \circ f \circ \varphi \quad (6.51)$$

and therefore the existence of the canonical immersion, which is equivalent to

$$f(x^1, \dots, x^m) = (x^1, \dots, x^m, 0, \dots, 0) \quad (6.52)$$

is guaranteed. $\square$

Since we have covered the definition of an immersion, we can now consider a stronger definition, which is known as an imbedding, or an embedding. We add the additional stipulation that the mapping itself must be injective, not just the differential. That is, the imbedded surface cannot intersect itself in such a way that two distinct points are “merged” into one. One may be able to visualize such an example of an immersion that is not an imbedding in their head. The definition of an imbedding is stated officially below:

Definition 189 (Imbedding). *If f in the previous definition is 1-1, then f is called an imbedding, and M is imbedded in $\overline{M}$.*

It is a fact that there can exist manifolds inside other manifolds. We call such manifolds submanifolds. A submanifold of a given manifold is a particular subset of a manifold such that the subset is the image of an imbedding, like below:

Definition 190 (Submanifold). *A submanifold of a manifold $\overline{M}$ is the image $f(M)$, where M is a manifold, and $f : M \rightarrow \overline{M}$ is an imbedding.*

Since we have explored the idea of immersing manifolds inside manifolds of equal or higher dimension, one might wish to ask if it is possible to apply the reverse of such a procedure. That is, to map a manifold to a manifold of lower dimension. In fact, it is possible; this is known as a submersion. The submersion is simply a mapping between manifolds such that the differential is surjective, rather than injective necessarily, for all points in the manifold.

Definition 191 (Submersion). *A mapping $f : M \rightarrow N$ is considered to be a submersion if we have that df_p is surjective for all $p \in M$.*

There exists a mirror version of the local immersion theorem specifically for submersions, aptly named the local submersion theorem. The statement is exactly as one knowledgeable of the local immersion theorem would expect. Consider the following theorem, presented **without** proof:

Theorem 62 (Local Submersion Theorem). *If $f : M \rightarrow N$ is a submersion at $p \in M$, then there exist coordinates around p and $f(p)$ such that we have*

$$f(x^1, \dots, x^m) = (x^1, \dots, x^n) \quad (6.53)$$

such that $m = \dim M$, $n = \dim N$, and $m \geq n$. This submersion is known as the canonical submersion.

The proof of this is similar to the proof of the Local Immersion Theorem and will be left as an exercise to the reader.

Since we have explored the role of coordinates in our study of manifolds, it would benefit us to examine one other aspect

of coordinates. We can use coordinates to take a “slice” of a manifold, in the same way that one may be able to take a two-dimensional slice of a three-dimensional manifold, or perhaps a one-dimensional slice of a surface. The number of dimensions does not necessarily matter, so long that the dimension of the slice is less than the dimension of the manifold. We can do this by creating a coordinate system and setting certain coordinates to be constant. The reader should imagine a surface parametrized in two variables, where one variable is held constant. What sort of shape will result?

Definition 192 (Coordinate Slice). *A coordinate slice with dimension m of a manifold N is the set of points in some coordinate neighborhood U with coordinates $y^1, \dots, y^n$ given by*

$$\{p \mid p \in U, y^{m+1}(p) = c_{m+1}, \dots, y^n(p) = c_n\} \quad (6.54)$$

It essentially assigns all coordinates $y^{m+1}, \dots, y^n$ to various constants.

One final topic before we end the chapter on manifolds is that of a refinement. We can take an open cover of a manifold and refine such a cover. Intuitively speaking, a refinement makes the cover “finer,” with smaller sets in the cover, in the same way that smaller grains of sand would be considered finer than that of larger grains of sand. We examine the definition of a refinement below.

Definition 193 (Refinement). *An open cover $\mathcal{O}$ of a manifold M is a set of open subsets of M $U_1, \dots$ such that*

$$\bigcup_i U_i = M \quad (6.55)$$

An open cover $\mathcal{O}'$ is called a refinement of $\mathcal{O}$ if, for every $U \in \mathcal{O}'$, we have some $V \in \mathcal{O}$ such that $U \subset V$.

Some open covers exhibit a property known as local finiteness. That means that the number of sets sufficiently near any given point is finite. This is a tough concept to wrap one’s head around, so ponder this for a moment. The definition will be waiting for you below whenever you finish thinking about it.

Definition 194 (Local Finiteness). *A locally finite cover $\mathcal{O}$ of a manifold M is an open cover such that for all $p \in M$, there exists some open set $U \ni p$ such that $U \cap U_i \neq \emptyset$ for only finitely many U_i .*

In this chapter, we have seen a great many of mathematical beasts. We have seen the manifold. We have seen the fiber bundle and the vector field. We have examined the properties of the tangent and cotangent spaces, as well as their respective bundles. Now, we are ready to perform calculus on these manifolds, starting with the concept of covariant differentiation.

6.7 Exercises

Exercise 1: In the definition of a topological space, only an intersection of a finite number of open sets is an open set. Why might we specify this definition so as not to include an intersection of infinitely many open sets?

Exercise 2: Let's say we have two smooth manifolds M and N . The disjoint union $M \cup N$ obtains the disjoint union topology in which a set $U \subset M \cup N$ is open if and only if $U = U_m \cup U_n$, where U_m is open in M and U_n is open in N . Prove that $M \cup N$ is also a smooth manifold with the given topology.

Exercise 3: Building on the previous exercise, if M and N are n -dimensional, what would the dimension of $M \cup N$ be? Justify your answer.

Exercise 4: Consider the set of all 2×2 orthogonal matrices $O(2)$. Is this set a smooth manifold? What about the set of all 3×3 orthogonal matrices $O(3)$?

Exercise 5: Let X be an n -dimensional manifold. Show that the Cartesian product $X \times X$ is also a manifold. What is its

dimension?

Exercise 6: The projective plane $\mathbb{P}^2$ is the set of lines passing through the origin in $\mathbb{R}^3$. Show that $\mathbb{P}^2$ is a surface; that is, a 2-dimensional manifold.

Exercise 7: If M, N, P are smooth manifolds, show that $(M \times N) \times P$ is diffeomorphic to $M \times (N \times P)$.

Exercise 8: How might we introduce a differentiable structure on $\mathbb{P}^n$?

Exercise 9: Prove the chain rule for functions of multiple variables in flat space.

Exercise 10: If we define $v + w$ as $(v + w)(f) = v(f) + w(f)$, where $v, w \in T_p M$ prove that $v + w \in T_p M$. That is, show that it is linear and obeys the product rule.

Exercise 11: Is the tangent space a manifold? Do you think that the tangent bundle is a manifold?

Exercise 12: Can you come up with a geometric explanation for the cotangent vectors? Consider the fact that it takes in a tangent vector and spits out a scalar number. What might the cotangent vector represent?

Exercise 13: Consider the inclusion map $i : M \rightarrow N$ for a manifold $N \subset M$, where M is also a manifold. Is di_p the inclusion map of $T_p M$ to $T_p N$? Show your thought process.

Exercise 14: Show that if $f : M \rightarrow N$ is a diffeomorphism, then $df_p : T_p M \rightarrow T_p N$ is an isomorphism and a linear map for each $p \in M$.

Exercise 15: Say that we have a C^∞ map $f : M \rightarrow N$. Given a $v \in T_p M$ and a function $h \in C^\infty(N)$, show that

$$v(h \circ f) = df(v)h \quad (6.56)$$

Use the fact that if $(y_1, \dots, y_n)$ is a chart for N and $(x_1, \dots, x_m)$ is a chart for M such that f takes one chart to the other, then

$$df \left(\frac{\partial}{\partial x_i} \right) = \frac{\partial y_j}{\partial x_i} \frac{\partial}{\partial y_j} \quad (6.57)$$

where $\partial/\partial x_i$ and $\partial/\partial y_i$ are the appropriate basis tangent vectors.

Exercise 16: Show that TM is a vector bundle. Show that T^*M is a vector bundle.

Exercise 17: Show that the trivial bundle $\epsilon^n(M)$ admits a vector space structure on each of its fibers.

Exercise 18: Show that equivalence of vector bundles is really an equivalence relation.

Exercise 19: Find a couple different vector bundles that are equivalent to the trivial bundle. The more you can find, the better.

Exercise 20: When we represent vector fields as a linear combination of ∂_i , why do we have no argument for the partial derivative? Why is it not $\partial_i x_i$, for instance, or something similar?

Exercise 21: Prove the existence and uniqueness of a vector field $Z = XY - YX$.

Exercise 22: Show that ∂_i on $\mathbb{R}^n$ are all linearly independent.

Exercise 23: Find an alternate proof for Theorem 60, without using linear independence of the coordinate frame field.

Exercise 24: What is an example of a tensor field on an arbitrary manifold? Describe the mapping that the tensor field admits.

Exercise 25: A section on a tangent bundle has a very particular name that you have seen before. What is it called? Justify your answer.

Exercise 26: Do all fiber bundles have global sections? Why or why not?

Exercise 27: Why is it true that there exists a local bundle if there is a homeomorphism that is a local trivialization of the fiber bundle?

Exercise 28: Find a particular global section for all vector bundles.

Exercise 29: Consider the Klein bottle, which is constructed by “taping” the sides of two Möbius strips together. Is it possible to find an imbedding of the Klein bottle in $\mathbb{R}^3$? If not, what is the lowest n such that the Klein bottle can be imbedded in $\mathbb{R}^n$?

Exercise 30: Consider the projective plane $\mathbb{P}^2$, which is the set

of all lines through the origin in $\mathbb{R}^3$. Is it possible to find an imbedding of $\mathbb{P}^2$ in $\mathbb{R}^3$? (Remember that it has to be C^∞).

Exercise 31: Prove the Local Submersion Theorem.

Exercise 32: Is it possible for a mapping between two manifolds to be both an immersion and a submersion? Explain why or why not.

Challenge 1: Consider a topological manifold M .

(a) Show that if the dimension of M is 1, 2, or 3, then there exists a smooth structure on M . Do some topological manifolds of higher dimension admit no smooth structures? Is it possible for some topological manifolds to admit non-equivalent smooth structures? If so, what dimensions guarantee equivalent smooth structures? Prove all of your answers.

(b) Show that every one-dimensional smooth manifold is diffeomorphic to either $\mathbb{R}$ or S^1 , with their typical smooth structures. What about two or three dimensional smooth manifolds? What about higher dimensional manifolds?

(c) Consider a tensor field T . In a coordinate neighborhood U with a basis of vector fields ∂_i and a basis of dual vector fields dx^i , express the tensor field in terms of components of the tensor. Define a tensor field in terms of a tensor bundle. What is the relationship between the two objects?

(d) Show that any tensor density is the product of a tensor with some scalar density.

Challenge 2: Consider the following definitions:

A sheaf $\mathcal{F}$ on a topological space X assigns to each open set $U \subseteq X$ a set $\mathcal{F}(U)$, together with restriction maps

$$\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V), \quad V \subseteq U, \quad (6.58)$$

satisfying the following properties:

Definition 195 (Sheaf Axioms). *A presheaf $\mathcal{F}$ is a sheaf if:*

- (Identity) $\rho_{UU} = \text{id}$.
- (Compatibility) $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$ for $W \subseteq V \subseteq U$.
- (Gluing) If $\{U_i\}$ is an open cover of U , and sections $s_i \in \mathcal{F}(U_i)$ satisfy

$$\rho_{U_i, U_i \cap U_j}(s_i) = \rho_{U_j, U_i \cap U_j}(s_j), \quad (6.59)$$

then there exists a unique $s \in \mathcal{F}(U)$ such that $\rho_{U, U_i}(s) = s_i$.

(a) Show that the projection $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$, given by $\pi(x, y) = x$, defines a fiber bundle with fiber $\mathbb{R}$. Explicitly construct a local trivialization.

(b) Consider the map $\pi : S^1 \times \mathbb{R} \rightarrow S^1$. Show that this is a fiber bundle. Now consider the Möbius strip M with projection $\pi : M \rightarrow S^1$. Explain why M is locally trivial but not globally trivial.

(c) Let $E \rightarrow B$ be a fiber bundle. Define a *section* of the bundle over an open set $U \subseteq B$. Show that the assignment

$$U \mapsto \{\text{continuous sections } s : U \rightarrow E \text{ with } \pi \circ s = \text{id}_U\} \quad (6.60)$$

forms a presheaf.

(d) Prove that the presheaf of sections in (c) is in fact a sheaf.

(e) Let $\mathcal{F}$ be a sheaf on X . Define the stalk $\mathcal{F}_x$ at a point $x \in X$. Explain intuitively how stalks capture “local behavior,” and relate this idea to the local triviality condition of fiber bundles.

Definition 196 (Stalk). *The stalk of a sheaf $\mathcal{F}$ at $x \in X$ is defined as the direct limit*

$$\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U). \quad (6.61)$$

(f) Compare the following two ideas:

- A fiber bundle is locally a product space.

- A sheaf is locally determined by compatible data.

Formulate a precise analogy between these two statements. In what sense can sections of a bundle be viewed as elements of a sheaf?

CHAPTER 7

Covariant Differentiation

7.1 Affine Connections

In order to effectively and rigorously define the covariant derivative, we must first be introduced to the concept of an affine connection. The definition of an affine connection is rather abstract, but think of it as a sort of way to define a differentiation on vector fields. This will become particularly important once we define the covariant derivative later on in this chapter. The affine connection takes two different vector fields and maps them to a new vector field. It also follows certain properties that will become clear in the definition.

Definition 197 (Affine Connection). *Let $\mathcal{X}(M)$ be the set of all C^∞ vector fields on a smooth manifold M . An affine connection is a mapping*

$$\nabla : \mathcal{X}(M) \times \mathcal{X}(M) \rightarrow \mathcal{X}(M) \quad (7.1)$$

such that

$$(X, Y) \mapsto \nabla_X Y \quad (7.2)$$

Furthermore, the affine connection must satisfy the following conditions: given vector fields X, Y, Z and C^∞ functions f, g on M , we have

$$\nabla_X(Y + Z) = \nabla_X Y + \nabla_X Z \quad (7.3)$$

$$\nabla_{fX+gY}Z = f\nabla_X Z + g\nabla_Y Z \quad (7.4)$$

$$\nabla_X(fY) = f\nabla_X Y + X(f)Y \quad (7.5)$$

One may find it rather strange that the affine connection is linear over $C^\infty(M)$ in the first argument but not in the second argument. We will explore this interesting property below.

Remark 33. $\nabla_X Y$ is tensorial in X , but not Y . That is to say, we have, by the fundamental properties of the affine connection,

$$\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z \quad (7.6)$$

However,

$$\nabla_X(fY + gZ) \neq f\nabla_X Y + g\nabla_X Z \quad (7.7)$$

As it turns out, because of this, we have that $\nabla_X Y$ is a tensor of type $(1, 1)$, if we hold Y constant. Furthermore, if we hold X constant, the map $Y \mapsto \nabla_X Y$ is a first-order differential operator. This allows us to define differentiation on manifolds, as we will soon see.

The following theorem concerns the fact that the affine connection depends on the values of the vector field locally, and not globally. Consider the following theorem with proof below:

Theorem 63. *Let $p \in M$. Then $\nabla_X Y(p)$ depends only on the values of X and Y in some neighborhood of p .*

Proof. Suppose we have some vector fields Y_1 and Y_2 that are equal to each other on a neighborhood of p . Then we have that $Y_1 - Y_2 = 0$ on some neighborhood U of p . If we have $f \in C^\infty(M)$ be a function such that $f = 1$ near p and the support of f is a subset of U (that is, the set of points where $f \neq 0$ is a subset of U). Then we have that

$$Y_1 - Y_2 = f(Y_1 - Y_2) \quad (7.8)$$

If we apply ∇_X , we have that

$$\nabla_X(Y_1 - Y_2) = \nabla_X(f(Y_1 - Y_2)) = f\nabla_X(Y_1 - Y_2) + X(f)(Y_1 - Y_2) \quad (7.9)$$

Recall the fundamental properties of the affine connection. At p , we have $f(p) = 1$ and $X(f)(p) = 0$, while $Y_1 - Y_2 = 0$ near p , so we have

$$\nabla_X(Y_1 - Y_2)(p) = 0 \quad (7.10)$$

Therefore we have

$$\nabla_X Y_1(p) - \nabla_X Y_2(p) = 0 \implies \nabla_X Y_1(p) = \nabla_X Y_2(p) \quad (7.11)$$

□

This theorem sheds some light on the behavior of the affine connection. If we have two vector fields Y and Z that are equivalent on some neighborhood $U \subset M$, where M is a smooth manifold, then we have that $\nabla_X Y = \nabla_X Z$ on U . Note that this behavior may not extend to the global case.

Remark 34. *A refinement of the argument for Theorem 63 shows that $\nabla_X Y(p)$ depends only on $X(p)$ and the first-order behavior of Y near p . Thus an affine connection is a local first-order operator.*

We are aware that the affine connection is not tensorial in both arguments; that is to say, it is only tensorial in one argument. However, with a bit of creativity, we are able to conjure a mathematical object built from affine connections such that is tensorial in both arguments. Observe:

Theorem 64. *Let ∇ and $\tilde{\nabla}$ be two affine connections on M . Let*

$$A(X, Y) = \nabla_X Y - \tilde{\nabla}_X Y \quad (7.12)$$

Then A is tensorial in both arguments. Specifically, A is a tensor field of type $(1, 2)$.

Proof. We verify tensoriality in each argument. First, given $f \in C^\infty(M)$, we have

$$A(fX, Y) = \nabla_{fX} Y - \tilde{\nabla}_{fX} Y = f\nabla_X Y - f\tilde{\nabla}_X Y = fA(X, Y) \quad (7.13)$$

Clearly A is tensorial in the first argument. Then, we have

$$A(X, fY) = \nabla_X (fY) - \tilde{\nabla}_X (fY) \quad (7.14)$$

Using the properties of the affine connection, we have that this is equal to

$$(f\nabla_X Y + X(f)Y) - (f\tilde{\nabla}_X Y + X(f)Y) \quad (7.15)$$

$$= f\nabla_X Y - f\tilde{\nabla}_X Y = fA(X, Y) \quad (7.16)$$

Thus A is tensorial in both arguments. $\square$

We now consider the affine connection ∇ . We will find that we are able to extend the definition of an affine connection from vector fields to tensor fields, in such a way that preserves the properties of the affine connection as is. We present the formal statement below:

Theorem 65. *Let ∇ be an affine connection on M . There exists a unique extension of ∇ to all tensor fields on M such that*

1. *For functions $f \in C^\infty(M)$, we have that $\nabla_X f = X(f)$*
2. *For vector fields Y , the extension agrees with the given definition*
3. *The operator satisfies a Leibniz rule (product rule) with respect to tensor products*

If we consider the affine connection as a means of differentiation of vector fields, it only makes sense that the extension of the affine connection serves to be a means of differentiation of tensor fields.

Remark 35. *This result allows one to differentiate arbitrary tensor fields using only the axioms of an affine connection. The proof is obtained by defining the action on simple tensors and extending by linearity.*

We consider a more conceptual definition of the affine connection, that is equivalent to the more mathematically rigorous statements made earlier in our definition of the affine connection.

Remark 36. *An affine connection may be viewed as a rule that assigns to each pair of vector fields X, Y a new vector field $\nabla_X Y$, satisfying:*

- *Tensoriality in the first argument*
- *A Leibniz rule in the second argument*

This serves as reinforcement of a concept that we have derived from the raw axioms of the affine connection; namely, that it allows us to differentiate vector fields in a certain way. This will serve us well when we define such an object as the covariant derivative soon.

Remark 37. *In this sense, an affine connection provides a way of differentiating vector fields along vector fields, defined purely in terms of its algebraic properties.*

7.2 Christoffel Symbols

To begin our exposition of Christoffel symbols, we will first consider the case in two dimensions; that is, of a surface embedded in $\mathbb{R}^3$. Before we define a Christoffel symbol under these conditions, we must first introduce a couple of new terminology. Provided below is the definition of a surface patch. This should ring a bell as a particular case of a mapping from $\mathbb{R}^n$ to a manifold.

Definition 198 (Surface Patch). *A surface patch σ is a mapping from some open set $U \subset \mathbb{R}^2$ to $V \subset S \subset \mathbb{R}^3$, such that*

$$(u, v) \mapsto \sigma(u, v) \quad (7.17)$$

We can define upon such a surface patch an orientation N , which is a choice of a vector N normal to the tangent vectors of the surface σ_u and σ_v .

On a surface, and for manifolds too, in fact, there exist objects known as the first and second fundamental forms before. You are familiar with the first fundamental form—it also goes by the name of the metric tensor. The second fundamental form may be less well known. A very quick definition of these forms will be provided, as well as some other useful information, before we define the Christoffel symbols of surfaces.

Definition 199 (First and Second Fundamental Forms). *The first fundamental form should be familiar, as it is the two-dimensional metric tensor:*

$$I = Edu^2 + 2Fdudv + Gdv^2 = g_{11}du^2 + 2g_{12}dudv + g_{22}dv^2 \quad (7.18)$$

Note that we define $E = \langle \sigma_u, \sigma_u \rangle$, $F = \langle \sigma_u, \sigma_v \rangle = \langle \sigma_v, \sigma_u \rangle$, and $G = \langle \sigma_v, \sigma_v \rangle$.

$$II = edu^2 + 2fdudv + gdv^2 \quad (7.19)$$

Now, we have that

$$e = \langle \sigma_{uu}, N \rangle \quad (7.20)$$

$$f = \langle \sigma_{uv}, N \rangle \quad (7.21)$$

and

$$g = \langle \sigma_{vv}, N \rangle \quad (7.22)$$

The Christoffel symbol arises as a consequence of the affine connection of a given manifold—that is, if you have an affine connection, you have Christoffel symbols. These symbols essentially encapsulate how much the manifold “deviates” from flatness. This will become increasingly clear as we observe more formulas with Christoffel symbols in them. The first particular set of equations that we will look at concern the rate of change of the tangent vectors. Certainly there does exist a normal component to that rate of change, but if there exists a component to that rate of change tangential to the tangent vectors of the manifold, then the Christoffel symbols are nonzero. In that case, we see how the Christoffel symbols imply a sort of deviation from flatness.

Definition 200 (Christoffel Symbols in Two Dimensions). *The Christoffel symbols, Γ_{ij}^k , such that $i, j, k = \{1, 2\}$, are maps from $U \subset \mathbb{R}^2$ to $\mathbb{R}$ such that these equations are true:*

$$\sigma_{uu} = \Gamma_{11}^1 \sigma_u + \Gamma_{11}^2 \sigma_v + eN \quad (7.23)$$

$$\sigma_{uv} = \Gamma_{12}^1 \sigma_u + \Gamma_{12}^2 \sigma_v + fN \quad (7.24)$$

$$\sigma_{vu} = \Gamma_{21}^1 \sigma_u + \Gamma_{21}^2 \sigma_v + fN \quad (7.25)$$

$$\sigma_{vv} = \Gamma_{22}^1 \sigma_u + \Gamma_{22}^2 \sigma_v + gN \quad (7.26)$$

One important property of Christoffel symbols in surfaces embedded in $\mathbb{R}^3$ is that they are symmetric, in a sort. That is, we have that the bottom two indices can always be swapped with each other, and the value will remain the same. This may provide useful for various calculations. We will note what happens when we extend our vision to manifolds in n dimensions soon.

Theorem 66. *By this definition, under these conditions, we always have that $\Gamma_{21}^1 = \Gamma_{12}^1$ and $\Gamma_{21}^2 = \Gamma_{12}^2$.*

Proof. Note that $\sigma_{uv} = \sigma_{vu}$ through the commutativity of partial derivatives. This implies that

$$\Gamma_{12}^1 \sigma_u + \Gamma_{12}^2 \sigma_v + fN = \Gamma_{21}^1 \sigma_u + \Gamma_{21}^2 \sigma_v + fN \quad (7.27)$$

$$\implies \Gamma_{12}^1 \sigma_u + \Gamma_{12}^2 \sigma_v = \Gamma_{21}^1 \sigma_u + \Gamma_{21}^2 \sigma_v \quad (7.28)$$

Note that σ_u and σ_v are linearly independent. This implies that

$$\Gamma_{21}^1 = \Gamma_{12}^1 \quad (7.29)$$

and

$$\Gamma_{21}^2 = \Gamma_{12}^2 \quad (7.30)$$

as desired. $\square$

At this point, we can start proving certain elementary formulas for Christoffel symbols based on the setting of a surface embedded in $\mathbb{R}^3$. We consider the relationship between the Christoffel symbol and the first fundamental form and its indices' derivatives here.

Theorem 67. *The following identities are true.*

$$2(EG - F^2)\Gamma_{11}^1 = GE_u - 2FF_u + FE_v \quad (7.31)$$

$$2(EG - F^2)\Gamma_{11}^2 = 2EF_u - EE_v - FE_u \quad (7.32)$$

$$2(EG - F)^2\Gamma_{12}^1 = GE_v - FG_u \quad (7.33)$$

$$2(EG - F^2)\Gamma_{12}^2 = EG_u - FE_v \quad (7.34)$$

$$2(EG - F^2)\Gamma_{22}^1 = 2GF_v - GG_u - FG_v \quad (7.35)$$

$$2(EG - F^2)\Gamma_{22}^2 = EG_v - 2FF_v + FG_u \quad (7.36)$$

Proof. Consider that, by the product rule in calculus, we have

$$E_u = \langle \sigma_u, \sigma_u \rangle_u = \langle \sigma_{uu}, \sigma_u \rangle + \langle \sigma_u, \sigma_{uu} \rangle = 2\langle \sigma_{uu}, \sigma_u \rangle \quad (7.37)$$

$$\implies \langle \sigma_{uu}, \sigma_u \rangle = \frac{1}{2}E_u \quad (7.38)$$

Then, by the definition of Christoffel symbols in two dimensions, we have that

$$\langle \sigma_{uu}, \sigma_u \rangle = \langle \Gamma_{11}^1 \sigma_u + \Gamma_{11}^2 \sigma_v + eN, \sigma_u \rangle \quad (7.39)$$

$$= \Gamma_{11}^1 \langle \sigma_u, \sigma_u \rangle + \Gamma_{11}^2 \langle \sigma_v, \sigma_u \rangle + e \langle N, \sigma_u \rangle = \Gamma_{11}^1 E + \Gamma_{11}^2 F + 0 \quad (7.40)$$

$$= \Gamma_{11}^1 E + \Gamma_{11}^2 F \quad (7.41)$$

This gives us

$$\Gamma_{11}^1 E + \Gamma_{11}^2 F = \frac{1}{2} E_u \quad (7.42)$$

Furthermore, we examine what happens when we calculate the following expression:

$$F_u - \frac{1}{2} E_v = \langle \sigma_u, \sigma_v \rangle_u - \frac{1}{2} \langle \sigma_u, \sigma_u \rangle_v \quad (7.43)$$

$$= \langle \sigma_{uu}, \sigma_v \rangle + \langle \sigma_u, \sigma_{uv} \rangle - \langle \sigma_u, \sigma_{uv} \rangle = \langle \sigma_{uu}, \sigma_v \rangle \quad (7.44)$$

Furthermore, we obtain from the definition of Christoffel symbols

$$\langle \sigma_{uu}, \sigma_v \rangle = \langle \Gamma_{11}^1 \sigma_u + \Gamma_{11}^2 \sigma_v + eN, \sigma_v \rangle \quad (7.45)$$

$$= \Gamma_{11}^1 \langle \sigma_u, \sigma_v \rangle + \Gamma_{11}^2 \langle \sigma_v, \sigma_v \rangle + 0 = \Gamma_{11}^1 F + \Gamma_{11}^2 G \quad (7.46)$$

This gives us

$$\Gamma_{11}^1 F + \Gamma_{11}^2 G = F_u - \frac{1}{2} E_v \quad (7.47)$$

Now, multiply equation 7.42 by G :

$$\Gamma_{11}^1 EG + \Gamma_{11}^2 FG = \frac{1}{2} E_u G \quad (7.48)$$

Then multiply equation 7.47 by F :

$$\Gamma_{11}^1 F^2 + \Gamma_{11}^2 FG = FF_u - \frac{1}{2} E_v F \quad (7.49)$$

We subtract equation 7.49 from equation 7.48:

$$\Gamma_{11}^1 (EG - F^2) = \frac{1}{2} E_u G - FF_u + \frac{1}{2} E_v F \quad (7.50)$$

Multiplying by 2 gives us the desired result for the first equation:

$$2(EG - F^2)\Gamma_{11}^1 = GE_u - 2FF_u + FE_v \quad (7.51)$$

As you can see, the work required for this proof is rather tedious, although simple, and I do not think it would be helpful at all to work through the minutia of the entire proof for all six equations. Hence, I will leave the verification of the other identities as an exercise for the reader. $\square$

There are certain characteristics of the affine connection which may be of interest to the reader, for it will help us gain a deeper understanding of the Christoffel symbol. The first condition that we are interested in is that of compatibility with the Riemannian metric; some affine connections may be compatible with the Riemannian metric, but not all of them.

Definition 201 (Compatibility). *We say that a connection ∇ on a given Riemannian manifold M is compatible with the Riemannian metric $\langle \cdot, \cdot \rangle$ if the following equation holds for all vector fields $X, Y, Z \in \mathcal{X}(M)$:*

$$X\langle Y, Z \rangle = \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle \quad (7.52)$$

Similarly, an affine connection may be symmetric, but it may also not be symmetric. An affine connection that is both symmetric and compatible with the Riemannian metric is called a Riemannian connection, or Levi-Civita connection.

Definition 202 (Symmetric Connection). *A connection ∇ is symmetric if, for all $X, Y \in \mathcal{X}(M)$,*

$$\nabla_X Y - \nabla_Y X = [X, Y] \quad (7.53)$$

We note that, if such an affine connection is compatible with the Riemannian metric and symmetric, then we have that

$$X\langle Y, Z \rangle = \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle \quad (7.54)$$

$$Y\langle Z, X \rangle = \langle \nabla_Y Z, X \rangle + \langle Z, \nabla_Y X \rangle \quad (7.55)$$

$$Z\langle X, Y \rangle = \langle \nabla_Z X, Y \rangle + \langle X, \nabla_Z Y \rangle \quad (7.56)$$

If we have (7.54) + (7.55) - (7.56), we obtain

$$\begin{aligned} & X\langle Y, Z \rangle + Y\langle Z, X \rangle - Z\langle X, Y \rangle \\ &= \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle + \langle \nabla_Y Z, X \rangle \\ &+ \langle Z, \nabla_Y X \rangle - \langle \nabla_Z X, Y \rangle - \langle X, \nabla_Z Y \rangle \end{aligned} \quad (7.57)$$

$$\begin{aligned} &= \langle \nabla_X Y, Z \rangle - \langle Z, \nabla_Y X \rangle + 2\langle Z, \nabla_Y X \rangle + \langle \nabla_Y Z, X \rangle \\ &- \langle X, \nabla_Z Y \rangle + \langle Y, \nabla_X Z \rangle - \langle \nabla_Z X, Y \rangle \end{aligned} \quad (7.58)$$

$$\begin{aligned} &= \langle \nabla_X Y, Z \rangle - \langle \nabla_Y X, Z \rangle + \langle \nabla_Y Z, X \rangle - \langle \nabla_Z Y, X \rangle \\ &+ \langle \nabla_X Z, Y \rangle - \langle \nabla_Z X, Y \rangle + 2\langle Z, \nabla_Y X \rangle \end{aligned} \quad (7.59)$$

By the definition of symmetry, we have the following:

$$\begin{aligned} & X\langle Y, Z \rangle + Y\langle Z, X \rangle - Z\langle X, Y \rangle \\ &= \langle [X, Y], Z \rangle + \langle [Y, Z], X \rangle + \langle [X, Z], Y \rangle + 2\langle Z, \nabla_Y X \rangle \end{aligned} \quad (7.60)$$

Isolating $\langle Z, \nabla_Y X \rangle$, we obtain

$$\begin{aligned} \langle Z, \nabla_Y X \rangle &= \frac{1}{2}(X\langle Y, Z \rangle + Y\langle Z, X \rangle - Z\langle X, Y \rangle \\ &- \langle [X, Y], Z \rangle - \langle [Y, Z], X \rangle - \langle [X, Z], Y \rangle) \end{aligned} \quad (7.61)$$

Let us shelve this equation for now and define the Christoffel symbol in general. As expected based on what has been said before, the Christoffel symbol can be defined using the affine connection. In fact, we have the following as the definition:

Definition 203 (Christoffel Symbol). *We define the Christoffel symbol Γ_{ij}^k by the relation $\nabla_{e_i} e_j = \Gamma_{ij}^k e_k$.*

One can ponder the similarities between the two definitions of the Christoffel symbol given in this textbook; indeed, they are very much connected. However, if we have a Riemannian connection (that is, a connection both symmetric and compatible with the Riemannian metric), then the following theorem holds true, which gives us an excellent and straightforward, if not tedious, way to calculate each Christoffel symbol.

Theorem 68. *The formula for the Christoffel symbol in terms of the metric and its derivatives is as follows:*

$$\Gamma_{ij}^m = \frac{1}{2}(\partial_i g_{jk} + \partial_j g_{ki} - \partial_k g_{ij})g^{km} \quad (7.62)$$

Proof. Consider equation 7.61. Let $Z = e_k$, $Y = e_i$, and $X = e_j$. We are given that $\nabla_Y X = \nabla_{e_i} e_j = \Gamma_{ij}^l e_l$. We have that

$$\langle Z, \nabla_Y X \rangle = \langle e_k, \Gamma_{ij}^l e_l \rangle = \Gamma_{ij}^l \langle e_k, e_l \rangle = \Gamma_{ij}^l g_{kl} \quad (7.63)$$

By equation 7.61, we know that

$$\begin{aligned} \Gamma_{ij}^l g_{kl} &= \frac{1}{2}(e_j \langle e_i, e_k \rangle + e_i \langle e_k, e_j \rangle - e_k \langle e_j, e_i \rangle \\ &\quad - \langle [e_j, e_i], e_k \rangle - \langle [e_i, e_k], e_j \rangle - \langle [e_j, e_k], e_i \rangle) \end{aligned} \quad (7.64)$$

However, we know that, identifying $e_i = \partial_i$, since partial derivatives commute, we therefore have that $[e_i, e_j] = 0$, and so we have

$$\Gamma_{ij}^l g_{kl} = \frac{1}{2}(e_j \langle e_i, e_k \rangle + e_i \langle e_k, e_j \rangle - e_k \langle e_j, e_i \rangle) \quad (7.65)$$

Recall that $\langle e_i, e_j \rangle = g_{ij}$. Therefore,

$$\Gamma_{ij}^l g_{kl} = \frac{1}{2}(e_i g_{jk} + e_j g_{ik} - e_k g_{ij}) = \frac{1}{2}(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}) \quad (7.66)$$

Consider that $g_{kl} g^{km} = \delta_l^m$. Therefore,

$$\Gamma_{ij}^l g_{kl} g^{km} = \Gamma_{ij}^l \delta_l^m = \Gamma_{ij}^m = \frac{1}{2}(\partial_i g_{jk} + \partial_j g_{ki} - \partial_k g_{ij})g^{km} \quad (7.67)$$

We have achieved the desired result. $\square$

One would be forgiven for thinking that the Christoffel symbol is a tensor, for it has indices, both upstairs and downstairs, it has to do with curvature and other properties of manifolds somewhat like the metric tensor (and soon some other tensors that we will learn about), and, after all, this is a book about tensors. But it is not a tensor! The reader should figure out why so, especially after reading the rest of this section.

Remark 38. *The Christoffel symbol Γ_{ij}^k is not a tensor. The verification of this claim will be left up to the reader in the exercises.*

Up until now, we have taken for granted that the connection is Riemannian, when considering Christoffel symbols. But if we do not have a Riemannian connection, there are certain properties of Christoffel symbols that may be lost. For instance, the property of being torsion-free is guaranteed under a Riemannian connection, but may not necessarily be guaranteed with a regular affine connection. A torsion-free Christoffel symbol is symmetric, in a way; you can swap the bottom two indices and it will remain the same.

Definition 204 (Torsion-Free). *A connection ∇ is torsion-free if its Christoffel symbols satisfy the relation*

$$\Gamma_{ij}^k = \Gamma_{(ij)}^k = \frac{1}{2}(\Gamma_{ij}^k + \Gamma_{ji}^k) \quad (7.68)$$

Now, in the interest of the physicist reading this book, let us concern ourselves with the calculation of Christoffel symbols. Provided below is a fairly simple example of how to calculate Christoffel symbols. We consider the plane with polar coordinates particularly, and use the formula presented in Theorem 68. Pay close attention to what is happening—you should be able to figure out why the Christoffel symbol is not a tensor based on this example.

Theorem 69. *In the plane with polar coordinates, such that the metric is equal to $g_{rr} = 1$, $g_{r\theta} = 0$, $g_{\theta\theta} = r^2$ (the variables are r, θ , in that order), then we have that*

$$\Gamma_{rr}^r = \Gamma_{r\theta}^r = \Gamma_{rr}^\theta = \Gamma_{\theta\theta}^\theta = 0 \quad (7.69)$$

$$\Gamma_{\theta\theta}^r = -r \quad (7.70)$$

$$\Gamma_{r\theta}^\theta = \frac{1}{r} \quad (7.71)$$

Proof. The proof is unfortunately a lot of tedious calculus and algebra. I will selfishly leave the reader to do most of the work.

However, some examples will be provided on how to calculate Christoffel symbols:

$$\Gamma_{rr}^r = \frac{1}{2}((\partial_r g_{rr} + \partial_r g_{rr} - \partial_r g_{rr})g^{rr} + (\partial_r g_{r\theta} + \partial_r g_{\theta r} - \partial_\theta g_{rr})g^{\theta r}) \quad (7.72)$$

$$= \frac{1}{2}((0 + 0 - 0)1 + (0 + 0 - 0)0) = 0 \quad (7.73)$$

Then, we have

$$\Gamma_{r\theta}^r = \frac{1}{2}((\partial_r g_{r\theta} + \partial_\theta g_{rr} - \partial_r g_{r\theta})g^{rr} + (\partial_r g_{\theta\theta} + \partial_\theta g_{r\theta} - \partial_\theta g_{r\theta})g^{r\theta}) \quad (7.74)$$

$$= \frac{1}{2}((0 + 0 - 0)1 + (2r + 0 - 0)0) = 0 \quad (7.75)$$

Then, we have

$$\Gamma_{rr}^\theta = \frac{1}{2}((\partial_r g_{rr} + \partial_r g_{rr} - \partial_r g_{rr})g^{r\theta} + (\partial_r g_{r\theta} + \partial_r g_{r\theta} - \partial_\theta g_{rr})g^{\theta\theta}) \quad (7.76)$$

$$= \frac{1}{2} \left((0 + 0 - 0)0 + (0 + 0 - 0)\frac{1}{r^2} \right) = 0 \quad (7.77)$$

You should be able to gain some sort of idea of how to calculate Christoffel symbols from these examples. From there, after a little bit of calculus, we see that the values of the Christoffel symbols match those in the theorem, for the plane with polar coordinates. $\square$

We have spent a lot of time with these Christoffel symbols. But there exists another kind of Christoffel symbol; in fact, there exists exactly two kinds of Christoffel symbols. One is the kind that we know about, but the other can easily be derived from the first one. Let us see below for further information.

Remark 39. *There exist two kinds of Christoffel symbols. The Christoffel symbol of the second kind, Γ_{ij}^k , is the one that you are familiar with. The Christoffel symbols of the first kind, $\Gamma_{kij} = g_{kl}\Gamma_{ij}^l$, can be found by lowering the indices appropriately.*

7.3 The Covariant Derivative

The covariant derivative is best described with the idea that we want to have a version of the derivative vector that is tangent to the manifold; that is, the covariant derivative is essentially the derivative vector from the perspective of an object living in the manifold. For instance, when we travel along the Earth's equator, we think that we are going straight, when instead, we are actually taking a curved path in space. The former encapsulates the covariant derivative; the latter encapsulates the regular derivative. Before we define the covariant derivative properly, however, (and we will define it multiple times in different contexts), let us consider a vector field along a curve.

Definition 205 (Vector Field Along a Curve). *If we have a manifold M and a regular curve $\gamma : [0, a] \rightarrow M$ in the manifold M , then a vector field V along γ is considered to be a mapping $V : [0, a] \rightarrow T_p M$ for all $t \in [0, a]$.*

At this point, we are now able to define the covariant derivative. The covariant derivative, as mentioned before, is essentially the projection of the regular derivative onto the tangent space of a manifold containing the curve.

Definition 206 (Covariant Derivative). *We have that the covariant derivative $\frac{DV}{dt}$ or $\frac{D}{dt}V$ of a vector field V along a curve γ is the uniquely defined vector field along γ such that*

$$\left. \frac{DV}{dt} \right|_{t=t_0} = \text{proj}_{T_{\gamma(t_0)}M} \left. \frac{dV}{dt} \right|_{t=t_0} \quad (7.78)$$

The covariant derivative allows us to define the concept of parallelism mathematically. We consider a vector field to be parallel if the covariant derivative of the vector field vanishes; that is, if the arrows of the vector field are parallel with each other relative to the manifold that the vector field exists as part of the tangent space of.

Definition 207 (Parallel). *We have that a vector field V is*

considered to be parallel if we have that

$$\frac{DV}{dt} = 0 \quad (7.79)$$

on all $[0, a]$.

Now we consider some of the properties of covariant derivatives. The covariant derivative inherits the following properties thanks to the properties of the regular derivative. We consider the sum of vector fields, the scalar multiple of a vector field (where the scalar is a function), and the inner product of two vector fields.

Lemma 10. *The following properties of the covariant derivative hold.*

- If we have two vector fields V, W along a curve $\gamma : [0, a] \rightarrow M$, then

$$\frac{D}{dt}(V + W) = \frac{DV}{dt} + \frac{DW}{dt} \quad (7.80)$$

- If we have a smooth function $f : [0, a] \rightarrow \mathbb{R}$, then

$$\frac{D}{dt}(fV) = f'V + f \frac{DV}{dt} \quad (7.81)$$

- We have that

$$\frac{d}{dt}\langle V, W \rangle = \left\langle \frac{DV}{dt}, W \right\rangle + \left\langle V, \frac{DW}{dt} \right\rangle \quad (7.82)$$

Proof. The proof will be left to the reader. The reader should use the properties of regular derivatives to complete this proof. $\square$

As it turns out, the covariant derivative retains the above properties when we generalize to higher dimensions. We can describe the covariant derivative in terms of an affine connection, like so:

Theorem 70 (Covariant Derivative on a Manifold). *Given a manifold M equipped with an affine connection ∇ , if we have a differentiable curve $\gamma : I \rightarrow M$ and a vector field V on M , then there exists a unique vector field known as the covariant derivative, $\frac{DV}{dt}$ along γ . We have that, if W is a vector field along c ,*

$$\frac{D}{dt}(V + W) = \frac{DV}{dt} + \frac{DW}{dt} \quad (7.83)$$

and if f is a function on I , we have

$$\frac{D}{dt}(fV) = \frac{df}{dt}V + f\frac{DV}{dt} \quad (7.84)$$

Furthermore, if $V(t) = X(\gamma(t))$, where $X \in \mathcal{X}(M)$, we have

$$\frac{DV}{dt} = \nabla_{d\gamma/dt}X \quad (7.85)$$

Proof. This is an existence and uniqueness theorem, meaning that this theorem asserts the existence of some mathematical object as well as the uniqueness of this object under given conditions. Therefore, we need to prove both existence and uniqueness. We will start with uniqueness. Let us assume that such a covariant derivative exists, satisfying all properties listed. We find a system of coordinates that can describe some of the points on the curve γ ; that is, consider some $\mathbf{x} : U \subset \mathbb{R}^n \rightarrow M$ such that $\gamma(I) \cap \mathbf{x}(U) \neq \emptyset$. We can therefore express each $\gamma(t)$ as $(x^1(t), x^2(t), \dots, x^n(t))$, where $t \in I$. We can also define tangent vectors like so:

$$X_i = \frac{\partial}{\partial x^i} \quad (7.86)$$

Since there are n tangent vectors and all are linearly independent, it stands to reason that any vector in $T_pM = \mathbb{R}^n$ can be written as a linear combination of the tangent vectors listed. Hence, we can write

$$V = a^i X_i \quad (7.87)$$

Since we have that $\frac{D}{dt}(V + W) = \frac{DV}{dt} + \frac{DW}{dt}$ and $\frac{D}{dt}(fV) = \frac{df}{dt}V + f\frac{DV}{dt}$, it follows that

$$\frac{DV}{dt} = \frac{da^i}{dt}X_i + a^i\frac{DX_i}{dt} \quad (7.88)$$

Now, we know that $\frac{DX_i}{dt} = \nabla_{d\gamma/dt} X_i$, by the final property of the covariant derivative. We understand that $\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z$, and we necessarily have

$$\frac{d\gamma}{dt} = \frac{dx^i}{dt} X_i \quad (7.89)$$

so

$$\nabla_{d\gamma/dt} X_i = \nabla_{\frac{dx^j}{dt} X_j} X_i = \frac{dx^j}{dt} \nabla_{X_j} X_i \quad (7.90)$$

Recall that j is being summed over in the last expression. We therefore have that

$$\frac{DV}{dt} = \frac{da^i}{dt} X_i + \frac{dx^j}{dt} a^i \nabla_{X_j} X_i \quad (7.91)$$

Because the covariant derivative must adhere to this formula, the covariant derivative may only take on one value. Hence, it is impossible for two or more distinct covariant derivatives to exist. Therefore, if the covariant derivative exists, then it must be unique.

To prove existence, suppose that we are able to define the covariant derivative in one coordinate chart, $\mathbf{x}(U)$ using equation 7.91. It will be left as an exercise to the reader to prove that all properties exhibited by the covariant derivative can be verified through computation with that equation. However, it must be brought to attention that one coordinate chart does not necessarily define existence over the whole manifold, for the chart may not cover the whole manifold. If we define the covariant derivative in another coordinate neighborhood, $\mathbf{y}(W)$ with the same equation, it is clear that the covariant derivatives for both neighborhoods must be equivalent in the intersection, for we have already proven uniqueness before. Repeat this process indefinitely, and we are able to define the covariant derivative over the whole manifold with its requisite properties. It is clear, then, that it exists. $\square$

This is not the end of our generalizing of the covariant derivative, however. We can utilize Koszul connections to consider not

just vector fields, but also sections of fiber bundles in general, which we covered before in Chapter 6.

We denote $\Gamma(M, E)$ to be the space of all *smooth* sections of the vector bundle $E \rightarrow M$. The covariant derivative has a more general definition, where E may not necessarily be $\mathcal{X}(M)$. We consider the following definition:

Definition 208 (Koszul Connection). *A Koszul connection is a map $\nabla : \mathcal{X}(M) \times \Gamma(M, E) \rightarrow \Gamma(M, E)$ such that*

- $\nabla_{fX}(s) = f\nabla_X s$ for all $f \in C^\infty(M)$, $X \in \mathcal{X}(M)$, and $s \in \Gamma(M, E)$.
- $\nabla_{X+Y}s = \nabla_X s + \nabla_Y s$ for all $X, Y \in \mathcal{X}(M)$ and $s \in \Gamma(M, E)$.
- $\nabla_X(s_1 + s_2) = \nabla_X s_1 + \nabla_X s_2$ for all $X \in \mathcal{X}(M)$ and $s_1, s_2 \in \Gamma(M, E)$.
- $\nabla_X(fs) = (Xf)s + f\nabla_X s$ for all $f \in C^\infty(M)$, $X \in \mathcal{X}(M)$ and $s \in \Gamma(M, E)$.

The Koszul connection allows us to define what is known as a “natural covariant derivative,” which is a Koszul connection on an open set with some given properties that may seem familiar from our previous studies.

Definition 209 (Natural Covariant Derivative). *A natural covariant derivative ∇ gives each open set $U \subset M$ a map $\nabla^U : \mathcal{X}(M) \times \Gamma(M, E) \rightarrow \Gamma(M, E)$ such that*

- ∇^U is a Koszul connection on the restricted bundle $E|_U \rightarrow U$.
- We have that, for some open set $V \subset U$, $(\nabla_X^U \sigma)|_V = \nabla_{X|_V}^V \sigma|_V$.
- The value $(\nabla_X^U \sigma)(p)$ depends solely on the value of X .

We are now able to study some of the properties of the Koszul connection. Many of the properties are very advanced in nature, but we shall explore some of the more basic properties. We begin with the following theorem:

Theorem 71. *If we have a Koszul connection $\nabla : \mathcal{X}(M) \times \Gamma(M, E) \rightarrow \Gamma(M, E)$ for some vector bundle $\pi : E \rightarrow M$, then if we have for some neighborhood $U \subset M$ either $X|_U = 0$ or $\sigma|_U = 0$, then it must follow that*

$$(\nabla_X \sigma)(p) = 0 \quad (7.92)$$

for all $p \in U$.

Proof. The proof is left as an exercise for the reader. $\square$

Now let us define the connection form. The connection form arises naturally from taking the covariant derivative of a basis of sections. Recall the Einstein summation formula as presented in the formula below.

Definition 210 (Connection Form). *A connection form ω_j^i is a form such that, for a basis of sections e_i and $X \in \mathcal{X}$, we have*

$$\nabla_X e_j = \omega_j^i(X) e_i \quad (7.93)$$

We can represent the covariant derivative of any section in terms of these connection forms; all it requires is a calculation. We will not go through the calculation in this text.

Theorem 72. *If we can represent $s \in \Gamma(M, E)$ as a linear combination $s = s^i e_i$, then*

$$\nabla_X s = (X s^i + \omega_j^i(X) s^j) e_i \quad (7.94)$$

(Do not forget the Einstein summation convention!)

Let us come back from abstraction and Koszul connections and consider the covariant derivative in coordinates. We begin with the covariant derivative of a vector, which readily allows itself to be shown in terms of the partial derivative and the Christoffel symbol, as seen in the previous section.

Definition 211 (Covariant Derivative in Coordinates). *We define the coordinate derivative of a vector field in terms of the partial derivative and the Christoffel symbol:*

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\lambda}^\nu V^\lambda \quad (7.95)$$

Note that the covariant derivative transforms covariantly; that is, the covariant derivative of a tensor must also be a tensor. We also wish for the covariant derivative to contain some additional properties.

Remark 40. *The covariant derivative contains a couple additional properties:*

1. $\nabla_\mu(T^\nu{}_{\nu\rho}) = (\nabla T)^\nu{}_{\nu\rho}$
2. $\nabla_\mu\phi = \partial_\mu\phi$

These properties will allow us to define the covariant derivative of a 1-form, like so:

Definition 212 (Covariant Derivative of a 1-Form). *The covariant derivative of a 1-form is defined as*

$$\nabla_\mu\omega_\nu = \partial_\mu\omega_\nu - \Gamma_{\mu\nu}^\lambda\omega_\lambda \quad (7.96)$$

One will be able to derive the covariant derivative of a general tensor from the above formulas.

Definition 213 (Covariant Derivative of a Tensor). *The covariant derivative of a tensor generally is defined as*

$$\begin{aligned} \nabla_\mu T^{\nu_1\nu_2\dots\nu_m}{}_{\rho_1\rho_2\dots\rho_n} &= \partial_\mu T^{\nu_1\nu_2\dots\nu_m}{}_{\rho_1\rho_2\dots\rho_n} \\ &+ \Gamma_{\mu\lambda}^{\nu_1} T^{\lambda\nu_2\dots\nu_m}{}_{\rho_1\rho_2\dots\rho_n} + \Gamma_{\mu\lambda}^{\nu_2} T^{\nu_1\lambda\dots\nu_m}{}_{\rho_1\rho_2\dots\rho_n} + \dots \\ &- \Gamma_{\mu\rho_1}^\lambda T^{\nu_1\nu_2\dots\nu_m}{}_{\lambda\rho_2\dots\rho_n} - \Gamma_{\mu\rho_2}^\lambda T^{\nu_1\nu_2\dots\nu_m}{}_{\rho_1\lambda\dots\rho_n} - \dots \end{aligned} \quad (7.97)$$

7.4 Parallel Transport

Now we examine the concept of parallel transport. We recall that a vector field V on a smooth curve γ is parallel if the covariant derivative of the vector field vanishes on the entire interval of the curve. We can use this concept to transport a vector from one point on a manifold to another in such a way that both vectors are in the same parallel vector field.

Definition 214 (Parallel Transport). *If we have a manifold M and a smooth curve $\gamma : [a, b] \rightarrow M$, parallel transport along the curve γ is a function $P_\gamma : T_{\gamma(a)}M \rightarrow T_{\gamma(b)}M$ which sends each vector $v_a \in T_{\gamma(a)}M$ to the vector $v_b \in T_{\gamma(b)}M$ such that v_a and v_b are both elements of a parallel vector field along γ at $t = a$ and $t = b$ respectively. If we have a piecewise smooth curve γ which can be split up into smooth curves γ_i , $i = [1, n]$, then the parallel transport along γ is the function $P_\gamma : T_{\gamma(a)}M \rightarrow T_{\gamma(b)}M$ defined by taking the parallel transport of the vector for each smooth segment:*

$$P_\gamma = P_{\gamma_n} \circ P_{\gamma_{n-1}} \circ \cdots \circ P_{\gamma_2} \circ P_{\gamma_1} \quad (7.98)$$

We will prove soon that there is a unique parallel vector field along a curve given a tangent vector at an endpoint of the curve.

We can define a specific type of parallel transport known as the holonomy, which is when a vector is parallel transported along a closed curve.

Definition 215 (Holonomy). *In the above definition, if γ is a closed curve, or if $\gamma(a) = \gamma(b) \in M$, then we can label the parallel transport function $P_\gamma : T_{\gamma(a)}M \rightarrow T_{\gamma(a)}M$ as the holonomy around γ .*

Now we will use our knowledge of affine connections to prove the claim we proposed earlier.

Theorem 73. *If we have a manifold M equipped with an affine connection ∇ , a curve $\gamma : I \rightarrow M$, where $I = [a, b]$, and a vector $v_0 \in T_{\gamma(a)}M$, then there exists a unique parallel vector field V along the curve γ , where $V(a) = v_0$.*

Proof. We will begin by showing that this theorem is true if $\gamma(I)$ is contained in one coordinate neighborhood $\mathbf{x}(U)$, where $\mathbf{x} : U \subset \mathbb{R}^n \rightarrow M$. Suppose

$$\mathbf{x}^{-1}(\gamma(t)) = (x_1(t), \dots, x_n(t)) \quad (7.99)$$

and

$$v_0 = v_0^j \partial_j \quad (7.100)$$

Then it must be true that

$$0 = \frac{DV}{dt} = \frac{dv^j}{dt} \partial_j + \frac{dx_i}{dt} v^j \nabla_{\partial_i} \partial_j \quad (7.101)$$

Recall that

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k \quad (7.102)$$

Thus, we have

$$0 = \left(\frac{dv^k}{dt} + v^j \frac{dx_i}{dt} \Gamma_{ij}^k \right) \partial_k \quad (7.103)$$

We assert that the system of differential equations

$$0 = \frac{dv^k}{dt} + \Gamma_{ij}^k v^j \frac{dx_i}{dt} \quad (7.104)$$

has a unique solution given initial conditions $v^k(a) = v_0^k$, if such a solution exists. Furthermore, a solution must exist for any $t \in I$, as the differential equation is linear. Therefore, the parallel vector field exists and is unique if the curve is contained in one coordinate neighborhood. Now, suppose the curve was not contained in one coordinate neighborhood. The curve must be covered by some number of coordinate neighborhoods. Now, $\gamma(I)$ is compact, as I is a closed and bounded interval, and so, it is a standard result from topology that a finite number of coordinate neighborhoods may be able to cover such a curve. In each of these coordinate neighborhoods, we have shown that we can define a parallel vector field on each. Because such a vector field must be unique, when the intersections are not empty, then it follows that the vector fields must coincide. Hence, it can be shown easily from there that V exists and is unique on the curve. $\square$

We can define the parallel transport of a tensor as well. We begin by considering the covariant derivative in coordinates:

Remark 41. *Given a curve represented in coordinates by $x^\mu(\lambda)$ for $\mu = 1, \dots, n$, we have that*

$$\frac{D}{d\lambda} = \frac{dx^\mu}{d\lambda} \nabla_\mu \quad (7.105)$$

where ∇ is the covariant derivative in coordinates, or the affine connection ∇_{X_μ} .

7.5 Geodesics

7.6 Introduction to Curvature and Torsion

7.7 Exercises

Exercise 1: Based on the results of Theorem 65, come up with a formula for

$$\nabla_X(Y \otimes Z) \quad (7.106)$$

Exercise 2: Explain why A is a tensor field of type $(1, 2)$. Explain why ∇ is a tensor field of type $(1, 1)$ under certain conditions. [Hint: Consider that A is viewed as a map

$$A : \mathcal{X}(M) \times \mathcal{X}(M) \rightarrow \mathcal{X}(M) \quad (7.107)$$

and ∇ under specific conditions is a map

$$\nabla : \mathcal{X}(M) \rightarrow \mathcal{X}(M) \quad (7.108)$$

Note that the former at a given point $p \in M$ is equivalent to

$$T_p M \times T_p M \rightarrow T_p M \implies T_p M \times T_p M \times T_p^* M \rightarrow \mathbb{F} \quad (7.109)$$

You can come up with something similar for ∇ . Use this as a springboard for your explanation.]

Exercise 3: Let ∇ be symmetric, that is,

$$\nabla_X Y - \nabla_Y X = [X, Y] \quad (7.110)$$

where $[X, Y](f) = X(Y(f)) - Y(X(f))$. Find

$$\nabla_{\partial_i} \partial_j - \nabla_{\partial_j} \partial_i \quad (7.111)$$

where $i \neq j$.

Exercise 4: Suppose that M is a smooth manifold with an affine connection ∇ . Suppose that $c : I \rightarrow M$ is a C^∞ curve and X_0 is a tangent vector to M at $p = c(t_0)$, where $t_0 \in I$. Show that there exists a unique vector field X along c such that $X(p) = X_0$ and $\nabla_{c'} X = 0$ for all $t \in I$.

Exercise 5: Find all Christoffel symbols for the paraboloid $(x, y, z) = (u, v, u^2 + v^2)$.

Exercise 6: Verify the rest of Theorem 67.

Exercise 7: Let $E = 1$ and $F = 0$. Find all six Christoffel symbols in terms of G and its derivatives.

Exercise 8: Show that the Christoffel symbols Γ_{ij}^k in $\mathbb{R}^n$ with the Euclidean metric are all equal to 0.

Exercise 9: The half-plane with the Lobatchevski metric is defined as the set

$$\mathbb{H}^2 = \{(x, y) \in \mathbb{R}^2; y > 0\} \quad (7.112)$$

such that $g_{11} = g_{22} = \frac{1}{y^2}$ and $g_{12} = 0$. Calculate all six Christoffel symbols Γ_{ij}^k .

Exercise 10: Show that the Christoffel symbol Γ_{ij}^k is not a tensor.

Exercise 11: Complete the rest of Theorem 69.

Exercise 12: Show that

$$\Gamma_{\mu\nu}^{\mu} = \frac{1}{\sqrt{|g|}} \partial_{\nu} \sqrt{|g|} \quad (7.113)$$

and therefore, express

$$\nabla_{\mu} V^{\mu} = \partial_{\mu} V^{\mu} + \Gamma_{\mu\nu}^{\mu} V^{\nu} \quad (7.114)$$

in terms of $\sqrt{|g|}$, V^{μ} , and their derivatives.

Exercise 13: Find the Christoffel symbols of the first kind in the plane with polar coordinates.

Exercise 14: Verify Lemma 10, on the elementary properties of the covariant derivative. Use the properties of the traditional derivative that you are familiar with.

Exercise 15: Using the following formula,

$$\frac{DV}{dt} = \frac{da^i}{dt} X_i + \frac{dx^j}{dt} a^i \nabla_{X_j} X_i \quad (7.115)$$

verify the following equations under the necessary assumptions:

$$\frac{D}{dt}(V + W) = \frac{DV}{dt} + \frac{DW}{dt} \quad (7.116)$$

$$\frac{D}{dt}(fV) = \frac{df}{dt} V + f \frac{DV}{dt} \quad (7.117)$$

$$\frac{DV}{dt} = \nabla_{d\gamma/dt} X \quad (7.118)$$

Exercise 16: Can you think of a curve $\gamma(t)$ and a vector field V such that $\frac{DV}{dt} = \frac{dV}{dt}$? The less trivial the better.

Exercise 17: Prove that if we have a Koszul connection $\nabla : \mathcal{X}(M) \times \Gamma(M, E) \rightarrow \Gamma(M, E)$ for some vector bundle $\pi : E \rightarrow M$, then if we have for some neighborhood $U \subset M$ either $X|_U = 0$ or $\sigma|_U = 0$, then it must follow that

$$(\nabla_X \sigma)(p) = 0 \quad (7.119)$$

for all $p \in U$.

Exercise 18: Show that $\nabla_\mu \delta_\sigma^\lambda = 0$.

Exercise 19: Does the holonomy of a tangent vector along a smooth curve always result in the exact same tangent vector? Prove the claim if so, or find a counterexample if not.

Exercise 20: Verify that

$$\frac{D}{d\lambda} = \frac{dx^\mu}{d\lambda} \nabla_\mu \quad (7.120)$$

based on the properties of affine connections.

Exercise 21: Show that if the connection is metric-compatible, then

$$\frac{D}{d\lambda} g_{\mu\nu} = 0 \quad (7.121)$$

Conclude that the covariant derivative of the inner product of vectors v^μ and w^ν vanishes when the vectors are parallel transported along a curve $x^\rho(\lambda)$.

CHAPTER 8

Riemannian Geometry and Curvature Tensors

8.1 Metric Connection

8.2 Levi-Civita Connection

8.3 Riemann Curvature Tensor

8.4 Ricci Tensor

8.5 Scalar Curvature

8.6 Bianchi Identities

8.7 Ricci Decomposition

8.8 Applications to General Relativity

8.9 Exercises

CHAPTER 9

Tensor Calculus in Coordinates

9.1 Index Gymnastics

9.2 Variation of Tensor Fields

9.3 The Lie Derivative

9.4 The Exterior Derivative

9.5 Differential Forms

9.6 The Hodge Dual

9.7 Exercises

Part III

Applications

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Continuum and Solid Mechanics

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10.2 Strain Tensor

10.3 Deformation Gradient

10.4 Constitutive Relations

10.5 Isotropy and Anisotropy

10.6 Electromagnetic Tensors

10.7 Elasticity

10.8 Magneto-Elasticity

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Fluid Mechanics and Turbulence

11.1 Velocity Gradient Tensor

11.2 Rate of Strain

11.3 Vorticity Tensor

11.4 Reynolds Stress Tensor

11.5 Turbulence Modeling

11.6 Fluid Flows

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CHAPTER 12

General Relativity

12.1 Stress-Energy Tensor

12.2 Einstein Tensor

12.3 Einstein Field Equations

12.4 Metric Signature Conventions

12.5 Worldlines

12.6 Curvature Coupling

12.7 Schwarzschild Metric

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CHAPTER 13

Advanced Applications

13.1 Tensor Decompositions in Data Science

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13.3 Tensors in Quantum Physics

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Advanced Geometry

14.1 Characteristic Classes

14.2 Tensor Fields in Topology

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14.6 More on Differential Forms

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APPENDIX A

Notation Conventions

A.1 Einstein Summation Convention

A.2 Index Positions

APPENDIX B

Common Tensors

APPENDIX C

Linear Algebra Prerequisites

C.1 Systems of Linear Equations

C.2 Matrices

C.3 Eigenvalues and Eigenvectors

C.4 The Inner Product

APPENDIX D

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